MASTERMIND SCHOLARS EDUCATIONAL CENTRE

(MSEC)

"Dream, Believe and Achieve Excellence"

BIOLOGY

PART 1

THE SKELETAL SYSTEM

Skeletons, Muscles and Movement

Mastermind Notes • Diagrams • 40 Objectives • 7 Theory Questions • Answers

MASTERMIND NOTES

Types of skeleton • Bone & cartilage • Axial skeleton • Vertebrae • Limbs & girdles • Joints & muscles

Name: …… Class: ……

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LEARNING OUTCOMES

By the end of this topic, the learner should be able to:

1. Define **locomotion** and name the systems that bring about movement in animals.
2. Describe the three **types of skeleton** (exoskeleton, endoskeleton, hydrostatic skeleton) with examples, advantages and disadvantages.
3. Describe the **structure of bone** (Haversian system) and **cartilage**, and state the differences and similarities between them.
4. Describe the **axial skeleton**: skull, vertebral column, ribs and sternum.
5. Identify and describe each **type of vertebra** (cervical, atlas, axis, thoracic, lumbar, sacral, caudal) and relate its features to its function.
6. Describe the **appendicular skeleton**: fore limb, hind limb, pectoral girdle and pelvic girdle.
7. Classify **joints**, describe a synovial joint, and explain how **antagonistic muscles** bring about movement.
8. State the **functions of the mammalian skeleton**.

KEY IDEAS

- Movement is brought about by the **skeletal, muscular and nervous systems** working together.
- There are three types of skeleton: **exoskeleton** (outside), **endoskeleton** (inside) and **hydrostatic skeleton** (fluid under pressure).
- The mammalian skeleton is an **endoskeleton** made of **bone** and **cartilage** — both are connective tissues.
- It is divided into the **axial skeleton** (skull, vertebral column, ribs, sternum) and the **appendicular skeleton** (limbs and limb girdles).
- Bones meet at **joints**; muscles pull on bones through **tendons** in **antagonistic pairs** to cause movement.

PART 1: MOVEMENT, LOCOMOTION AND SUPPORTING STRUCTURES

All living things can move at least part of their body. **Movement** is a change in position of part of an organism (e.g. bending the arm). **Locomotion** is the movement of the **whole body** from one place to another (e.g. walking, swimming, flying).

1.1 Systems that bring about movement

Movement in animals is achieved **collectively** by:

- **Skeletal system** — provides a rigid framework and levers.
- **Muscular system** — contracts to pull on the skeleton.
- **Nervous system** — sends impulses that coordinate and control muscle contraction.

In other words, movement depends on **skeletal, muscular and nervous tissues**.

1.2 What is a skeleton?

The **skeleton** is the supporting structural framework of an animal, which gives the body shape, protects delicate organs and provides surfaces for muscle attachment for movement and locomotion.

1.3 Supporting structures (skeletal materials) in animals

Supporting material	**Found in**
Body fluid (coelomic fluid)	Earthworm and other soft-bodied animals
Cuticle (made of chitin)	Arthropods — insects, crustaceans, spiders
Bone	Vertebrates
Cartilage	Vertebrates (entire skeleton in cartilaginous fish)
Muscles (muscle tone)	All mammals — help to hold organs in position
Also: shells (calcium carbonate)	Snails, clams (molluscs)

NOTE

Do not confuse **supporting structures** (the *materials* a skeleton is made of — chitin, bone, cartilage, fluid) with **types of skeleton** (exo-, endo-, hydrostatic). Each type of skeleton is built from one or more supporting materials.

PART 2: TYPES OF SKELETON

There are **three** types of skeleton: **exoskeleton**, **endoskeleton** and **hydrostatic skeleton**.

2.1 Exoskeleton

An **exoskeleton** is a hard, non-living skeleton found **outside** the body, covering it. *Example:* arthropods such as insects (grasshopper, cockroach) and crustaceans (crab, prawn) have an exoskeleton made of **chitin**. In crustaceans it is hardened further with calcium salts.

Advantages of the exoskeleton

1. Protects the body against **mechanical injury**.
2. Its waxy layer **prevents desiccation** (drying out) — a key adaptation for life on land.
3. Provides a surface for the **attachment of muscles** (on its inner side).
4. Gives the body shape and support; the jointed parts act as levers.

Disadvantages of the exoskeleton

1. It **restricts growth** — the animal can only grow by shedding the old skeleton (**ecdysis** or **moulting**).
2. During moulting, the animal is soft and **vulnerable** to predators and water loss.
3. It becomes very **heavy** as the animal gets bigger, which limits the size of arthropods.

2.2 Endoskeleton

An **endoskeleton** is a hard, living skeleton found **inside** the body. All vertebrates have an endoskeleton made mainly of **bone** with some **cartilage** — **except** the cartilaginous fishes (dogfish, sharks, rays), whose skeleton is made of **cartilage only**.

In mammals, cartilage is found in the **embryo**, the **pinna** (earlobe), the **tip of the nose**, **parts of the ribs**, **between the vertebrae** and at the **ends of bones at joints**.

Ossification

Cartilage is a softer tissue and **bone is a harder tissue** made of calcium and phosphorus. As a mammal grows, cartilage is transformed into bone by the **addition of mineral deposits of calcium and phosphorus** — a process called **ossification**. Ossification makes the skeleton harder but **more brittle** (more likely to break).

THINK & ANSWER

Q: Why is the skeleton of adult mammals such as humans more fragile and more likely to fracture than that of a young individual?

A: The skeleton of a young mammal contains much **cartilage**, which is soft and flexible. With age, ossification replaces cartilage with **bone**, which is harder but **brittle**. In old age, bone also loses minerals faster than they are replaced, so bones become weaker and break more easily.

2.3 Hydrostatic Skeleton

A **hydrostatic skeleton** is a type of skeleton in which the **muscles of the body wall contract against internal fluid** (coelomic fluid) held under pressure. *Examples:* earthworm, maggot, caterpillar, mosquito larva, jellyfish, starfish and sea anemone. It is common among **lower (simple) organisms**.

Structure

- An outer **body wall** with a **muscle layer**.
- The body wall surrounds a **fluid-filled coelom** (coelomic fluid). Fluid cannot be compressed, so it transmits the force of muscle contraction.
- The muscle layer is made up of **longitudinal muscles** and **circular muscles**, which act **antagonistically**.

Movement in the earthworm (or maggot)

- Contraction of **longitudinal muscles** → the segment becomes **shorter and fatter**.
- Contraction of **circular muscles** → the segment becomes **longer and thinner**.
- Waves of these contractions (**peristalsis**) pass along the body. **Chaetae** (bristles) grip the soil so the worm pushes forward and does not slip back.

Fig. 1: Movement of the earthworm (hydrostatic skeleton)

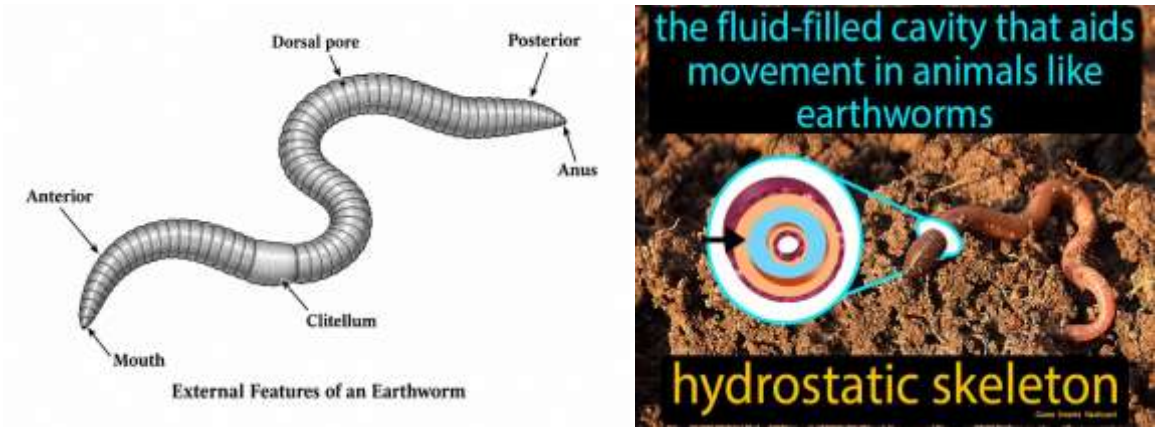


Fig. 1: Movement of the earthworm (hydrostatic skeleton)

Advantages of the hydrostatic skeleton

1. **Movement:** the fluid shape allows the animal to move easily by swimming and burrowing.
2. **Escape from danger:** the flexible shape lets it squeeze into narrow places to escape predators.
3. **Repair:** it heals more easily than bone because lost fluid is easily refilled.
4. **Protection:** the fluid cushions internal organs.

Disadvantages of the hydrostatic skeleton

1. Limited attachment for body organs.
2. Such animals are restricted mainly to **aquatic or moist habitats**, because water is needed to refill the skeletal fluid.
3. Terrestrial members **lack strength** and cannot support a large body on land.

THINK & ANSWER

Q: "The penis in humans is a hydrostatic organ." Explain this statement.

A: The penis has no bone. It becomes firm and rigid when its **spongy (erectile) tissue fills with blood**, which is held under pressure. Like the coelomic fluid of an earthworm, this **fluid under pressure** gives the organ its shape and stiffness — the principle of a hydrostatic skeleton.

2.4 Differences between Endoskeleton and Exoskeleton

Endoskeleton	Exoskeleton
Located **inside** the body, internal to the muscles.	Located **outside** the body, external to the muscles.
Made of **living tissue** (bone, cartilage).	Made of **non-living** material (chitin).
Grows with the organism, so no ecdysis.	Cannot grow; must be **shed (ecdysis)** for growth.
Muscles are attached to its **outer** surface.	Muscles are attached to its **inner** surface.
Relatively light; allows **large body size**.	Heavy; **limits body size**.
Found in vertebrates (e.g. man, rabbit, fish).	Found in arthropods (e.g. insects, crabs).

PART 3: SKELETAL TISSUES IN MAMMALS

The mammalian skeleton is an **endoskeleton** made up of **bone** and **cartilage**. Both are **connective tissues**.

> **RECALL**
> Mammals have four main types of tissue: **nervous**, **muscular**, **connective** and **epithelial**. **Connective tissue** consists of living cells embedded in a non-living organic and/or inorganic material called the **extracellular matrix**. Connective tissues include **blood**, **bone** and **cartilage**. (Blood is made of blood cells in a matrix of water, mineral salts, glucose, amino acids and blood proteins.)

3.1 Bone Tissue

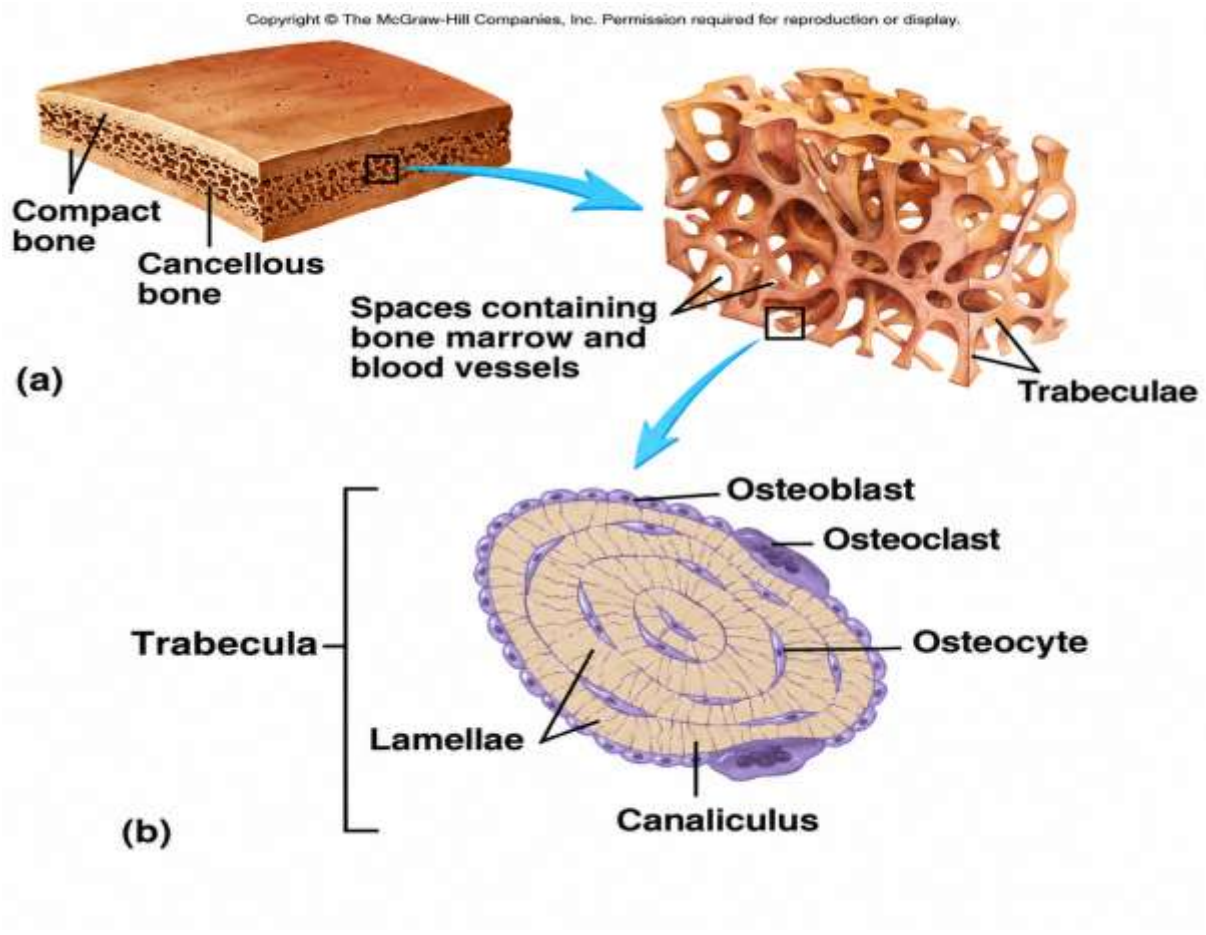
In bone, the extracellular matrix is made of:

- **Inorganic materials** — calcium salts: **calcium phosphate** and **calcium carbonate**. The main mineral crystal in bone and teeth is **hydroxyapatite** (a form of calcium phosphate). These give bone its **hardness**.
- **Organic materials** — a protein called **collagen**, which gives bone some **toughness and slight flexibility**.

Bone cells

Cell	Role
Osteoblasts	Precursor (young) cells that **secrete the matrix** and so form **new bone**.
Osteocytes	Mature bone cells: osteoblasts that have become **fully enclosed** by the matrix they secreted. They **maintain** the bone matrix.
Osteoclasts	Cells that **break down (resorb)** old or damaged bone and release its minerals into the blood.

Resorption and remodelling: Osteoclasts degrade old bone tissue and release the minerals to the blood; osteoblasts then lay down new bone tissue to replace it. This continuous process keeps bone strong and helps repair fractures.

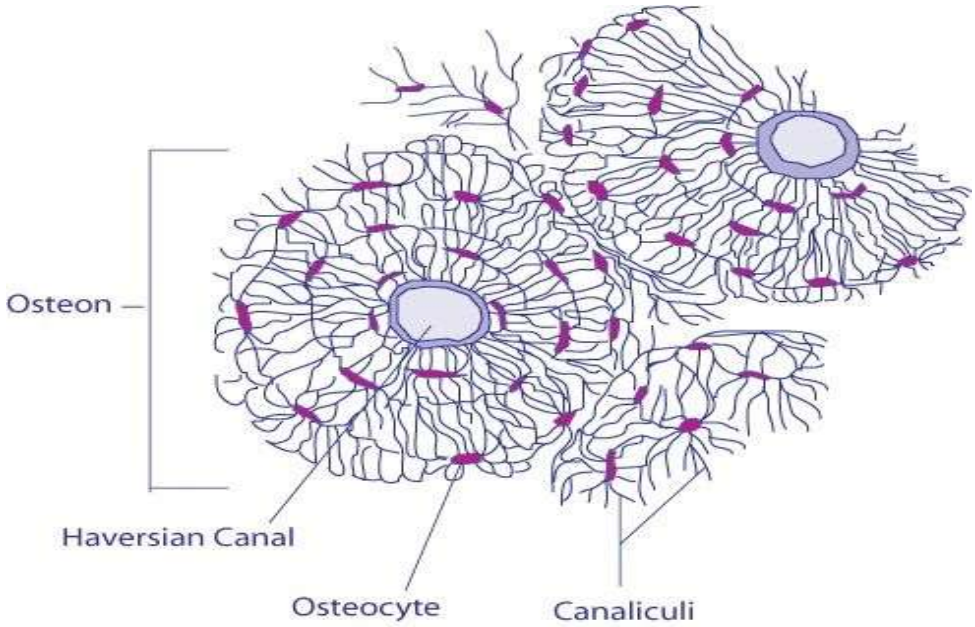


3.2 Structure of Bone Tissue (the Haversian System)

Bone is a **tough, hard and non-elastic** connective tissue made of living cells embedded in a non-living matrix. The basic structural unit of compact bone is the **Haversian system**, also called the **osteon**. Bone is made up of many repeated Haversian systems.

- Each Haversian system has a central canal called the **Haversian canal**, which contains **blood vessels and nerves**.
- The canal is surrounded by **concentric rings (layers) of matrix** called **lamellae** (singular: lamella).
- Between successive lamellae are small spaces called **lacunae** (singular: lacuna), each containing a bone cell (**osteocyte**).
- Tiny channels called **canaliculi** (singular: canaliculus) connect the lacunae to one another and to the central canal. They carry nutrients and oxygen to the osteocytes and remove wastes.

Fig. 2: The Haversian systems (osteons) — transverse section of compact bone



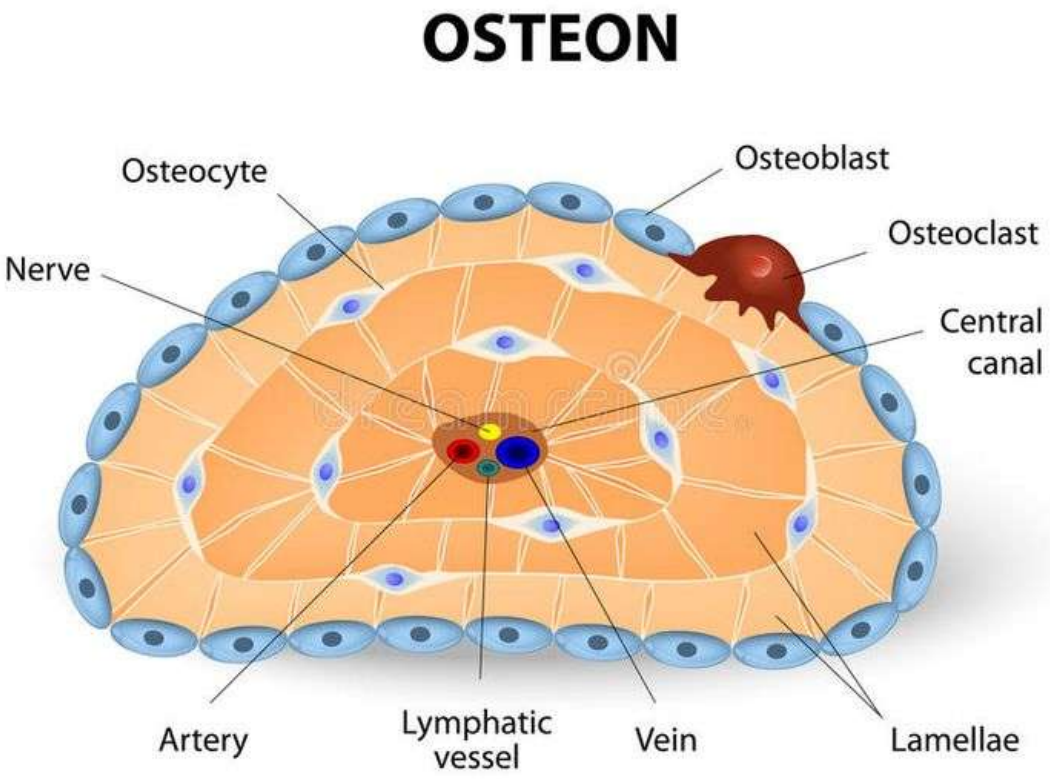


Fig. 2: The Haversian systems (osteons) — transverse section of compact bone

SUMMARY FOR EXAMS — Describing bone tissue

1. Bone is a tough, hard, non-elastic connective tissue.
2. It is made of living cells in an organic (collagen) and inorganic (calcium phosphate, calcium carbonate) matrix.
3. The matrix is arranged in concentric layers called lamellae.
4. Its basic unit is the Haversian system (osteon), with a central Haversian canal carrying blood vessels and nerves.
5. Osteocytes lie in lacunae between the lamellae.
6. Canaliculi connect the lacunae to each other and to the Haversian canal.

EXAM TIP

Key components of the osteon: Haversian canal, lamellae, lacunae, osteocytes, canaliculi. Remember the spelling is **Haversian** (after Clopton Havers) — not "Harvesian".

3.3 Structure of Cartilage

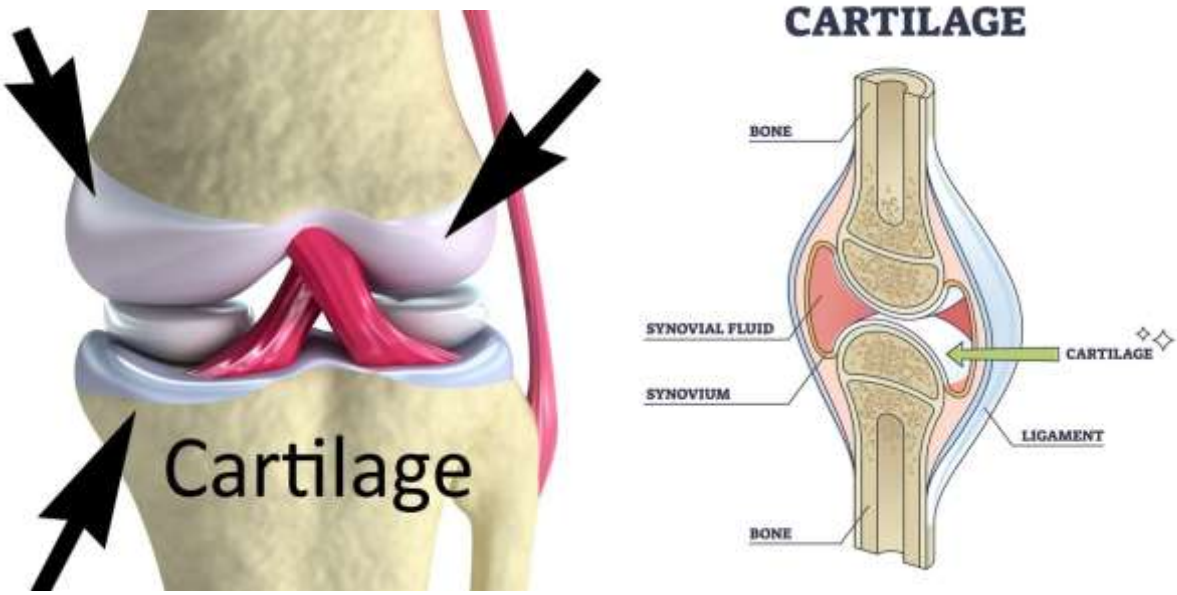
Cartilage is a **tough but flexible** connective tissue with great **tensile strength** (it resists pulling).

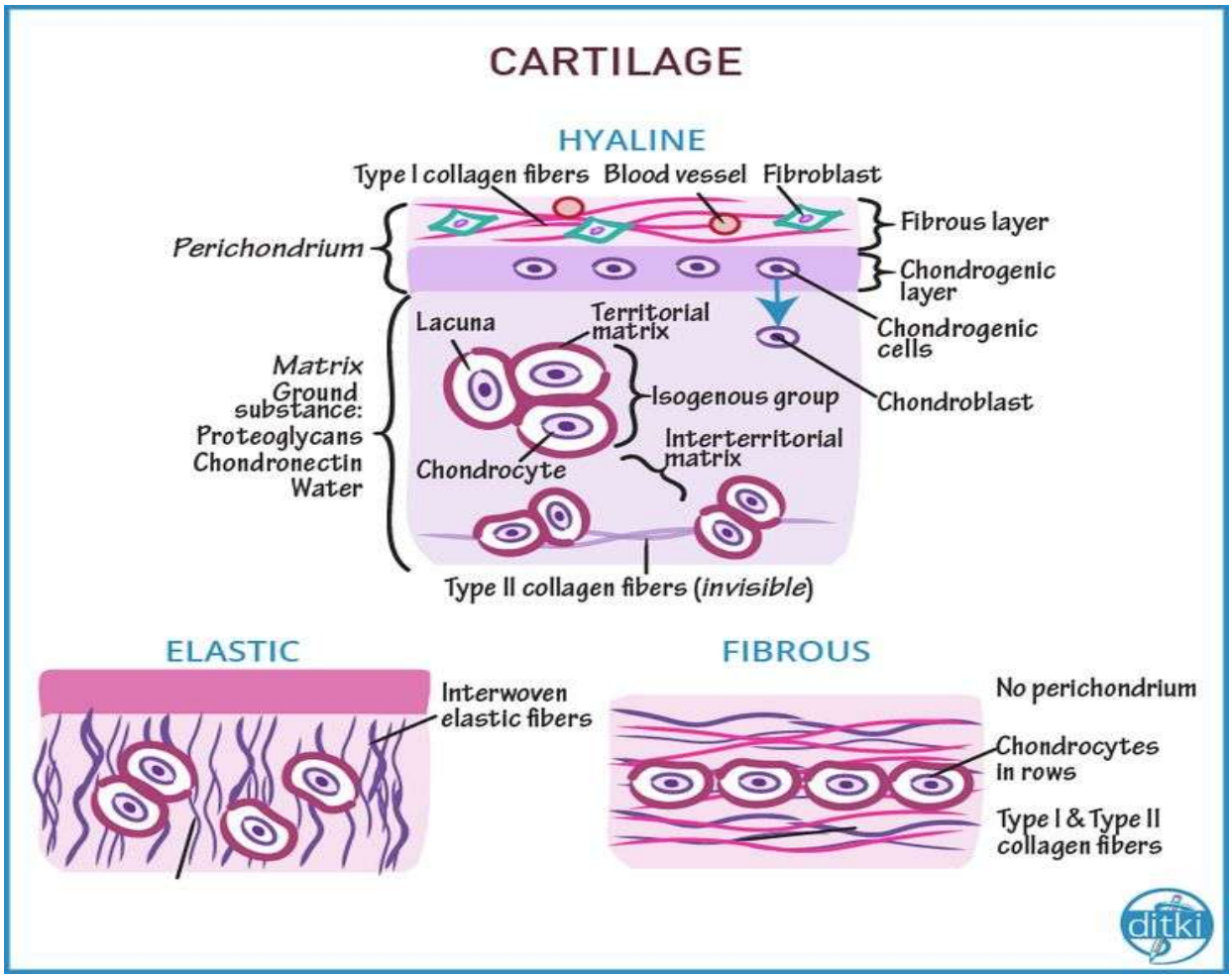
- It is made of living cells embedded in an extracellular matrix called **chondrin**.
- The matrix is mainly **carbohydrates and protein fibres (collagen)**. It is purely **organic** — there are no calcium deposits.
- In the matrix are spaces called **lacunae**, which contain cartilage cells called **chondrocytes**.
- **Chondroblasts** secrete the organic matrix; once enclosed in it, they become **chondrocytes**.
- Cartilage is **avascular** — it has **no blood vessels and no nerves**. It obtains nutrients by **diffusion** through the matrix, so its **repair is slow**.

3.4 Location of Cartilage in Mammals

Location	Type of cartilage
Embryo (most of the skeleton at first)	Hyaline
Tip of the nose	Hyaline
Parts of the ribs (costal cartilage joining ribs to sternum)	Hyaline
Ends of long bones at joints (articular cartilage)	Hyaline

Fig. 3: Structure of cartilage





Location	Type of cartilage
Larynx and trachea (C-shaped rings)	Hyaline
Intervertebral discs (between the vertebrae) and pubic symphysis	Fibrocartilage
Pinna (earlobe) and epiglottis	Elastic cartilage

Types of cartilage

- **Hyaline cartilage** — the most common; smooth and glassy. *Found in:* nose, trachea, larynx, ends of bones.
- **Fibrocartilage** — has many dense collagen fibres, so it is the toughest; absorbs shock. *Found in:* intervertebral discs.
- **Elastic cartilage** — contains elastic fibres, so it is very flexible and springs back. *Found in:* pinna, epiglottis.

Functions of cartilage

1. Provides **support** while still allowing flexibility (e.g. keeps the trachea open, gives the pinna shape).
2. Reduces **friction** between bones at joints.
3. Acts as a **shock absorber** (intervertebral discs).
4. Forms the skeleton of the embryo, which is later ossified; allows growth of long bones.

3.5 Bone versus Cartilage

Similarities

- Both are **connective tissues** with living cells in an **extracellular matrix**.
- Both contain **collagen** fibres and have **tensile strength**.
- In both, cells are found in spaces called **lacunae**.
- Both give **support** to the body.

Differences

Bone	Cartilage
Hard, rigid and non-elastic.	Softer, flexible and elastic.
Has **blood vessels and nerves** (in Haversian canals).	**No** blood vessels and nerves (avascular).
Matrix is **organic and inorganic**.	Matrix is **purely organic** (chondrin).
Has calcium deposits.	No calcium deposits.
Less collagen fibres.	More collagen fibres (hence elastic).
Porous (has canals).	Non-porous.
Cells: osteocytes; organised in Haversian systems.	Cells: chondrocytes; no Haversian systems.

Bone	Cartilage
Repair is faster (good blood supply).	Repair is slow (nutrients by diffusion).

PART 4: THE MAMMALIAN SKELETON — OVERVIEW

The mammalian skeleton is made up of two divisions:

- **(a) Axial skeleton** — the skull, vertebral column, ribs and sternum. It forms the main axis of the body.
- **(b) Appendicular skeleton** — the limbs and the limb girdles (pectoral and pelvic).

Fig. 4: The mammalian (human) skeleton — front view

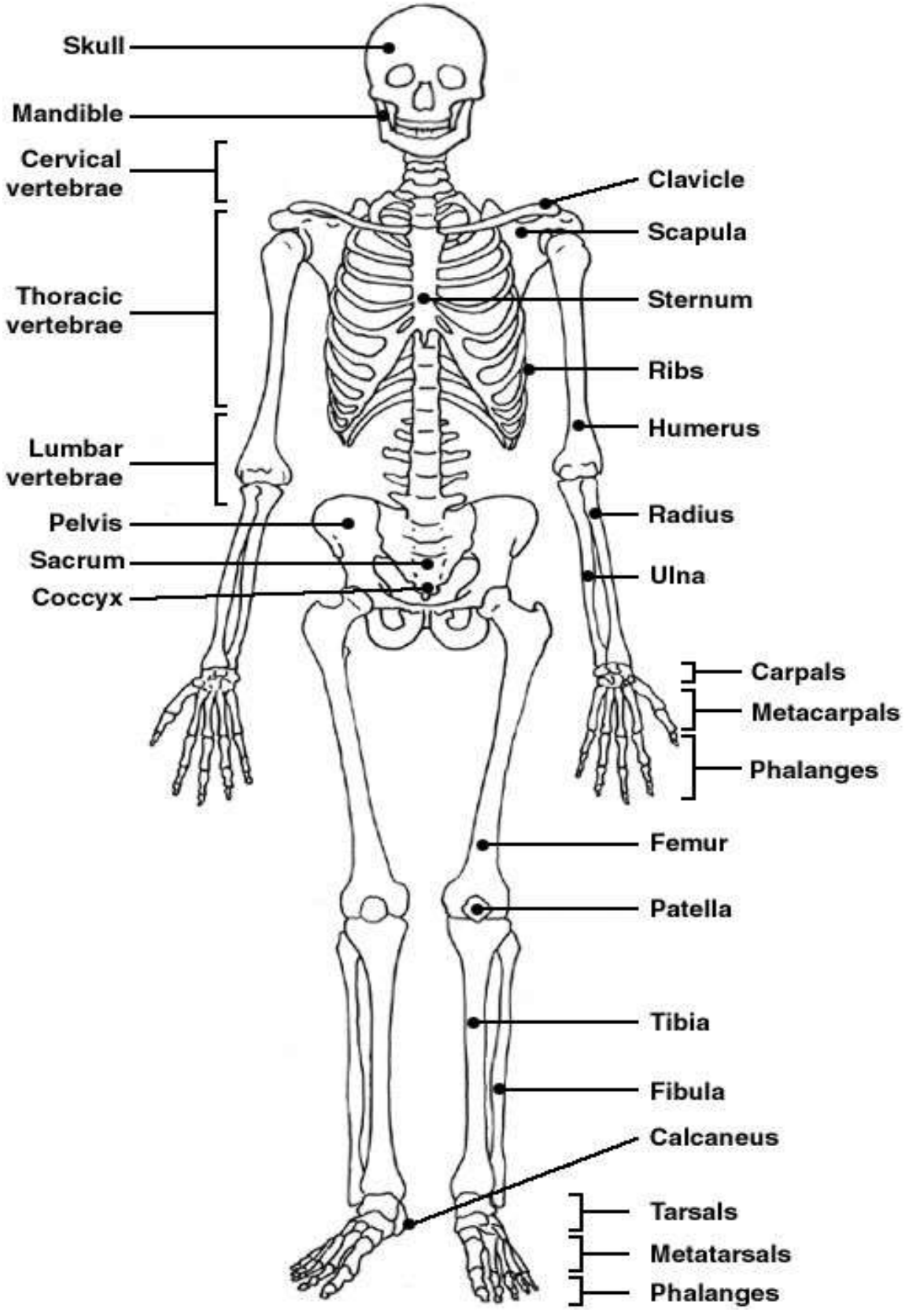


Fig. 4: The mammalian (human) skeleton — front view

Division	Parts	Main bones
Axial skeleton	Skull	Cranium, facial bones, upper and lower jaws
	Vertebral column	Cervical, thoracic, lumbar, sacral, caudal vertebrae
	Rib cage	Ribs and sternum
Appendicular skeleton	Fore limb	Humerus, radius, ulna, carpals, metacarpals, phalanges
	Hind limb	Femur, patella, tibia, fibula, tarsals, metatarsals, phalanges
	Girdles	Pectoral (scapula, clavicle); pelvic (ilium, ischium, pubis)

PART 5: THE AXIAL SKELETON

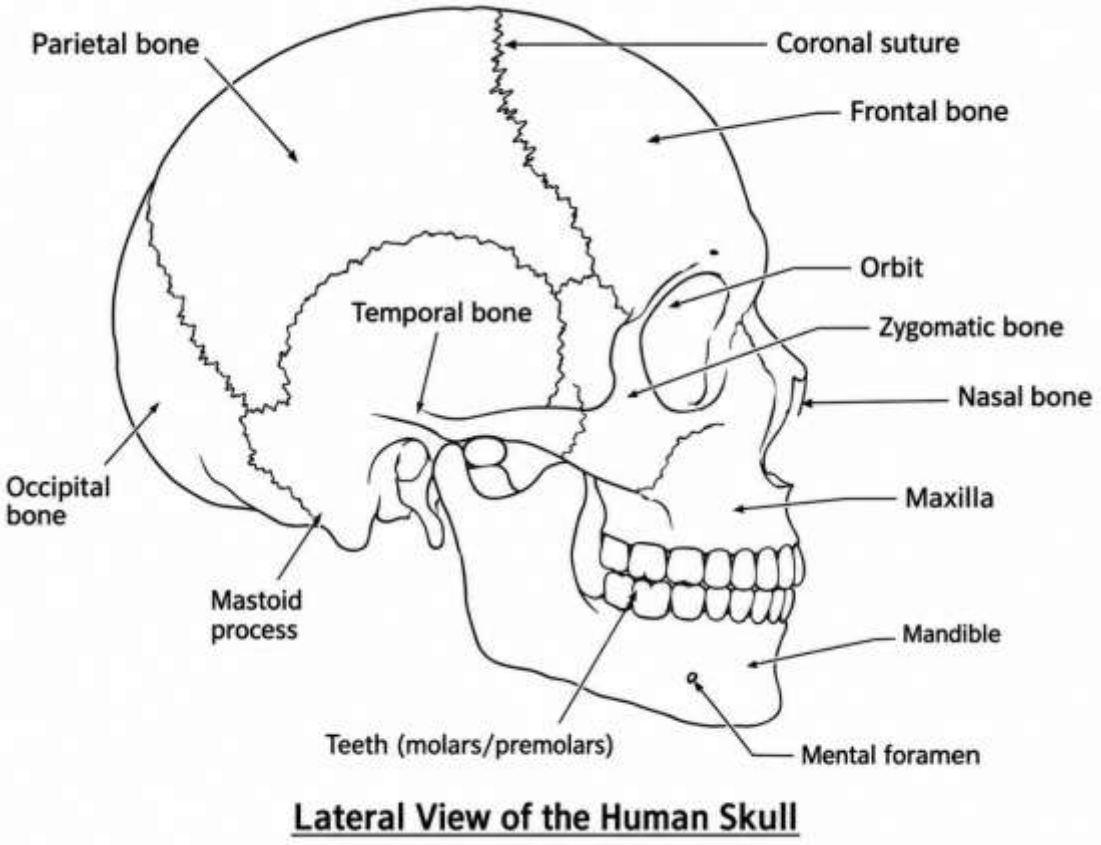
5.1 The Skull

The skull is the bone of the head. It is divided into three parts:

1. **The cranium (brain case):** holds and protects the brain. It is made of several wide, flat bones joined together by irregular, immovable joints called **sutures**.
2. **The facial skeleton:** supports the eyes, nose and the muscles of the cheek. It contains the **orbits (eye sockets)** and **nasal cavity**.
3. **The jaws:** the **upper jaw** (fixed to the cranium) and **lower jaw (mandible)**, which bear the teeth. The lower jaw is the only freely movable bone of the skull.

At the back and base of the skull are two smooth knobs called **occipital condyles**, which articulate with the atlas. The spinal cord leaves the skull through a large hole called the **foramen magnum**.

Fig. 5: The human skull (side view)



Functions of the skull

1. Protects the **brain**, eyes, ears and nose (sense organs).
2. Supports the **teeth** and provides the jaws for chewing.
3. Gives **shape** to the head and face.
4. Provides surfaces for **attachment of muscles** of the face, jaw and neck.

5.2 The Vertebral Column

The vertebral column (backbone or spine) is made up of a series of bones called **vertebrae** (singular: vertebra), held together by **ligaments** and separated from each other by discs of cartilage called **intervertebral discs**.

NOTE

- The **ligaments** hold the vertebrae together, preventing dislocation, but still allow flexibility of the column.
- Each vertebra is separated from the next by an **intervertebral disc** (fibrocartilage), which **absorbs shock** and **prevents friction** between vertebrae.

Functions of the vertebral column

1. **Protects the spinal cord**, which runs through the neural canal.
2. **Supports** the body and head and keeps the body upright.
3. Allows **flexibility** — bending and twisting of the trunk.
4. Provides surfaces for **attachment of muscles** and ribs.
5. Supports and attaches the **girdles** (especially the pelvic girdle).

Regions of the vertebral column

There are **five types of vertebrae**, each modified for the function of the region where it is found.

Type of vertebra	Region of the body	Rat	Rabbit	Human
Cervical	Neck	7	7	7
Thoracic	Chest	13	12–13	12
Lumbar	Abdomen / lower back	6	6–7	5
Sacral	Hip	4	4	5 (fused → sacrum)
Caudal	Tail	30±	16	4 (fused → coccyx)
Total	5 types	**60±**	**45–47**	**33**

EXAM TIP

Almost all mammals — from mouse to giraffe — have **7 cervical vertebrae**. A very common exam question: *How many vertebrae are in the human vertebral column?* **Answer: 33** (7 + 12 + 5 + 5 + 4).

5.3 The Generalised (Typical) Vertebra

A **generalised vertebra** is a diagram showing the basic plan of all vertebrae — the features common to all of them.

Fig. 6: A generalised / typical vertebra (anterior view)

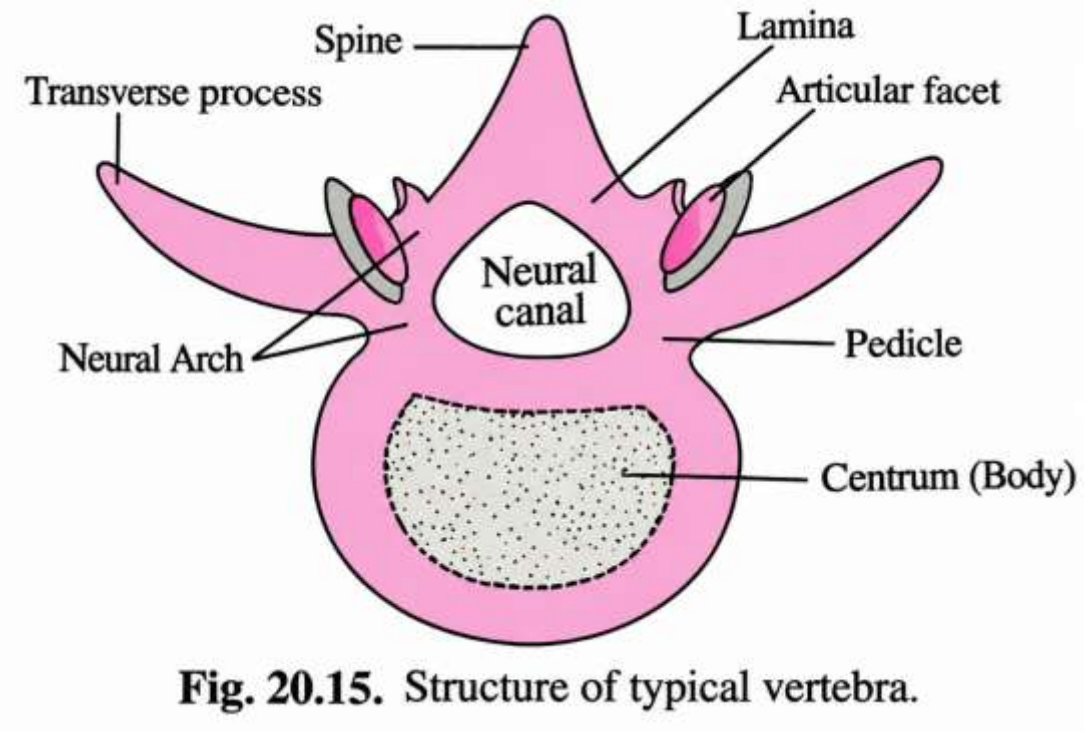


Fig. 20.15. Structure of typical vertebra.

Fig. 6: A generalised / typical vertebra (anterior view)

Feature	**Description**	**Function**
Centrum (pl. centra)	The solid body of the vertebra.	Withstands **compressional pressure**; bears the body's weight.
Neural arch	Arch of bone on top of the centrum.	Encloses and **protects the spinal cord**.
Neural canal	The canal enclosed by the neural arch and centrum.	Carries the **spinal cord**.
Neural spine	Projection at the top of the neural arch.	**Muscle attachment**.
Transverse processes	Side projections of the neural arch.	**Muscle attachment** (and rib attachment in thoracic).

Feature	Description	Function
Zygapophyses (articulating facets)	Prezygapophyses at the anterior end; postzygapophyses at the posterior end.	**Articulate** with neighbouring vertebrae (gliding joints).

Zygapophyses

There are two pairs of zygapophyses. The pair at the **anterior** end are the **prezygapophyses** — they point **upwards and inwards**. The pair at the **posterior** end are the **postzygapophyses** — they point **downwards and outwards**. The prezygapophyses of one vertebra fit against the postzygapophyses of the vertebra in front of it, forming a **gliding joint**.

Techniques to identify the anterior and posterior ends of a vertebra

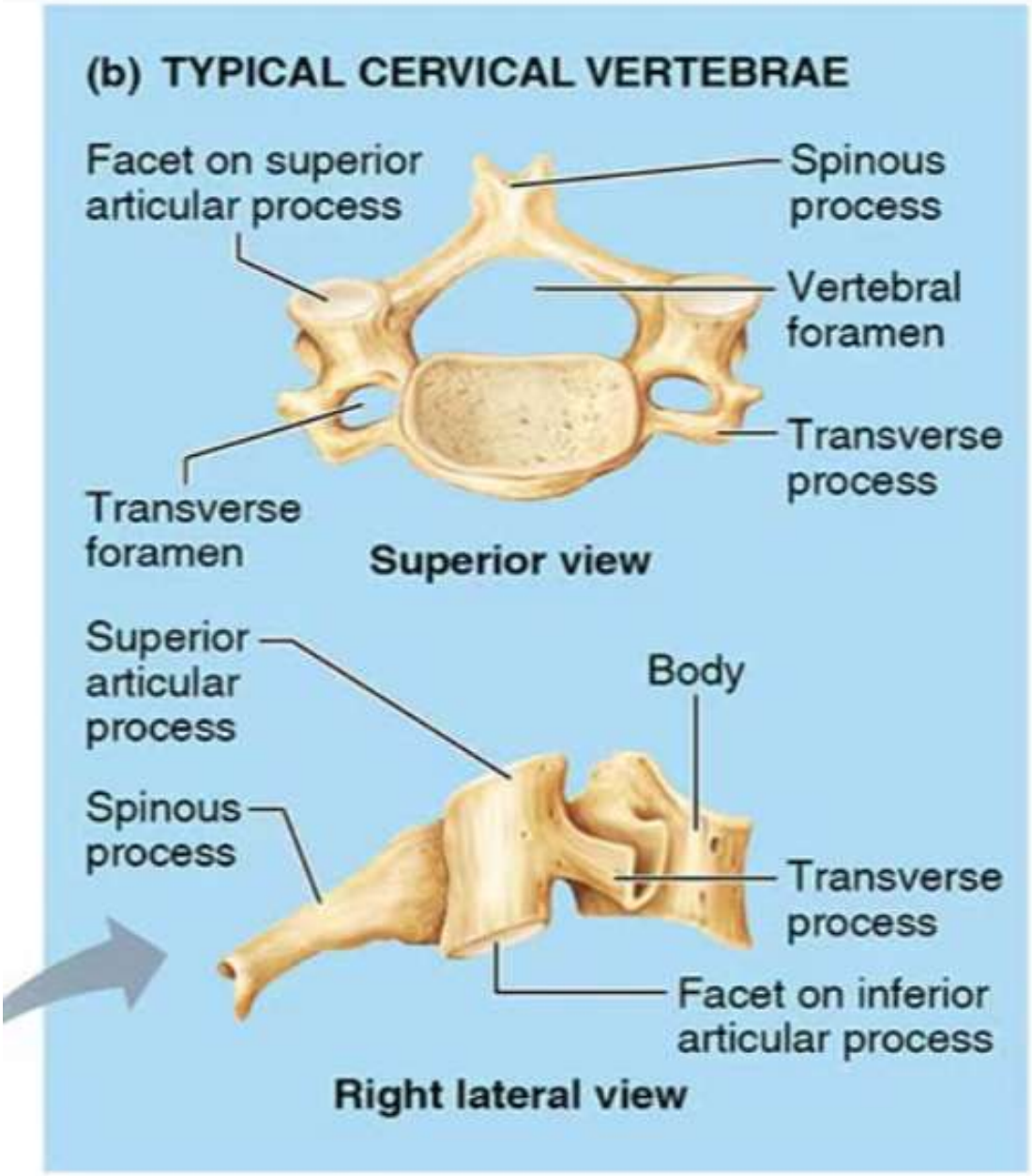
- The **neural spine** of all vertebrae except the lumbar points **backwards** (towards the posterior).
- The **prezygapophyses** (anterior) face upwards and inwards; the **postzygapophyses** (posterior) face downwards and outwards.

5.4 The Cervical Vertebrae (Neck)

A typical cervical vertebra differs from other vertebrae by having:

1. A **short** neural spine.
2. A **small** centrum.
3. A **large** neural canal.
4. A **vertebrarterial canal** — a small hole on each transverse process, through which the **vertebral artery** passes to the brain.
5. **Cervical ribs** — flattened, wing-like transverse processes.

cervical vertebra



General functions of cervical vertebrae

1. Carry the skull and allow **movement of the head**.
2. Protect **blood vessels** (vertebral arteries) and **nerves**.
3. Provide surfaces for **attachment of neck muscles**.

The **first two** cervical vertebrae are specially shaped to support the head and allow the skull to move on the vertebral column. They are the **atlas** and the **axis**.

5.5 The Atlas (First Cervical Vertebra)

The atlas is the **first cervical vertebra**; it directly carries the skull (like the Greek giant Atlas who carried the world).

Features of the atlas

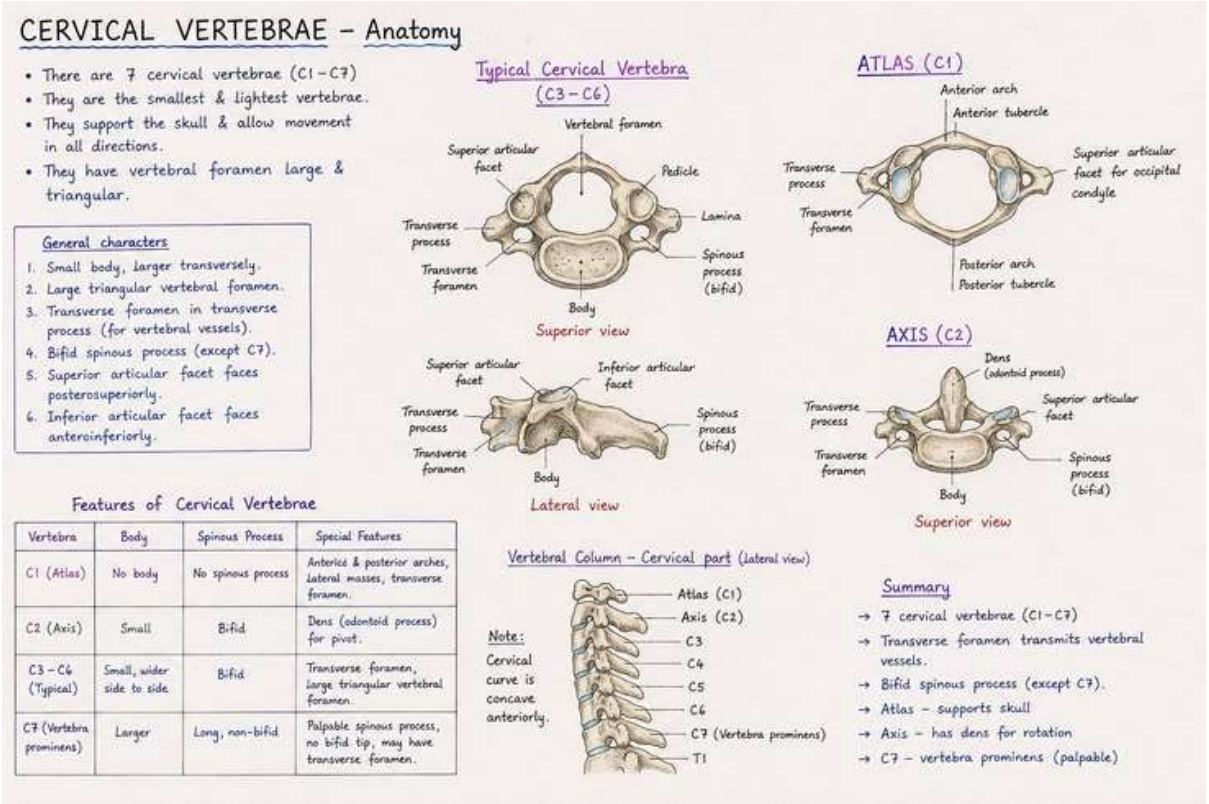
1. **Centrum reduced** (almost absent) — its centrum is fused to the axis as the odontoid process.
2. **Neural spine reduced** to a small crest.
3. **Prominent cervical ribs** — broad, flat, wing-like transverse processes for attachment of neck muscles.
4. **Large neural canal** — the upper part receives the odontoid process of the axis; the lower part carries the spinal cord.
5. **Vertebrarterial canals** present.
6. A pair of **large, concave anterior facets** that articulate with the **occipital condyles** of the skull to form a **hinge joint** for **nodding** ("yes") movement.

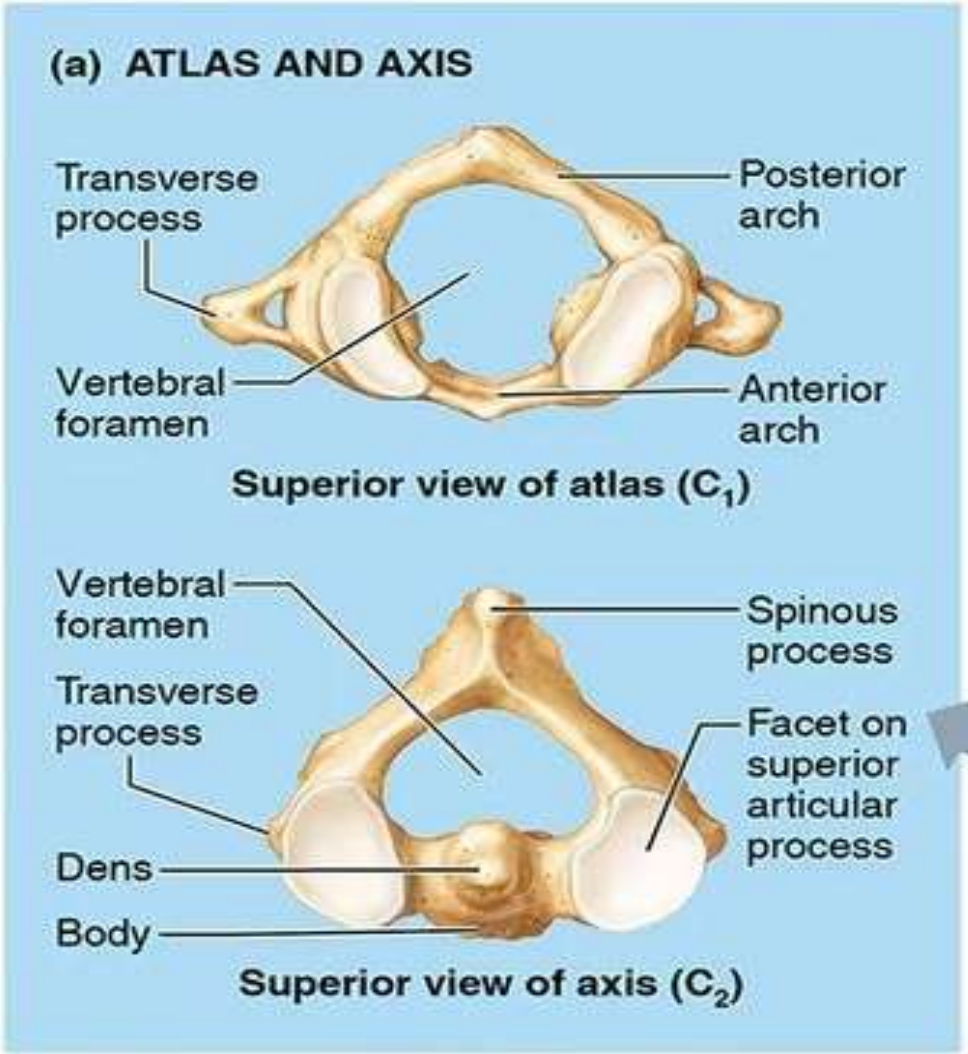
Functions of the atlas

1. Carries (supports) the skull.
2. Allows **nodding** (up-and-down) movement of the head.

5.6 The Axis (Second Cervical Vertebra)

The axis has a large centrum which projects forward to form a peg called the **odontoid process**.





Features of the axis

1. **Odontoid process** fits into the lower part of the neural canal of the atlas, forming a **pivot joint** that allows **rotation** of the skull ("no" movement).
2. **Large, broad neural spine** (flattened) for attachment of neck muscles.
3. **Small, flat** cervical ribs (transverse processes).
4. **No prezygapophyses** — instead there are flat articular surfaces for the atlas.
5. **Postzygapophyses** present at the posterior end.
6. Vertebrarterial canals present.

Function of the axis

Allows **rotational (side-to-side) movement** of the skull and atlas.

Differences between the atlas and the axis

Atlas	**Axis**
Odontoid process **absent**.	Odontoid process **present**.
Centrum reduced / absent.	Large centrum (extended as odontoid process).
Neural spine reduced to a small crest.	Large, broad neural spine.
Has **prezygapophyses** (large anterior facets for occipital condyles).	Has **no prezygapophyses**.
Transverse processes relatively **large and flat** (wing-like).	Transverse processes relatively **small**.
Forms a **hinge joint** with the skull → **nodding**.	Forms a **pivot joint** with the atlas → **rotation**.

5.7 The Thoracic Vertebrae (Chest)

Thoracic vertebrae are found in the chest region. Their main special function is to provide **attachment for the ribs**, forming the **rib cage**. Apart from the normal zygapophyses, a thoracic vertebra has **special facets for articulating with the ribs**:

- A **complete facet** (tubercular facet) on each **transverse process** — articulates with the **tubercle** of the rib.
- **Half facets** (capitular demifacets) on the **centrum** — articulate with the **head (capitulum)** of the rib. Two adjacent vertebrae together form one full facet for a rib head.

Features of a thoracic vertebra

1. **Long neural spine** pointing **backwards** — for attachment of back muscles.
2. **Large centrum**, heart-shaped in transverse view.
3. **Large, circular neural canal**.
4. **Short transverse processes** each bearing a **tubercular facet** for the rib.
5. **Demifacets** on the centrum for the head of the rib.
6. Pre- and postzygapophyses present.

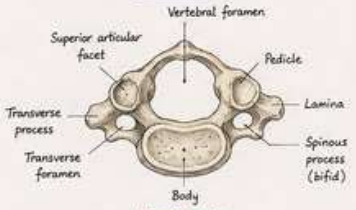
CERVICAL VERTEBRAE – Anatomy

- There are 7 cervical vertebrae (C1–C7)
- They are the smallest & lightest vertebrae.
- They support the skull & allow movement in all directions.
- They have vertebral foramen large & triangular.

General characters

1. Small body, larger transversely.
2. Large triangular vertebral foramen.
3. Transverse foramen in transverse process (for vertebral vessels).
4. Bifid spinous process (except C7).
5. Superior articular facet faces posterosuperiorly.
6. Inferior articular facet faces anteroinferiorly.

Typical Cervical Vertebra (C3–C6)

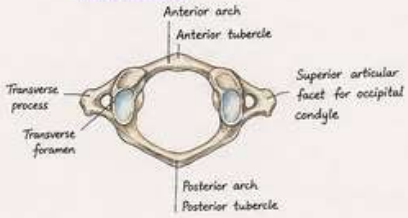


Superior view

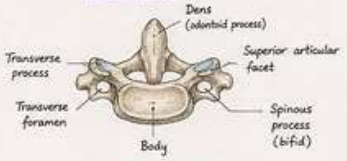


Lateral view

ATLAS (C1)



AXIS (C2)



Superior view

Features of Cervical Vertebrae

Vertebra	Body	Spinous Process	Special Features
C1 (Atlas)	No body	No spinous process	Anterior & posterior arches, lateral masses, transverse foramen.
C2 (Axis)	Small	Bifid	Dens (odontoid process) for pivot.
C3–C6 (Typical)	Small, wider side to side	Bifid	Transverse foramen, large triangular vertebral foramen.
C7 (Vertebra prominens)	Larger	Long, non-bifid	Palpable spinous process, no bifid tip, may have transverse foramen.

Vertebral Column – Cervical part (lateral view)



Note: Cervical curve is concave anteriorly.

Summary

→ 7 cervical vertebrae (C1–C7)
→ Transverse foramen transmits vertebral vessels.
→ Bifid spinous process (except C7).
→ Atlas – supports skull
→ Axis – has dens for rotation
→ C7 – vertebra prominens (palpable)

Fig. 11: Thoracic vertebra — anterior and lateral views (with rib attached)

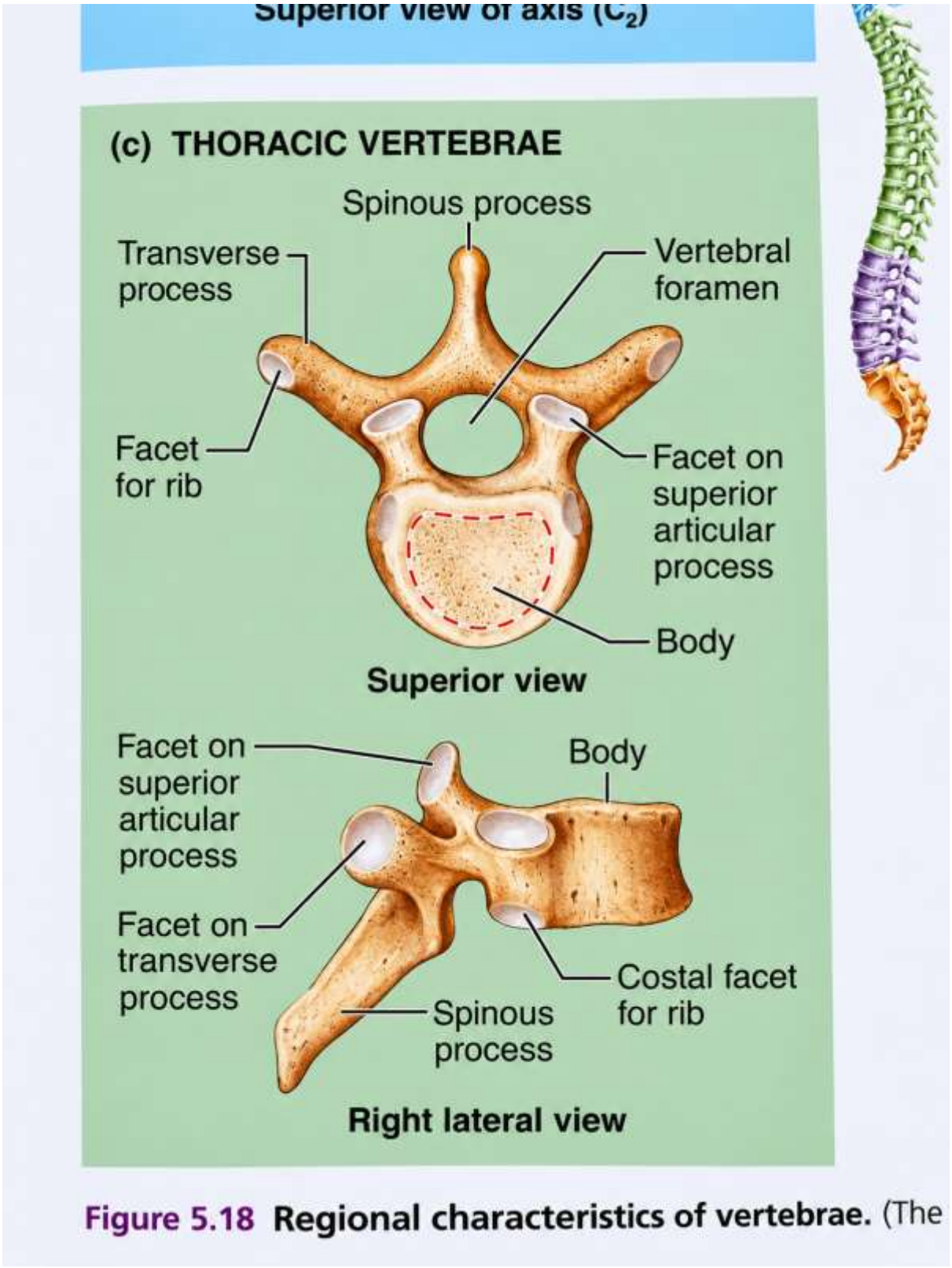


Fig. 11: Thoracic vertebra — anterior and lateral views (with rib attached)

Functions of thoracic vertebrae

1. Provide **articulation surfaces for the ribs** to form the rib cage.
2. The long neural spine provides **attachment for muscles** of the back and neck that support the head.
3. Protect the **spinal cord**; take part in **breathing** movements through the ribs.

5.8 The Lumbar Vertebrae (Abdomen / Lower Back)

The lumbar vertebra is the **largest** vertebra (with the largest centrum) and is subject to **great stress**. It is the most adapted (complex): it gives rigidity, yet permits bending, sideways movement and rotation of the trunk.

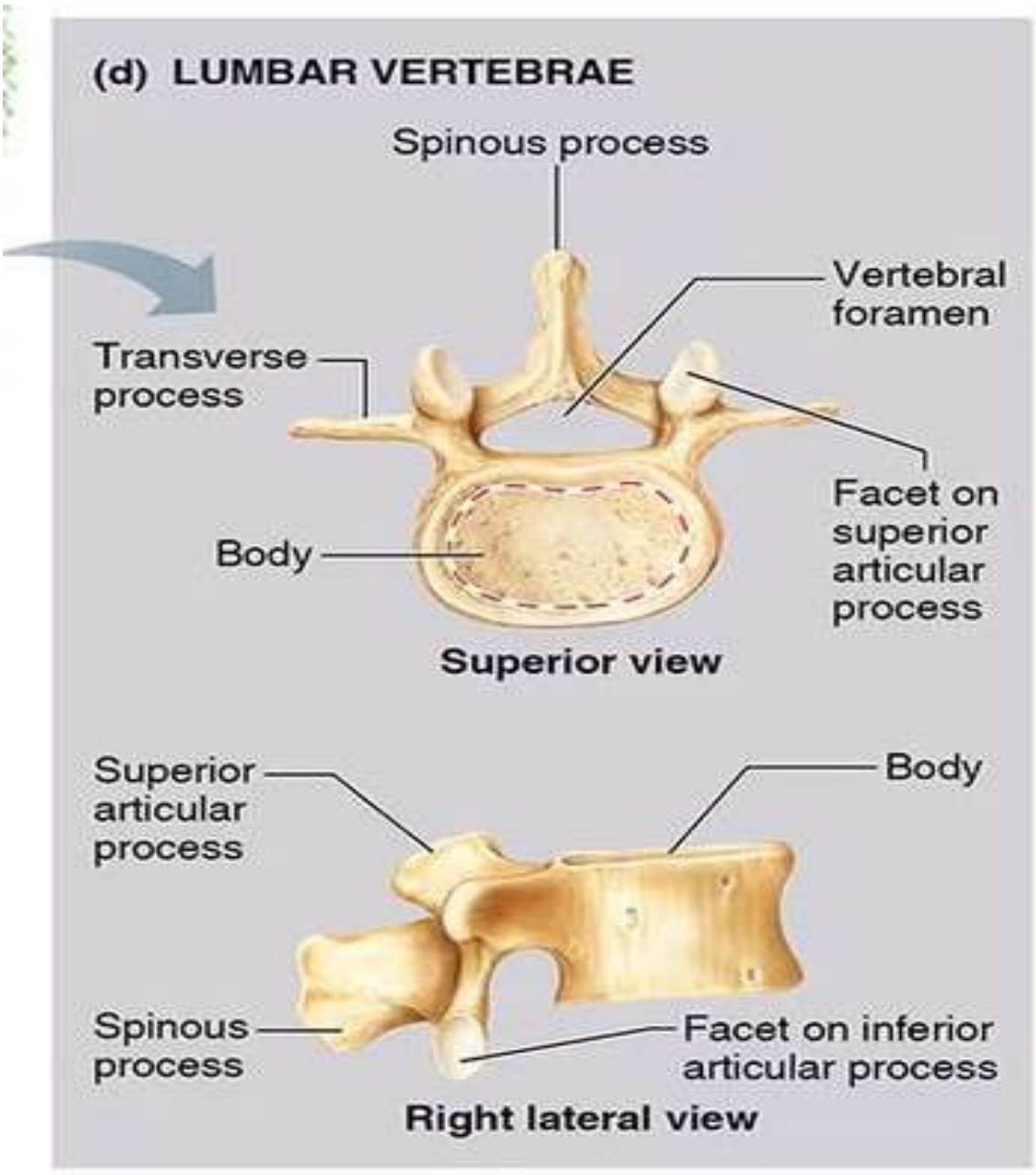


Fig. 12: Lateral view and anterior view of the mammalian lumbar vertebra

Features of the lumbar vertebra

1. **Large, thick centrum** — supports the weight of the upper body.
2. **Large, thick neural arch**.
3. **Neural spine** short, broad and projected **upwards and forwards**.
4. **Long, wide, well-developed transverse processes**, pointing **forwards and downwards** — longer than the neural spine.
5. A pair of **pre- and postzygapophyses** for articulation with neighbouring vertebrae.
6. A pair of **metapophyses** above the prezygapophyses and a pair of **anapophyses** below the postzygapophyses — for **muscle attachment**.
7. **Hypapophysis** — a projection below the centrum, found on the **first and second** lumbar vertebrae only (in the rabbit).

> **NOTE**
> **Metapophyses and anapophyses** are characteristic of the lumbar vertebra. They provide extra surfaces for muscle attachment, and they **interlock** to keep the column in position during stress.

Functions of lumbar vertebrae

1. **Support the upper half of the body** — the large centrum provides this ability.
2. Provide surfaces for **attachment of abdominal muscles**.
3. Support the **muscles of the lower back**.
4. Allow bending and twisting of the trunk.

Distinguishing features of the lumbar vertebra (quick recognition)

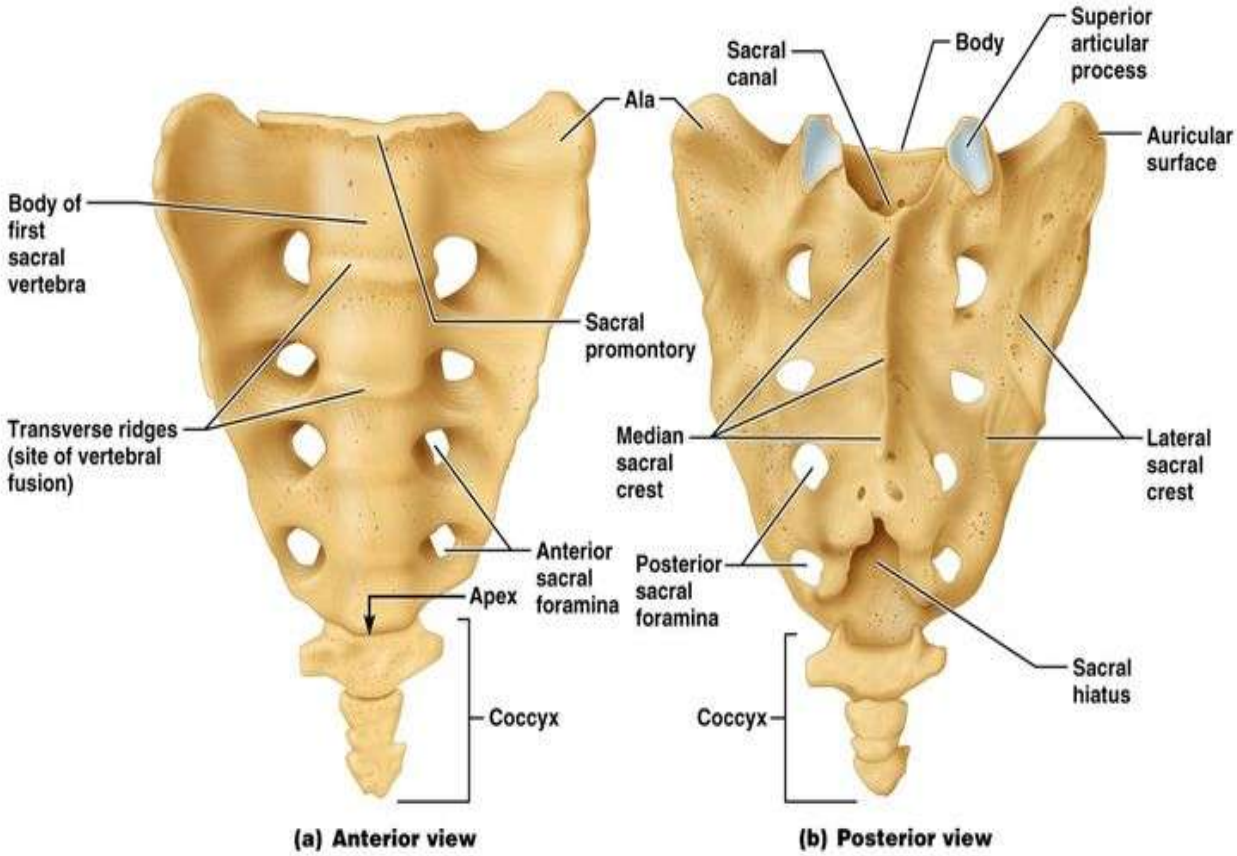
Well-developed transverse processes **larger than the neural spine**; neural spine pointing **forward**; **metapophyses and anapophyses**; **hypapophyses** in some.

5.9 The Sacral Vertebrae (Hip) — the Sacrum

The sacral vertebrae are **fused together** to form a single bone called the **sacrum** (4 in the rabbit, 5 in man). The pelvic girdle is firmly attached to it.

Features

1. Vertebrae **fused** into one bone (the sacrum).
2. The first sacral vertebra has **large, wing-like transverse processes** (sacral wings) which fuse with the **ilium** of the pelvic girdle.
3. **Large, broad centrum**; neural spines **short / reduced**.
4. **Small neural canal** — the spinal cord narrows here.
5. The sacrum tapers towards the tail end.



Functions

1. Provides a **firm attachment for the pelvic girdle**, transmitting the weight of the body to the hind limbs.
2. **Protects** the lower part of the spinal cord and pelvic organs.
3. Provides surfaces for muscle attachment.

5.10 The Caudal Vertebrae (Tail)

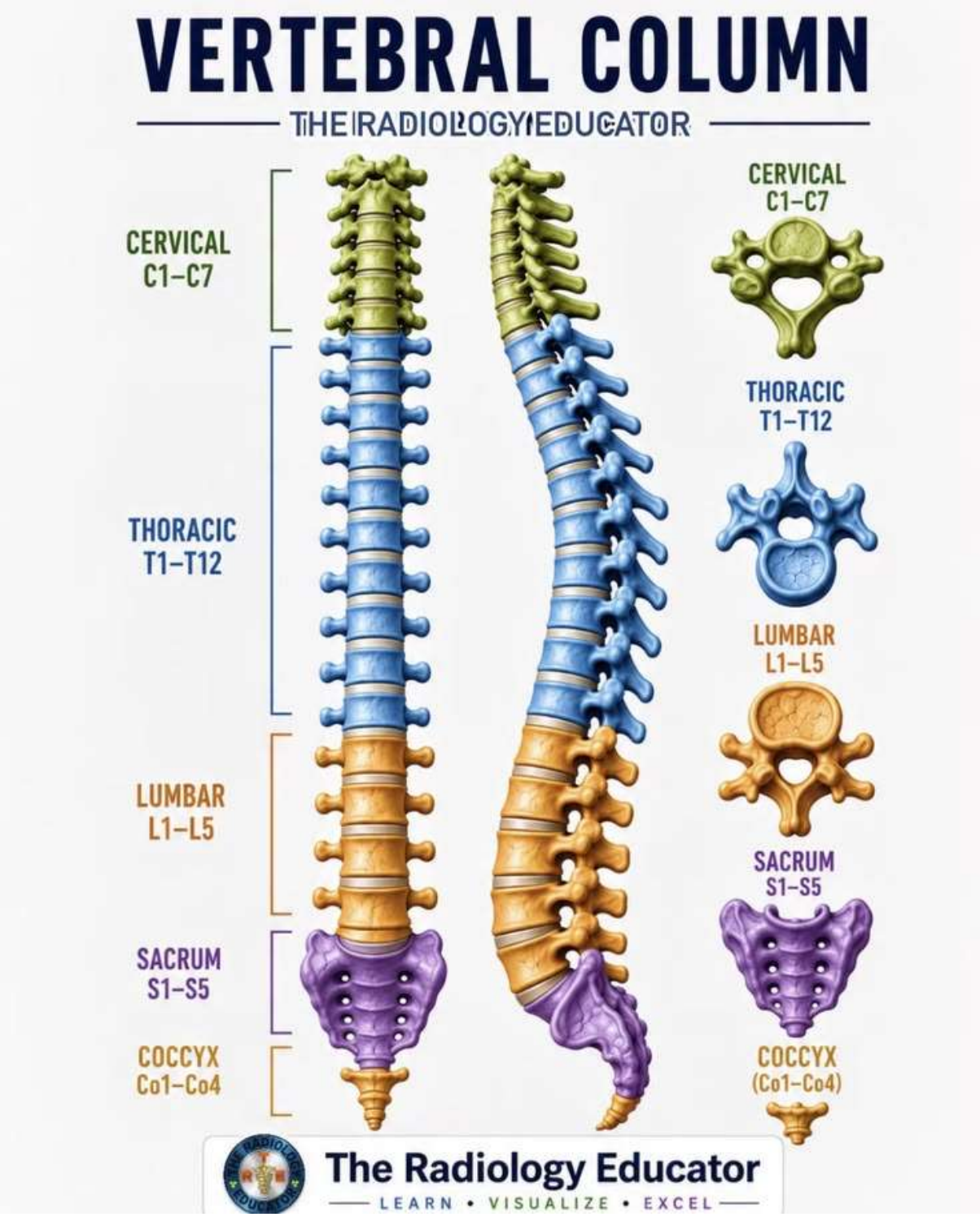
Caudal vertebrae form the **tail** (16 in the rabbit). In humans, 4 small caudal vertebrae are fused to form the **coccyx** (tailbone).

1. The vertebrae become **progressively smaller** towards the tip of the tail.
2. The first few have small neural arches and transverse processes; the last ones are reduced to **only a small centrum**.
3. The neural canal becomes smaller and disappears.

Functions: support the tail and provide attachment for tail muscles; in some mammals the tail helps in **balance**, **grasping** (monkeys) or swatting flies (cattle).

5.11 Summary Table: Distinguishing the Vertebrae

Vertebra	Key identifying features	Main function
Cervical	Vertebrarterial canal; cervical ribs; short neural spine; small centrum; large neural canal	Support and move the head; protect vertebral artery
Atlas	No centrum, no odontoid process; wing-like transverse processes; large facets for occipital condyles	Carries skull; nodding
Axis	Odontoid process; large flat neural spine; no prezygapophyses	Rotation of head
Thoracic	Long backward-pointing neural spine; facets for ribs on centrum and transverse process	Rib attachment; rib cage
Lumbar	Largest centrum; long forward-pointing transverse processes; metapophyses, anapophyses, hypapophysis	Support upper body; abdominal muscle attachment
Sacral	Fused into sacrum; wide sacral wings	Attachment of pelvic girdle
Caudal	Small; progressively reduced to just a centrum	Support tail



5.12 The Ribs and the Sternum (Rib Cage)

The **ribs** are curved, flat bones. Each rib articulates with a thoracic vertebra at the back: the **head (capitulum)** with the demifacets on the centra, and the **tubercle** with the facet on the transverse process. In front, most ribs are joined to the **sternum** by **costal cartilage**. Humans have **12 pairs** of ribs.

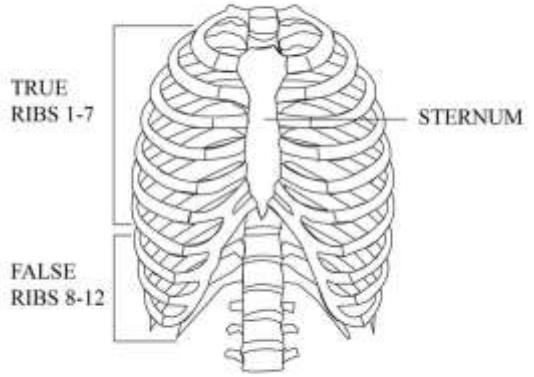
Type of rib	Number (human)	Attachment in front
True ribs	7 pairs (1st–7th)	Attached **directly** to the sternum by their own costal cartilage.
False ribs	3 pairs (8th–10th)	Attached **indirectly** — their cartilages join the cartilage of the rib above.
Floating ribs	2 pairs (11th–12th)	**Not attached** to the sternum at all; free in front.

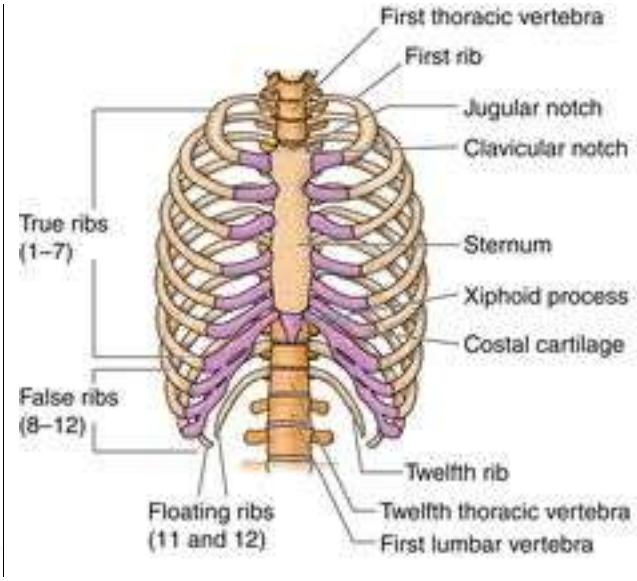
THINK & ANSWER

Q: How many false ribs are there in humans? How many floating ribs?

A: 3 pairs of false ribs and **2 pairs** of floating ribs. (Some texts count floating ribs as false ribs too, giving 5 pairs of false ribs — always show your breakdown.)

The **sternum (breastbone)** is a flat bone in the middle of the chest. It has three parts: the **manubrium** (top, joined to the clavicles), the **body (mesosternum)** and the **xiphoid process (xiphisternum)** at the bottom, which is cartilaginous.





Sternum (Breast Bone)

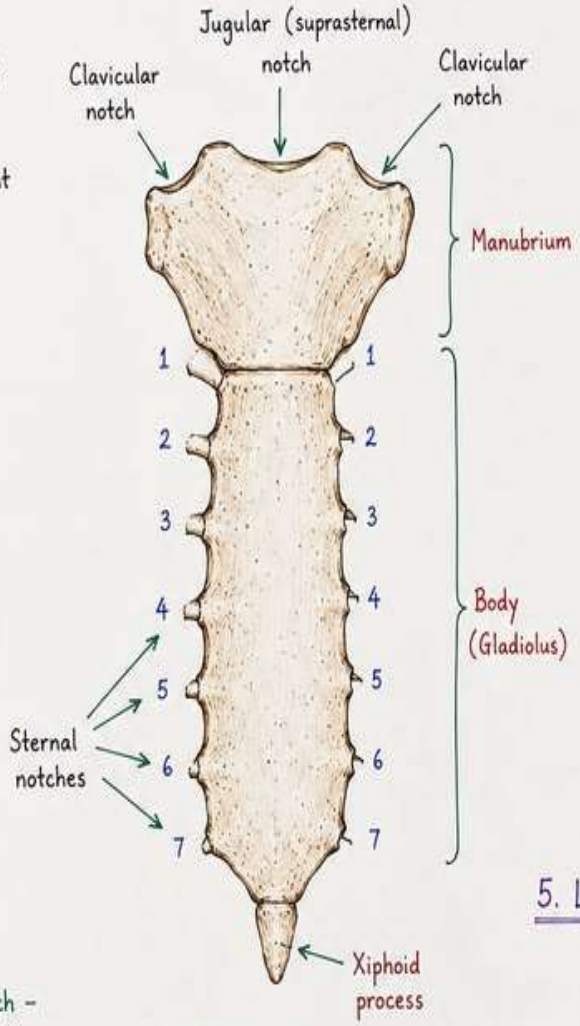
1. Overview

- Flat bone in the midline of anterior thoracic wall.
- It is a site of attachment for ribs and clavicles.
- Protects vital organs of thorax.

2. Parts

The sternum has 3 parts :

1. Manubrium
2. Body (Gladiolus)
3. Xiphoid process



Anterior view

3. Articulations

- With clavicles – Clavicular notches on manubrium.
- With ribs – Sternal notches on both sides.
- With costal cartilages – By synovial joints.
- Between parts – Manubriosternal (symphysis)
  Xiphisternal (synchondrosis) (at angle of Louis)

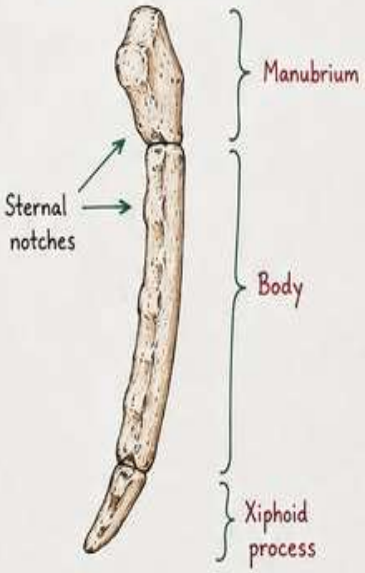
4. Important Landmarks

- Jugular (suprasternal) notch – At the superior border of manubrium.
- Clavicular notches – On either side for medial end of clavicle.
- Sternal angle (Angle of Louis) – At junction of manubrium & body (opposite 2nd costal cartilage). Important landmark in thorax.
- Costal notches – 7 on each side for attachment of costal cartilages of ribs 1–7.

Sternal angle (Angle of Louis) is palpable.

It lies at the level of intervertebral disc between T4 & T5 vertebrae.

5. Lateral View



Lateral view

Functions of the rib cage

1. **Protects** the heart, lungs and major blood vessels.
2. Plays a role in **breathing**: intercostal muscles move the ribs up/out and down/in to change the volume of the thorax.
3. Provides attachment for muscles of the chest, neck and back.

PART 6: THE APPENDICULAR SKELETON

The appendicular skeleton is made of the **limbs** and the **limb girdles**. Mammals have **two pairs of limbs** — fore limbs and hind limbs. **Limb girdles** are the bones that attach the limbs to the main axis of the body.

6.1 The Pentadactyl Limb

The limb bones of all vertebrates except fish conform to a basic plan called the **pentadactyl limb plan** — each limb **terminates in five digits**. Both the fore limb and the hind limb have the same three-part plan:

Part	**Fore limb**	**Hind limb**
Upper part (one bone)	Upper arm — **humerus**	Thigh — **femur**
Middle part (two bones)	Forearm — **radius** and **ulna**	Shank — **tibia** and **fibula**
Lower part	Hand — carpals, metacarpals, phalanges	Foot — tarsals, metatarsals, phalanges

6.2 The Fore Limb

(i) The upper arm — the humerus

The upper arm consists of a single long bone called the **humerus**.

- **Head:** the rounded upper end, which fits into the **glenoid cavity** of the scapula to form the **shoulder joint** (ball and socket).
- **Greater and lesser tuberosities:** projections near the head for attachment of shoulder muscles.
- **Bicipital groove:** a groove between the tuberosities through which the tendon of the **biceps** muscle passes.
- **Deltoid ridge:** a ridge on the shaft for attachment of the **deltoid** (shoulder) muscle.
- **Shaft:** the long middle part.
- **Trochlea:** the grooved, pulley-like lower end which articulates with the **ulna** at the **elbow** (hinge joint).
- **Supratrochlear foramen:** a hole just above the trochlea (in the rabbit) for the passage of a **nerve, artery and vein**.
- **Olecranon fossa** (at the **posterior** side): a depression into which the **olecranon process** of the ulna locks, preventing the arm from bending backwards when straightened.

- **Coronoid fossa** (at the **anterior** side): a depression into which the **coronoid process** of the ulna fits when the arm is bent (flexed).

HUMAN SKELETON

APPENDICULAR SKELETON = 126 BONES

A. PECTORAL (SHOULDER) GIRDLE = 4

1. Clavicle – 2
2. Scapula – 2



Clavicle (2)



Scapula (2)

B. UPPER LIMBS = 60 BONES (EACH SIDE = 30)

1. Humerus – 1
2. Radius – 1
3. Ulna – 1
4. Carpals – 8
5. Metacarpals – 5
6. Phalanges – 14



Humerus (1)



Radius (1)



Ulna (1)





Carpals (8)



Metacarpals (5)



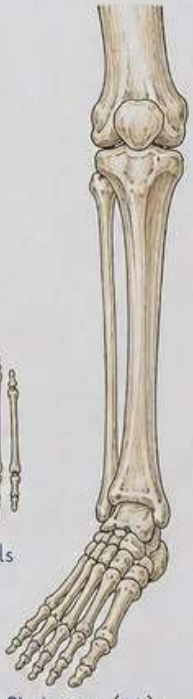
Phalanges (14)

C. PELVIC (HIP) GIRDLE = 2

1. Hip bone – 2



Hip bone (2)



D. LOWER LIMBS = 60 BONES (EACH SIDE = 30)

1. Femur – 1
2. Patella – 1
3. Tibia – 1
4. Fibula – 1
5. Tarsals – 7
6. Metatarsals – 5
7. Phalanges – 14



Femur (1)



Patella (1)



Tibia (1)



Fibula (1)



Tarsals (7)



Metatarsals (5)

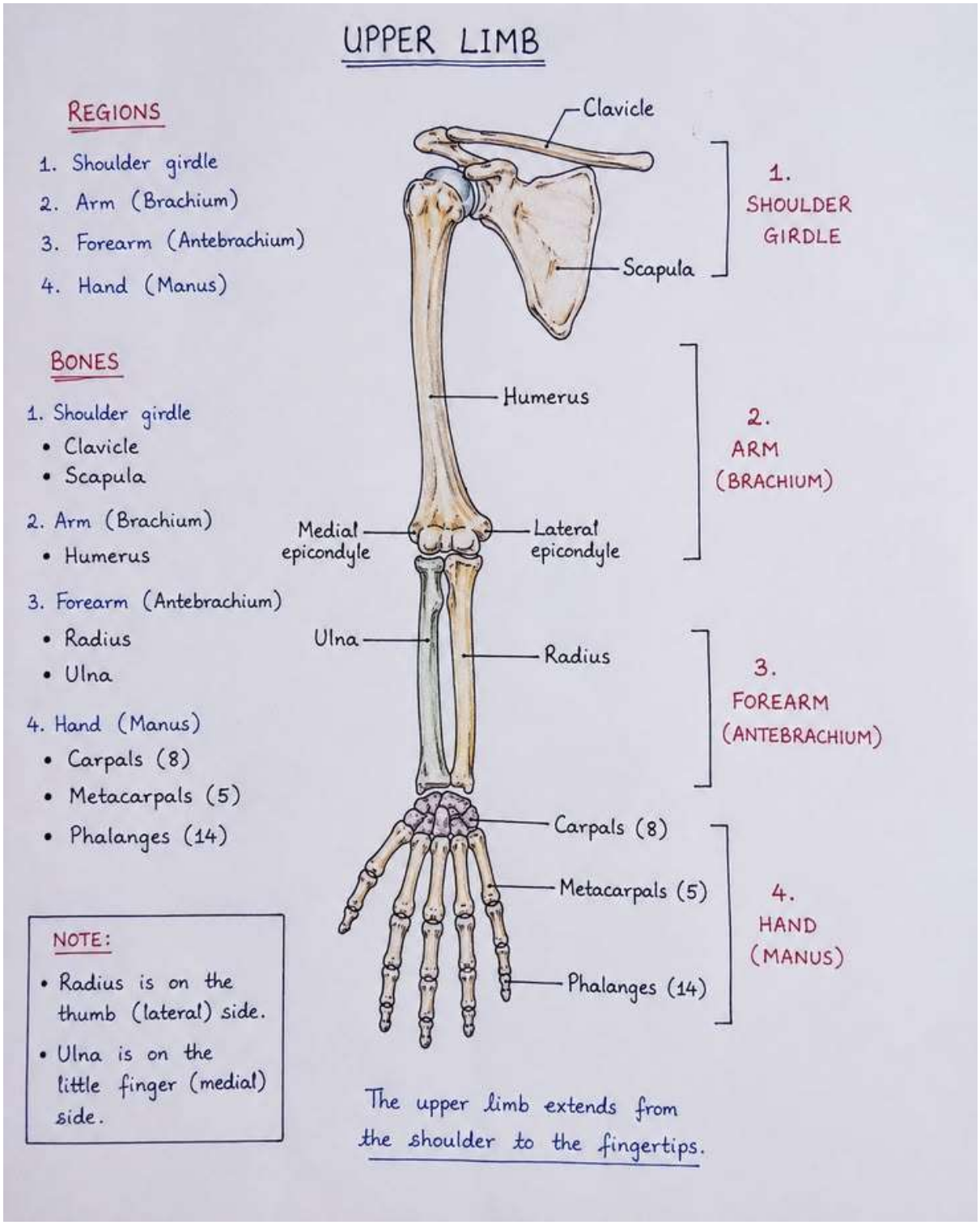
Phalanges (14)

(ii) The forearm — radius and ulna

The forearm (middle arm) is made of two long bones: the **ulna** and the **radius**.

- The **ulna is larger** than the radius.
- The ulna has a cavity called the **sigmoid notch (trochlear notch)**, which articulates with the trochlea of the humerus.
- Behind the sigmoid notch, the ulna extends to form the **olecranon process** (the point of the elbow).
- The lower border of the sigmoid notch forms a projection called the **coronoid process**.
- The **radius** lies alongside the ulna and allows the forearm and hand to rotate.

Structure	**Function**
Olecranon process	Fits into the olecranon fossa; **prevents the arm from bending backwards** (over-extension); attachment of the triceps.
Coronoid process	Fits into the coronoid fossa of the humerus when the arm is **bent / flexed**.
Sigmoid (trochlear) notch	Point of **articulation with the trochlea** of the humerus at the elbow.



(iii) The hand

The hand is made up of many small bones, which increases its **flexibility**:

- The **wrist** is made of small bones called **carpals**.
- The **palm** has **5** bones called **metacarpals**.

- Beyond the palm are the **phalanges**, which form the bones of the fingers and thumb. The **thumb has 2** phalanges; each of the other fingers has **3**.

6.3 The Hind Limb

The hind limb is also made of three parts: the **thigh**, the **lower leg (shank)** and the **foot**.

(i) The thigh — the femur

The thigh bone is a long bone called the **femur**. It is the **largest, longest and strongest bone** in the (human) body.

- **Head:** the rounded upper end, which articulates with the **acetabulum** of the pelvic girdle to form the **hip joint** (ball and socket).
- **Trochanters:** three projections near the head — the **greater**, **lesser** and **third** trochanters (third in the rabbit) — for attachment of **thigh muscles**.
- **Shaft:** the long middle part.
- **Condyles:** two knobs at the lower end, which articulate with the **tibia** at the **knee** (hinge joint).
- **Patella groove:** a groove between the condyles in which the **patella (knee cap)** fits.

Functions of the femur: (1) movement — it is a lever for walking, running and jumping; (2) supports the weight of the body; (3) its marrow produces blood cells.

(ii) The shank — tibia and fibula

- The shank has two bones, like the forearm: the **tibia** and the **fibula**. The **tibia is larger** than the fibula (in the rabbit the two are fused at the lower end).
- The tibia articulates with the femur to form the **knee joint**.
- A small bone, the **patella (knee cap)**, lies in front of the knee joint within the tendon of the thigh muscle. It protects the knee and **prevents the leg from bending forwards** at the knee — just as the olecranon process prevents the arm bending backwards.

(iii) The foot

- The **ankle** is made of bones called **tarsals** (7 in man, 6 in the rabbit). One tarsal elongates backwards to form the **heel bone (calcaneum)**, for attachment of the Achilles tendon.
- The **arch of the foot** is formed by the **metatarsals** — 5 in man, 4 in the rabbit.
- Each metatarsal articulates with the **phalanges** of a toe. The **big toe has 2** phalanges; each other toe has **3**.
- The big toe of the rabbit has **disappeared** (lost in evolution), so the rabbit hind foot has 4 toes ending in **claws**.

6.4 Comparison of the Fore Limb and Hind Limb

Fore limb	**Hind limb**
Three parts: upper arm, forearm, hand.	Three parts: thigh, shank, foot.
One bone in upper arm: **humerus**.	One bone in thigh: **femur**.
Two bones in forearm: **ulna** (larger) and **radius**.	Two bones in shank: **tibia** (larger) and **fibula**.
Wrist: **carpals** (8 in man; 9 in rabbit).	Ankle: **tarsals** (7 in man; 6 in rabbit).
Palm: 5 **metacarpals**.	Arch of foot: **metatarsals** (5 in man; 4 in rabbit).
Five digits (fingers and thumb): **phalanges**.	Five digits (toes): **phalanges** (4 in rabbit).
Thumb: 2 phalanges.	Big toe: 2 phalanges.
Humerus articulates with **glenoid cavity** (shoulder).	Femur articulates with **acetabulum** (hip).
Olecranon process stops backward bending at elbow.	**Patella** stops forward bending at knee.
Nails / claws / hooves.	Nails / claws / hooves.

6.5 The Limb Girdles

Limb girdles are bones which attach the limbs to the axial skeleton. There are two: the **pectoral (shoulder) girdle**, which attaches the fore limbs, and the **pelvic (hip) girdle**, which attaches the hind limbs.

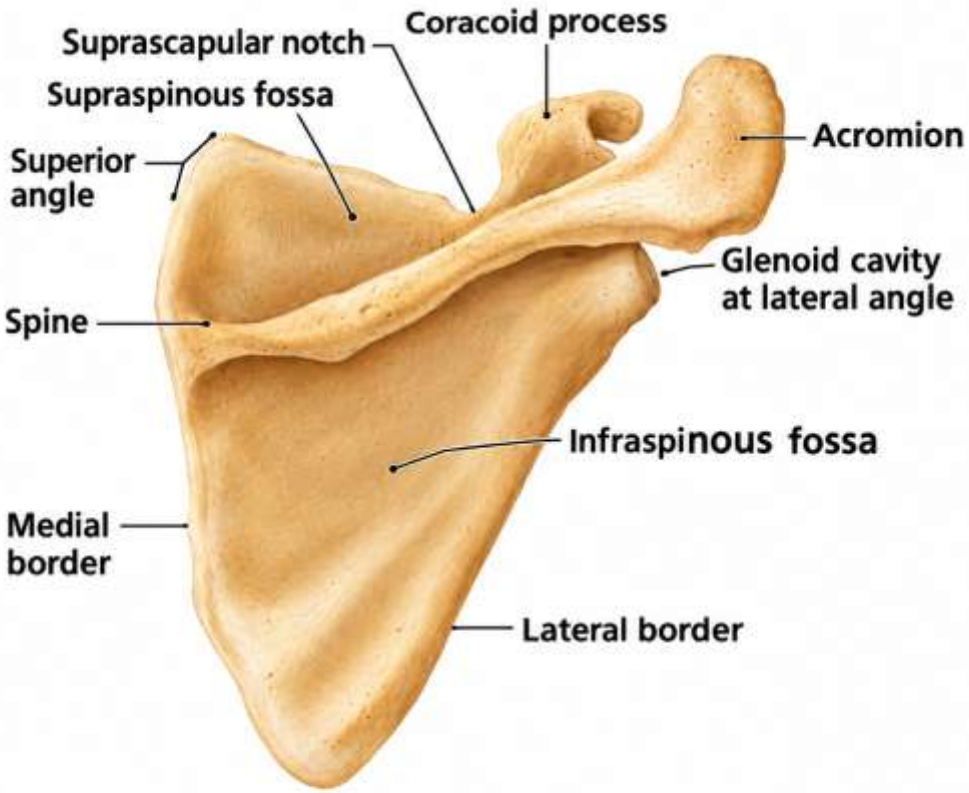
(a) The pectoral girdle

Each half of the pectoral girdle consists of the **scapula** (shoulder blade) and the **clavicle** (collar bone).

- **Scapula:** a flat, triangular bone lying over the ribs at the back. It has a ridge called the **spine**, which ends in the **acromion process** (and, in the rabbit, a backward-pointing **metacromion process**). At its lower end is a shallow depression, the **glenoid cavity**, into which the head of the humerus fits. A small hook, the **coracoid process**, provides muscle attachment.
- **Clavicle:** a thin rod-like bone joining the scapula to the sternum (manubrium). It is well developed in man but **reduced** in the rabbit.

The pectoral girdle is **not fused** to the vertebral column; it is held by muscles. This gives the shoulder a **wide range of movement** and helps absorb shock when landing on the fore limbs.

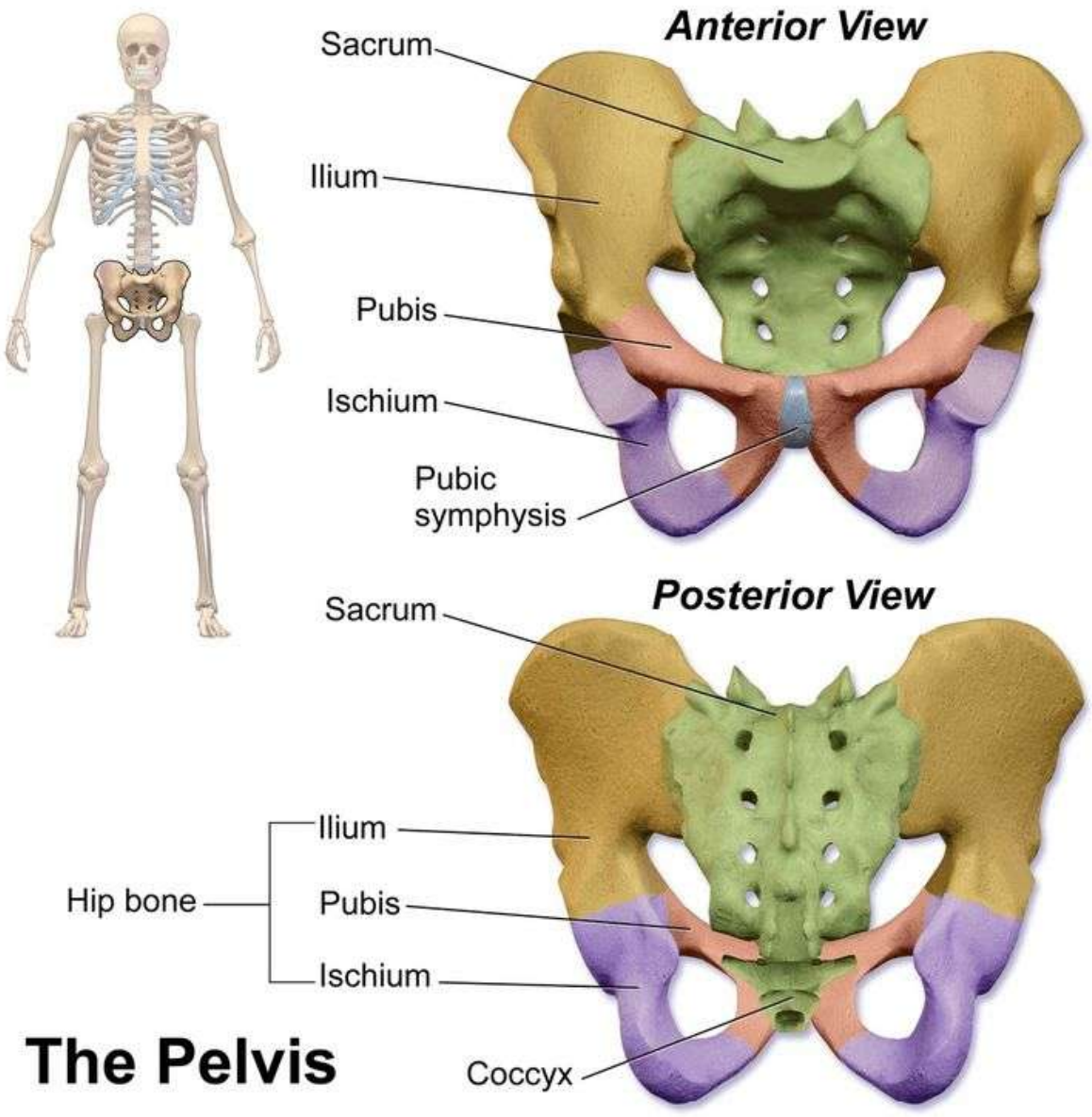
Fig. 24: The scapula of a mammal (lateral view)



(b) The pelvic girdle

The pelvic girdle is made of two halves (**innominate bones**). Each half is formed by the fusion of three bones: the **ilium**, **ischium** and **pubis**.

- The **ilium** is fused firmly to the **sacrum** — so the pelvic girdle is **rigid**, transmitting the body's weight to the hind limbs.
- Where the three bones meet is a deep cup, the **acetabulum**, into which the head of the femur fits (hip joint).
- A large hole between the ischium and pubis, the **obturator foramen**, lets nerves and blood vessels pass and makes the girdle lighter.
- The two halves are joined in front at the **pubic symphysis** (fibrocartilage).



Differences between the pectoral and pelvic girdles

Pectoral girdle	**Pelvic girdle**
Attaches the **fore limbs**.	Attaches the **hind limbs**.
Made of scapula and clavicle.	Made of ilium, ischium and pubis (fused).
Not fused to the vertebral column — held by muscles.	**Fused** to the sacrum.
Has a shallow **glenoid cavity**.	Has a deep **acetabulum**.
Flexible; allows wide range of movement.	Rigid; supports body weight and protects pelvic organs.
No obturator foramen.	Has an obturator foramen.

PART 7: JOINTS, MUSCLES AND MOVEMENT

7.1 Joints

A **joint** is a place where two or more bones meet. Joints are classified by the amount of movement they allow.

Type of joint	Movement	Examples
Immovable (fixed / fibrous)	No movement	Sutures between bones of the cranium; joint between sacrum and pelvic girdle
Slightly movable (cartilaginous)	Little movement	Between vertebrae (intervertebral discs); pubic symphysis; ribs and sternum
Freely movable (synovial)	Free movement	See the four types below

Types of synovial (freely movable) joints

Joint	Movement allowed	Examples
Ball and socket	In all planes — rotation through 360°	Shoulder (humerus + glenoid cavity); hip (femur + acetabulum)
Hinge	In one plane only (like a door)	Elbow, knee, finger joints; skull + atlas (nodding)
Pivot	Rotation of one bone around another	Atlas + axis (turning head); radius + ulna
Gliding	Sliding of flat surfaces over each other	Wrist (carpals), ankle (tarsals), zygapophyses of vertebrae

7.2 Structure of a Synovial Joint

- **Articular (hyaline) cartilage** covers the ends of the bones — reduces friction and absorbs shock.
- **Synovial membrane** lines the joint capsule and **secretes synovial fluid**.
- **Synovial fluid** fills the joint cavity — **lubricates** the joint and nourishes the cartilage.
- **Joint capsule / ligaments** — tough, **elastic** fibres that hold the bones together and prevent dislocation while allowing movement.

7.3 Muscles

Type	Also called	Features / location
Skeletal muscle	Striated / striped / voluntary	Attached to bones; under conscious control; contracts quickly but tires.
Smooth muscle	Unstriated / involuntary / visceral	Walls of gut, blood vessels, uterus; slow, sustained contraction.

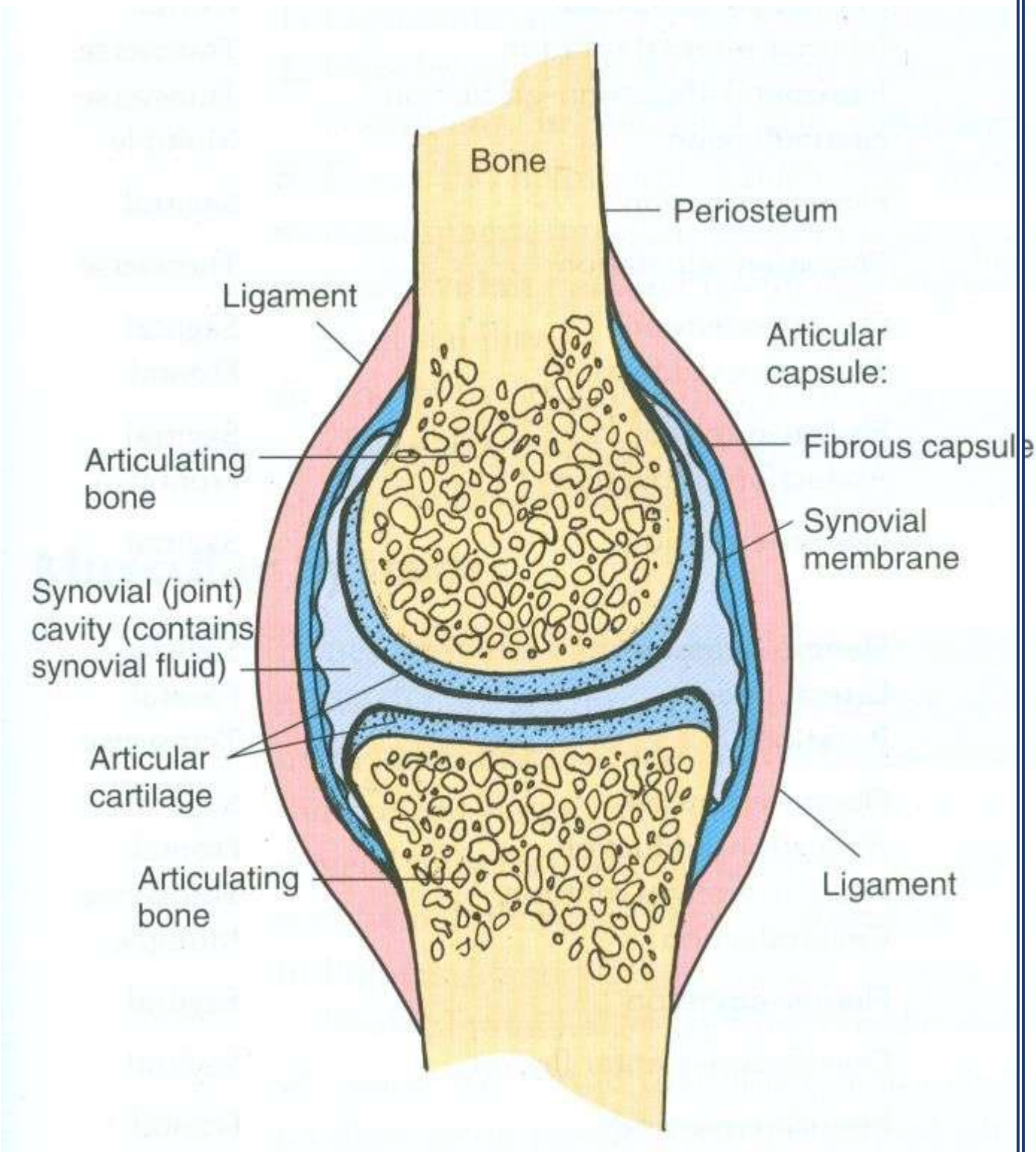
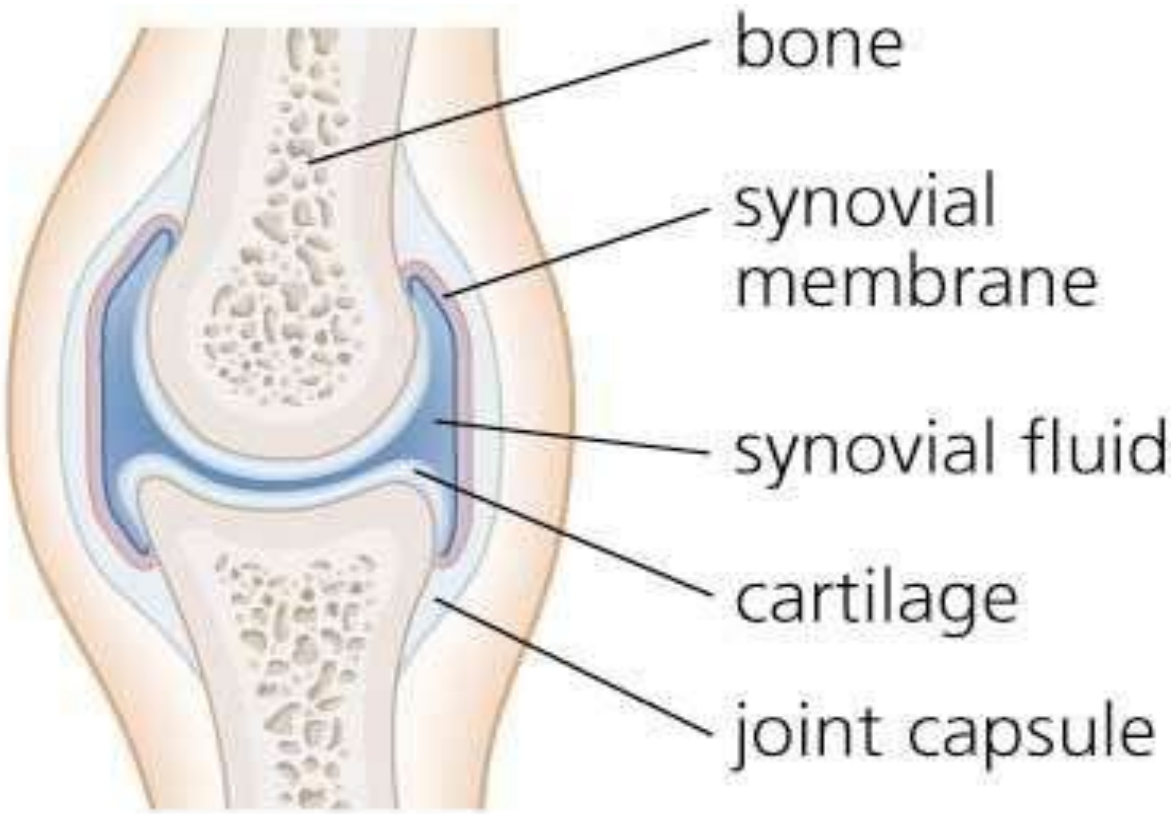
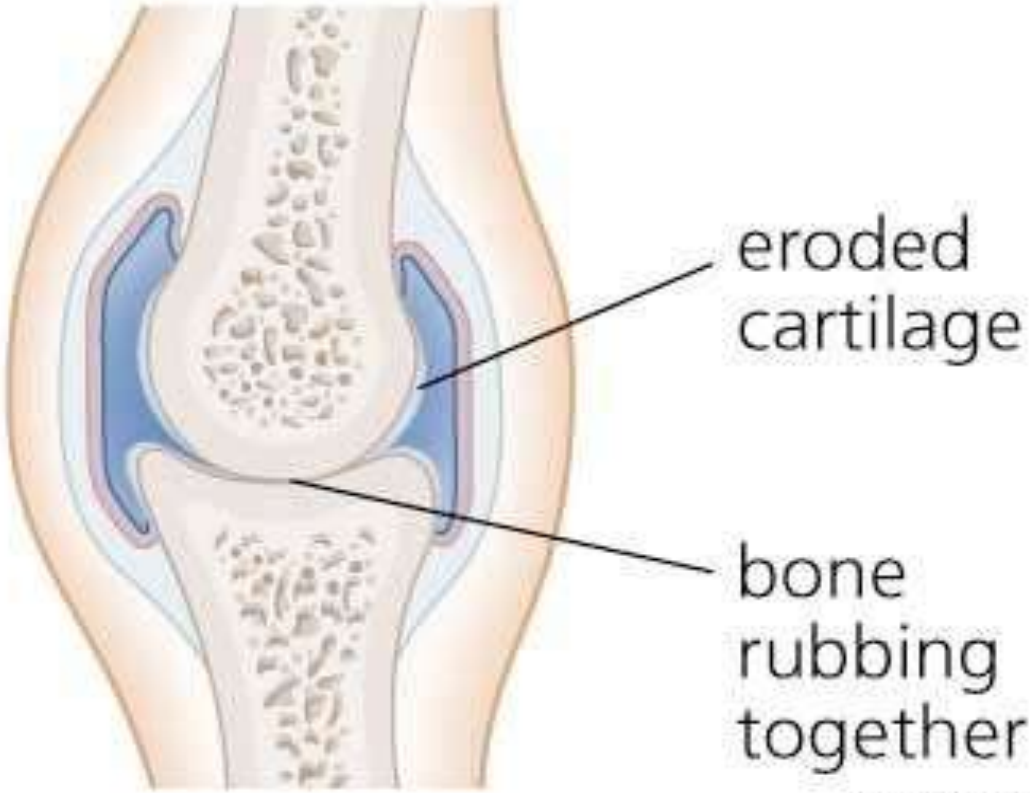


Fig. 26: Structure of a synovial joint (e.g. knee or elbow)





Precision Graphics

Type	Also called	Features / location
Cardiac muscle	Heart muscle	Only in the heart; contracts rhythmically without tiring (myogenic).

Tendons and ligaments

Tendon	Ligament
Joins **muscle to bone**.	Joins **bone to bone**.
Inelastic (non-stretchy) — transmits the full pull of the muscle.	**Elastic** (slightly stretchy) — allows movement at joints.
Made of white collagen fibres.	Made of yellow elastic fibres (and collagen).

7.4 Antagonistic Muscles and Movement of the Arm

Muscles can only **pull** (contract); they cannot push. So they work in **antagonistic pairs**: when one contracts, the other relaxes. The end of a muscle attached to a fixed bone is the **origin**; the end attached to the bone that moves is the **insertion**.

- **Bending (flexing) the arm:** the **biceps** (a **flexor**) contracts and the **triceps** relaxes. The biceps pulls on the radius, and the forearm is raised at the elbow.
- **Straightening (extending) the arm:** the **triceps** (an **extensor**) contracts and the **biceps** relaxes. The triceps pulls on the olecranon process of the ulna, and the arm straightens.

The bones act as **levers** and the joint (elbow) as the **fulcrum (pivot)**. Nerve impulses from the brain coordinate which muscle contracts.

PART 8: FUNCTIONS OF THE MAMMALIAN SKELETON

Function	Explanation / example
1. Support	Provides a rigid framework that holds up the body and supports soft organs.
2. Shape	Gives the body its definite shape and form.
3. Protection	Skull protects the brain; rib cage protects heart and lungs; vertebral column protects the spinal cord.
4. Movement / locomotion	Bones act as levers for muscles attached to them; joints allow movement.
5. Production of blood cells	Red bone marrow in long bones, ribs and sternum makes red and white blood cells and platelets.
6. Storage of minerals	Bone stores calcium and phosphorus, released to the blood when needed.
7. Breathing	Ribs and intercostal muscles help in ventilation of the lungs.

Function	Explanation / example
8. Hearing	The ear ossicles (malleus, incus, stapes) transmit and amplify sound vibrations.

EXAM TIP

When a question asks for "functions of the skeleton", give the **function + one example** each time (e.g. "Protection — the skull protects the brain"). A bare list of words usually earns only half marks.

QUICK REVISION GLOSSARY

Term	Meaning
Locomotion	Movement of the whole organism from place to place.
Ecdysis (moulting)	Shedding of the exoskeleton to allow growth.
Ossification	Hardening of cartilage into bone by deposition of calcium and phosphorus.
Osteon / Haversian system	Basic structural unit of compact bone.
Lamellae	Concentric rings of bone matrix.
Lacunae	Spaces in the matrix holding bone or cartilage cells.
Canaliculi	Tiny channels linking lacunae to each other and the Haversian canal.
Chondrin	The matrix of cartilage.
Zygapophyses	Articulating facets of vertebrae (pre- = anterior, post- = posterior).
Odontoid process	Peg on the axis around which the atlas and skull rotate.
Vertebrarterial canal	Hole in the transverse process of cervical vertebrae for the vertebral artery.
Glenoid cavity	Socket in the scapula for the head of the humerus.
Acetabulum	Socket in the pelvic girdle for the head of the femur.
Pentadactyl limb	Limb ending in five digits.
Antagonistic muscles	Pair of muscles whose actions oppose each other (e.g. biceps and triceps).

SECTION A: OBJECTIVE TEST (40 QUESTIONS)

Instructions: Choose the correct option (A, B, C or D) for each question.

1. The movement of a whole organism from one place to another is called

A. irritability
B. locomotion
C. respiration
D. growth

2. The skeleton of the earthworm is an example of

A. an endoskeleton
B. an exoskeleton
C. a cartilaginous skeleton
D. a hydrostatic skeleton

3. The exoskeleton of an insect is made mainly of

A. cartilage
B. bone
C. keratin
D. chitin

4. The skeleton of a shark is made of

A. cartilage only
B. chitin
C. bone and cartilage
D. bone only

5. The process by which cartilage is converted into bone is called

A. resorption
B. ossification
C. ecdysis
D. metamorphosis

6. The shedding of the exoskeleton to allow growth is known as

A. ecdysis
B. regeneration
C. hibernation
D. ossification

7. The main disadvantage of an exoskeleton is that it

A. provides surface for muscle attachment
B. restricts growth
C. protects against injury
D. prevents desiccation

8. Contraction of the circular muscles of an earthworm makes the body

A. shorter and fatter
B. stiff and immobile
C. longer and thinner
D. curl into a ball

9. Bone cells that secrete the bone matrix are

A. osteoblasts
B. osteoclasts
C. chondrocytes
D. osteocytes

10. Bone cells that break down (resorb) old bone tissue are

A. osteoblasts
B. osteoclasts
C. chondroblasts
D. osteocytes

11. The basic structural unit of compact bone is the

A. lamella
B. canaliculus
C. Haversian system
D. lacuna

12. The Haversian canal contains

A. cartilage
B. blood vessels and nerves
C. synovial fluid
D. chondrocytes

13. Tiny channels that connect the lacunae in bone are called

A. canaliculi
B. Haversian canals
C. lamellae
D. foramina

14. The protein fibres found in the matrix of both bone and cartilage are

A. collagen
B. myosin
C. chitin
D. keratin

15. Cartilage obtains its nutrients by

A. blood vessels within it
B. diffusion through the matrix
C. canaliculi
D. Haversian canals

16. The type of cartilage found in the pinna of the ear is

A. elastic cartilage
B. hyaline cartilage
C. fibrocartilage
D. calcified cartilage

17. Which of the following is NOT part of the axial skeleton?

A. Pectoral girdle
B. Sternum
C. Skull
D. Ribs

18. The immovable joints between the bones of the cranium are called

A. condyles
B. facets
C. sutures
D. ligaments

19. The total number of vertebrae in the human vertebral column is

A. 60
B. 45
C. 26
D. 33

20. The vertebrarterial canal is found in the

A. lumbar vertebra
B. cervical vertebra
C. thoracic vertebra
D. sacral vertebra

21. The odontoid process is a feature of the

A. axis
B. atlas
C. sacrum
D. lumbar vertebra

22. The joint between the skull and the atlas allows

A. rotation of the head
B. no movement
C. bending of the trunk
D. nodding of the head

23. The joint between the atlas and the axis is a

A. gliding joint
B. pivot joint
C. ball and socket joint
D. hinge joint

24. A vertebra with a long backward-pointing neural spine and facets for the ribs is a

A. caudal vertebra
B. lumbar vertebra
C. cervical vertebra
D. thoracic vertebra

25. Metapophyses and anapophyses are characteristic of the

A. cervical vertebra
B. atlas
C. lumbar vertebra
D. thoracic vertebra

26. The vertebrae which are fused and articulate with the pelvic girdle are the

A. thoracic vertebrae
B. sacral vertebrae
C. caudal vertebrae
D. lumbar vertebrae

27. The articulating facets of vertebrae are known as

A. transverse processes
B. zygapophyses
C. neural spines
D. centra

28. The spinal cord passes through the

A. Haversian canal
B. centrum
C. vertebrarterial canal
D. neural canal

29. Ribs that are not attached to the sternum are called

A. false ribs
B. true ribs
C. floating ribs
D. costal ribs

30. The bone of the upper arm is the

A. radius
B. humerus
C. femur
D. scapula

31. The olecranon process is found on the

A. ulna
B. humerus
C. radius
D. tibia

32. The structure which prevents the arm from bending backwards is the

A. olecranon process
B. patella
C. coronoid process
D. deltoid ridge

33. The head of the femur fits into the

A. acetabulum
B. sigmoid notch
C. glenoid cavity
D. obturator foramen

34. The head of the humerus articulates with the

A. trochlea
B. olecranon fossa
C. acetabulum
D. glenoid cavity

35. The longest and strongest bone in the human body is the

A. tibia
B. fibula
C. femur
D. humerus

36. The knee cap is also known as the

A. trochanter
B. patella
C. condyle
D. tarsal

37. The shoulder joint is an example of a

A. pivot joint
B. hinge joint
C. gliding joint
D. ball and socket joint

38. Tendons connect

A. muscle to bone
B. bone to bone
C. muscle to muscle
D. cartilage to bone

39. When the arm is bent at the elbow, the

A. biceps and triceps both contract
B. biceps contracts and the triceps relaxes
C. biceps and triceps both relax
D. triceps contracts and the biceps relaxes

40. The pelvic girdle is formed by the fusion of the

A. sacrum and coccyx
B. scapula and clavicle
C. ilium, ischium and pubis
D. femur and tibia

SECTION B: THEORY QUESTIONS

Instructions: Answer all questions. Draw diagrams neatly in pencil, label them fully and give each a title.

Question 1

(a) Distinguish between an exoskeleton, an endoskeleton and a hydrostatic skeleton, giving one example of an organism with each. **[6 marks]**

(b) State two advantages and two disadvantages of a hydrostatic skeleton. **[4 marks]**

(c) State four differences between an endoskeleton and an exoskeleton. **[4 marks]**

(d) Explain why the bones of old people fracture more easily than those of children. **[3 marks]**

Question 2

(a) Make a large, well-labelled diagram of the Haversian system of bone. **[8 marks]**

(b) Name the three types of bone cells and state the function of each. **[6 marks]**

(c) State four differences and two similarities between bone and cartilage. **[6 marks]**

Question 3

(a) Name the five regions of the mammalian vertebral column and state the number of vertebrae in each region in humans. **[5 marks]**

(b) Describe the features of a typical (generalised) vertebra and state the function of each feature. **[6 marks]**

(c) Explain how the atlas and axis are adapted to their functions. **[6 marks]**

(d) State three differences between the atlas and the axis. **[3 marks]**

Question 4

(a) Make a well-labelled diagram of the lateral view of the lumbar vertebra. **[8 marks]**

(b) State four features of the lumbar vertebra which fit it for its function. **[4 marks]**

(c) State four differences between a thoracic vertebra and a lumbar vertebra. **[4 marks]**

(d) Distinguish between true ribs, false ribs and floating ribs. **[3 marks]**

Question 5

(a) Make a labelled drawing of the anterior view of the humerus of a rabbit. **[7 marks]**

(b) State the function of each of the following: (i) olecranon process; (ii) coronoid process; (iii) sigmoid notch; (iv) supratrochlear foramen. **[4 marks]**

(c) State five similarities between the fore limb and hind limb of a mammal. **[5 marks]**

(d) State three differences between the pectoral and pelvic girdles. **[3 marks]**

Question 6

(a) Name the four types of freely movable joints and give one example of each. **[8 marks]**

(b) Describe the structure of a synovial joint. **[4 marks]**

(c) Explain how the biceps and triceps muscles bring about the bending and straightening of the arm. **[5 marks]**

(d) State three differences between tendons and ligaments. **[3 marks]**

Question 7 (Essay)

Describe **six** functions of the mammalian skeleton, giving an example in each case. **[12 marks]**

Explain how the structure of cartilage suits it for its functions in the body. **[8 marks]**

ANSWER SECTION (DETACH BEFORE ISSUING TO STUDENTS IF REQUIRED)

Answer Key — Objective Test

No.	Ans.	Reason	No.	Ans.	Reason
1	**B**	Whole body moves.	21	**A**	Peg for rotation.
2	**D**	Fluid under pressure.	22	**D**	Hinge joint.
3	**D**	Arthropod cuticle.	23	**B**	Rotation.
4	**A**	Cartilaginous fish.	24	**D**	Rib facets.
5	**B**	Ca and P deposited.	25	**C**	Lumbar only.
6	**A**	Moulting.	26	**B**	Sacrum.
7	**B**	Must moult to grow.	27	**B**	Articulating facets.
8	**C**	Circular muscles squeeze.	28	**D**	Spinal cord.
9	**A**	Build bone.	29	**C**	Last 2 pairs.
10	**B**	Break down bone.	30	**B**	Upper arm.
11	**C**	Osteon.	31	**A**	Elbow point.
12	**B**	Blood supply to bone.	32	**A**	Locks in olecranon fossa.
13	**A**	Link lacunae.	33	**A**	Hip socket.
14	**A**	Common protein.	34	**D**	Shoulder socket.
15	**B**	Avascular.	35	**C**	Longest bone.
16	**A**	Flexible, springs back.	36	**B**	Knee cap.
17	**A**	Appendicular.	37	**D**	Moves in all planes.
18	**C**	Fixed joints.	38	**A**	Tendon = muscle–bone.
19	**D**	7+12+5+5+4.	39	**B**	Flexor contracts.
20	**B**	For vertebral artery.	40	**C**	Innominate bone.

Answer distribution — A: 12 B: 13 C: 7 D: 8

Model Answers — Theory

Question 1

(a) Exoskeleton: hard, non-living skeleton outside the body, e.g. grasshopper (chitin). **Endoskeleton:** living skeleton inside the body made of bone and cartilage, e.g. man, rabbit. **Hydrostatic skeleton:** body-wall muscles contract against fluid under pressure in a coelom, e.g. earthworm.

(b) *Advantages (any two):* allows easy movement by burrowing/swimming; flexible shape lets animal squeeze into narrow places to escape; repairs easily as fluid is refilled; cushions

internal organs. *Disadvantages (any two):* limited attachment for organs; restricted to aquatic/moist habitats; lacks strength on land.

(c) Any four rows from the table in Section 2.4 (inside/outside; living/non-living; grows/must be shed; muscles attached outside/inside; allows large/limits size).

(d) Children's skeletons contain more flexible cartilage and bone rich in collagen. With age, ossification increases mineral content, making bone hard but brittle; in old age bone also loses minerals (osteoclast activity exceeds osteoblast activity), so bones become weak and fracture easily.

Question 2

(a) Diagram as Fig. 2. *Marks:* title (1); Haversian canal (1); lamellae (1); lacunae (1); osteocytes (1); canaliculi (1); accuracy/neatness of concentric rings (2).

(b) Osteoblasts — secrete bone matrix (form new bone). **Osteocytes** — mature bone cells enclosed in lacunae; maintain the matrix. **Osteoclasts** — break down/resorb old bone and release minerals to the blood.

(c) *Differences (any four):* bone hard/cartilage flexible; bone has blood vessels and nerves/cartilage avascular; bone matrix organic + inorganic/cartilage purely organic; bone has calcium deposits/cartilage none; bone porous/cartilage non-porous; bone has Haversian systems/cartilage none. *Similarities (any two):* both connective tissues; both have living cells in a matrix; both contain collagen; cells in lacunae in both; both give support.

Question 3

(a) Cervical — 7; thoracic — 12; lumbar — 5; sacral — 5 (fused); caudal/coccygeal — 4 (fused). Total 33.

(b) Centrum — bears weight, withstands compression; neural arch — protects spinal cord; neural canal — carries spinal cord; neural spine — muscle attachment; transverse processes — muscle attachment; pre- and postzygapophyses — articulate with neighbouring vertebrae.

(c) Atlas: large concave anterior facets receive the occipital condyles, forming a hinge joint for nodding; wide flat transverse processes give large surface for neck muscles that move the head; large neural canal receives the odontoid process. **Axis:** odontoid process projects into the atlas, forming a pivot for rotation of the head; large neural spine anchors muscles that turn the head; flat articular surfaces allow the atlas to turn smoothly.

(d) Any three rows of the atlas/axis table (odontoid absent/present; prezygapophyses present/absent; large/small transverse processes; reduced/large neural spine; nodding/rotation).

Question 4

(a) Diagram as Fig. 12 (lateral view). *Marks:* title (1); neural spine (1); metapophysis (1); anapophysis (1); pre- and postzygapophyses (1); centrum (1); transverse process (1); hypapophysis (1).

(b) Large, thick centrum to support weight of upper body; long transverse processes for attachment of abdominal and back muscles; metapophyses and anapophyses give extra muscle attachment and interlock under stress; large neural arch and zygapophyses allow bending and twisting.

(c) Thoracic: long backward-pointing neural spine / Lumbar: short forward-pointing spine; Thoracic: short transverse processes / Lumbar: long, forward-pointing transverse processes; Thoracic: facets for ribs / Lumbar: no rib facets; Thoracic: no metapophyses or anapophyses / Lumbar: present; Thoracic: smaller centrum / Lumbar: largest centrum.

(d) True ribs (7 pairs) attach directly to the sternum by their own costal cartilage; false ribs (3 pairs) attach indirectly through the cartilage of the rib above; floating ribs (2 pairs) are not attached to the sternum.

Question 5

(a) Diagram as Fig. 16. *Marks:* title (1); head (1); bicipital groove (1); deltoid ridge (1); shaft (1); coronoid fossa / supratrochlear foramen (1); trochlea (1).

(b) (i) Olecranon process — locks into the olecranon fossa and prevents the arm bending backwards; provides attachment for the triceps. (ii) Coronoid process — fits into the coronoid fossa when the arm is flexed. (iii) Sigmoid notch — articulates with the trochlea of the humerus at the elbow. (iv) Supratrochlear foramen — passage of nerve, artery and vein.

(c) Both are pentadactyl; both have three parts; one bone in the upper part (humerus/femur); two bones in the middle part (radius-ulna/tibia-fibula), one larger than the other; both have small bones at wrist/ankle, five metacarpals/metatarsals and phalanges; first digit has 2 phalanges; both end in nails/claws/hooves; both articulate with a girdle through a ball-and-socket joint.

(d) Any three rows from the girdles table (fore/hind limbs; scapula + clavicle/ilium + ischium + pubis; not fused/fused to sacrum; glenoid cavity/acetabulum; obturator foramen absent/present).

Question 6

(a) Ball and socket — shoulder or hip; hinge — elbow or knee; pivot — atlas and axis; gliding — wrist, ankle, or between zygapophyses.

(b) Ends of bones covered by articular (hyaline) cartilage to reduce friction; joint enclosed by a capsule with ligaments that hold the bones together; the capsule is lined by a synovial membrane which secretes synovial fluid; synovial fluid fills the cavity and lubricates the joint.

(c) Muscles work in antagonistic pairs and can only pull. To bend the arm, the biceps (flexor) contracts, pulling on the radius, while the triceps relaxes; the forearm rises about the elbow (fulcrum). To straighten, the triceps (extensor) contracts, pulling on the olecranon process of the ulna, while the biceps relaxes. Nerve impulses coordinate the contractions.

(d) Tendon joins muscle to bone / ligament joins bone to bone; tendon inelastic / ligament elastic; tendon of white collagen fibres / ligament of yellow elastic fibres.

Question 7

Award 2 marks per function (1 for the function, 1 for the example). Support (framework holds up body and soft organs); shape (gives body its form); protection (skull — brain; ribs — heart and lungs; vertebrae — spinal cord); movement (bones act as levers for muscles, e.g. humerus in arm); blood cell production (red marrow of femur, ribs, sternum); mineral storage (calcium and phosphorus); breathing (ribs and intercostal muscles); hearing (ear ossicles).

Cartilage: its **collagen fibres** give tensile strength, so it supports without breaking (e.g. rings keep the trachea open); its **flexible matrix** bends and springs back, giving shape to the pinna and nose; its **smooth surface** at the ends of bones reduces friction at joints; **fibrocartilage** in intervertebral discs absorbs shock; being **avascular**, it can sit between bones under pressure without blood vessels being crushed, getting nutrients by diffusion.

MASTERMIND SCHOLARS EDUCATIONAL CENTRE

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"Dream, Believe and Achieve Excellence"

PART 2

GIRDLES, JOINTS & MUSCLES

The Skeletal System continued — How Mammals Move

Mastermind Notes • Diagrams • 40 Objectives • Study Questions • 7 Theory Questions • Answers

MASTERMIND NOTES

Pectoral & pelvic girdles • Joints • Types of muscle • Antagonistic muscles • Sliding filament model • Bone injuries

Name: …… Class: ……

Programme Coordinator: Douglas Gyapong

HOW TO USE PART 2

This is the continuation of Chapter 9 (Part 1: Types of skeleton, bone and cartilage, axial skeleton, vertebrae and limbs). Figure numbers continue from Part 1, which ended at Fig. 27, so both parts can be bound together.

LEARNING OUTCOMES

By the end of Part 2, the learner should be able to:

1. Describe the structure and functions of the **pectoral girdle** and the **pelvic girdle**, and state their differences.
2. Identify the bone associated with any named feature of the mammalian skeleton.
3. Define a **joint**, classify joints and give examples of each type.
4. Describe the structure of a **synovial joint** and explain the functions of its parts, using the **shoulder** and **elbow** joints as examples.
5. Describe the three types of muscle — **skeletal, cardiac and smooth** — and state their characteristics, locations and functions.
6. Explain how **antagonistic muscles** (biceps and triceps) bend and straighten the arm.
7. Describe the structure of skeletal muscle and the **sarcomere**, and explain the **sliding filament model** of muscle contraction.
8. Describe common injuries and disorders of bones, joints and muscles, and their first aid.

KEY IDEAS

- **Limb girdles** attach the limbs to the axial skeleton: the **pectoral girdle** is loosely held by muscles (flexible); the **pelvic girdle** is fused to the sacrum (rigid).
- A **joint** is where two or more bones meet. **Synovial joints** (gliding, pivot, hinge, ball and socket) are freely movable.
- Muscles can only **pull**, so skeletal muscles work in **antagonistic pairs** — a **flexor** bends a limb and an **extensor** straightens it.
- Muscle contracts when **actin filaments slide over myosin filaments**, shortening each **sarcomere**. This needs **calcium ions** and **ATP**.

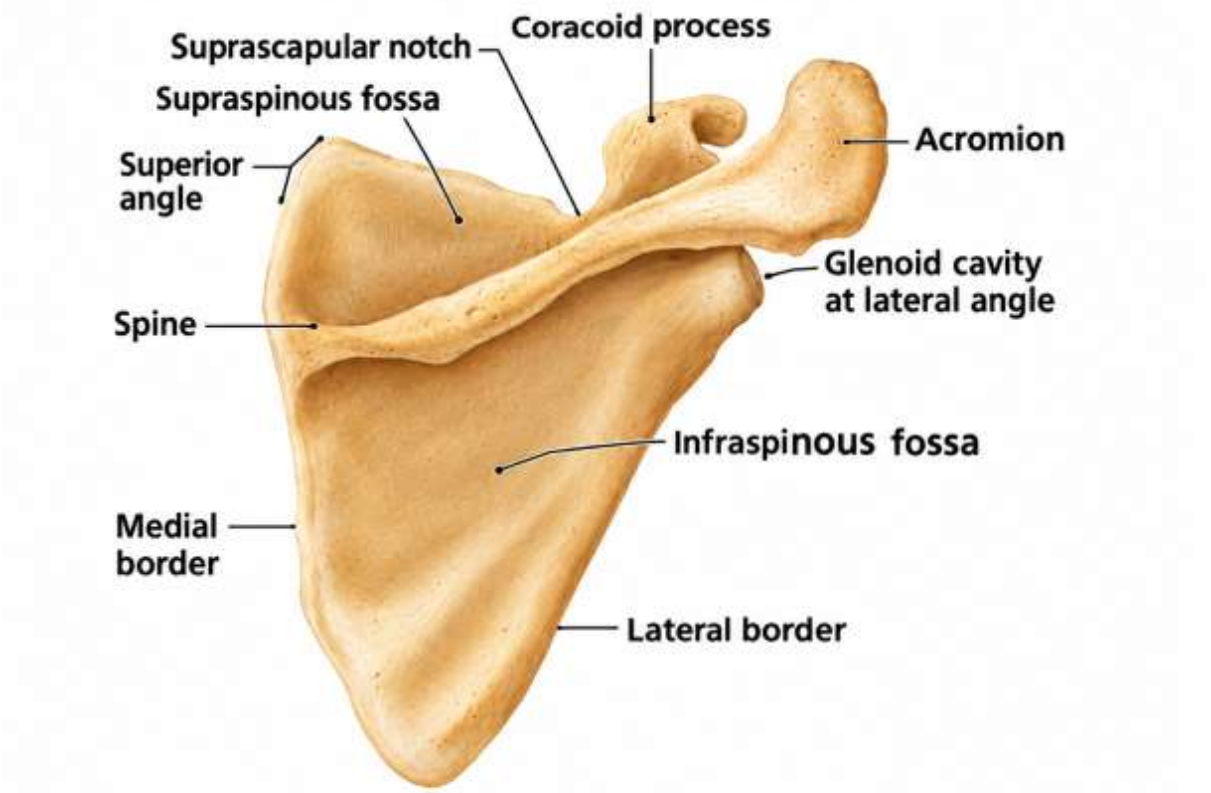
PART 9: THE LIMB GIRDLES

Limb girdles are bones which attach the limbs to the axial skeleton. There are two girdles: the **pectoral (shoulder) girdle**, which attaches the fore limbs, and the **pelvic (hip) girdle**, which attaches the hind limbs.

9.1 The Pectoral Girdle

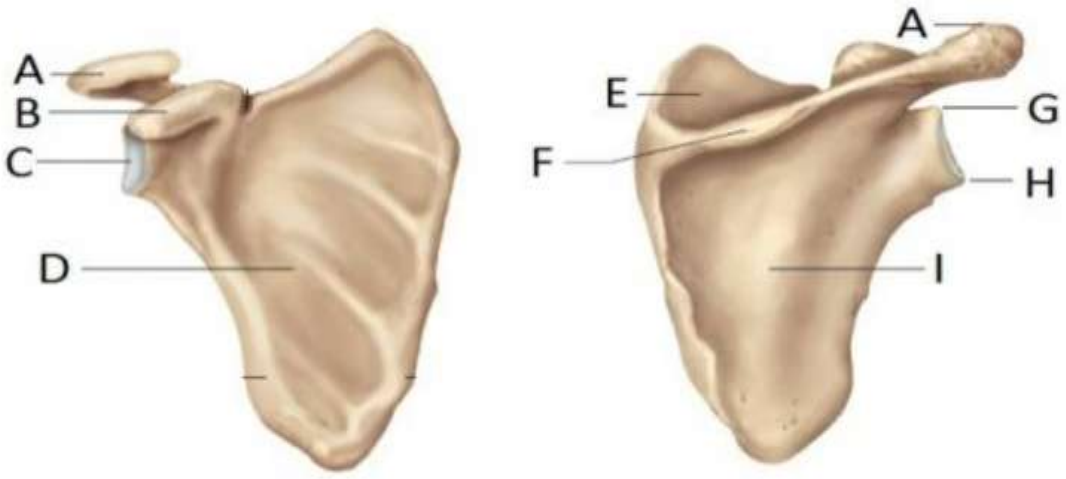
The pectoral girdle is made up of bones which attach the **fore limbs** to the axial skeleton. It consists of **two halves (left and right)**. Each half consists of three bones: the **scapula** (shoulder blade), the **coracoid** and the **clavicle** (collar bone).

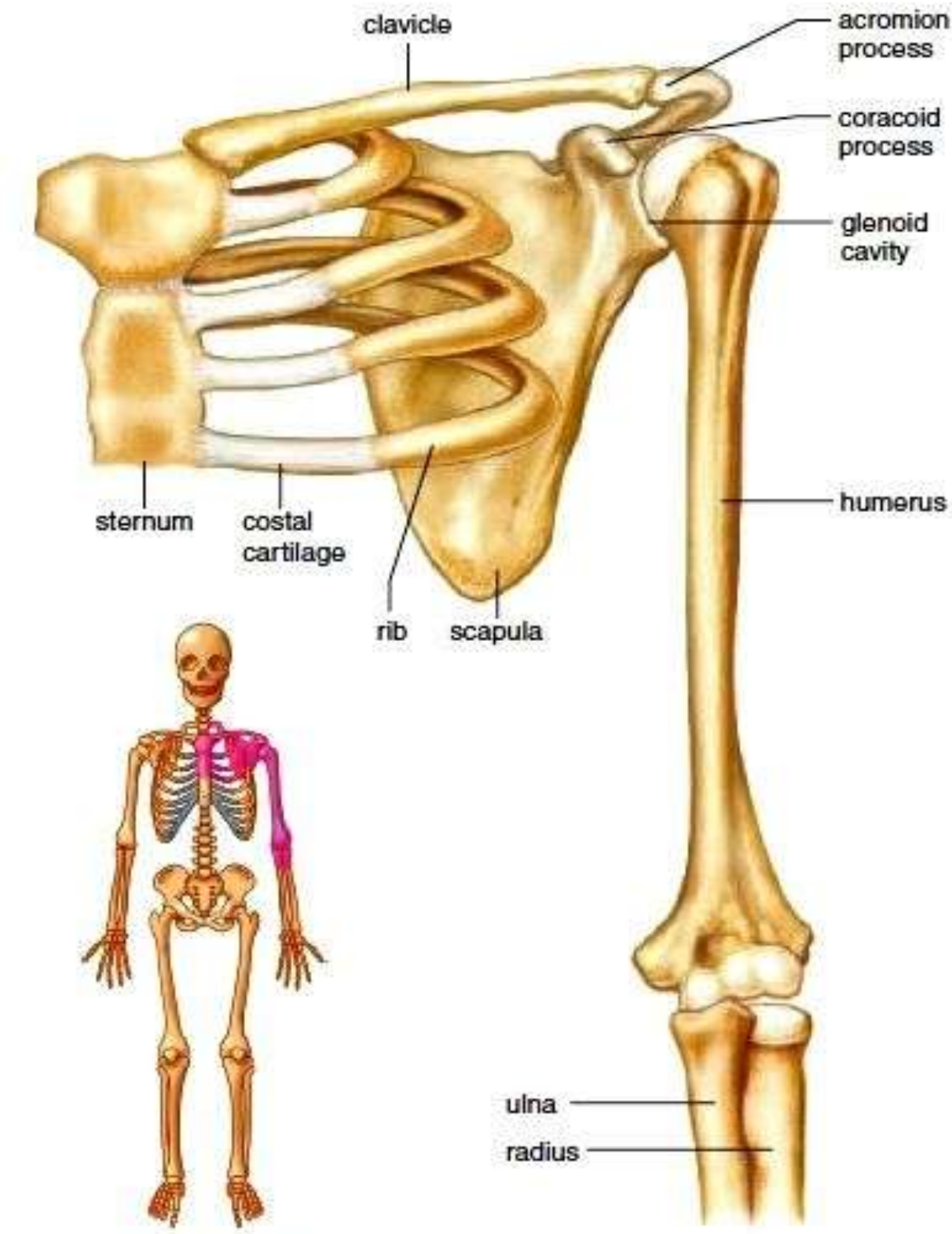
- The **scapula** is a **flat, triangular** bone lying over the ribs at the back of the thorax.
- The **coracoid** is a smaller bone which is **fused to the scapula at its apex**. In mammals it is reduced to a small hook called the **coracoid process**.
- At the apex of the scapula, near where the coracoid fuses, is a shallow depression called the **glenoid cavity**. The **head of the humerus articulates** here to form the **shoulder joint**.
- Close to the glenoid cavity are projections called the **acromion process** and the **metacromion process** (well developed in the rabbit). Both are for **muscle attachment**; the acromion also provides a surface for **articulation with the clavicle**.
- A ridge called the **spine** runs along the outer surface of the scapula for muscle attachment. The upper edge of the blade is the **suprascapula** (a cartilaginous border in the rabbit).
- One end of the **clavicle** joins the scapula at the acromion process; the other end is held **ventrally** by muscles and ligaments to the **sternum**.
- The base of the scapula is held **dorsally** by muscles to the vertebral column.



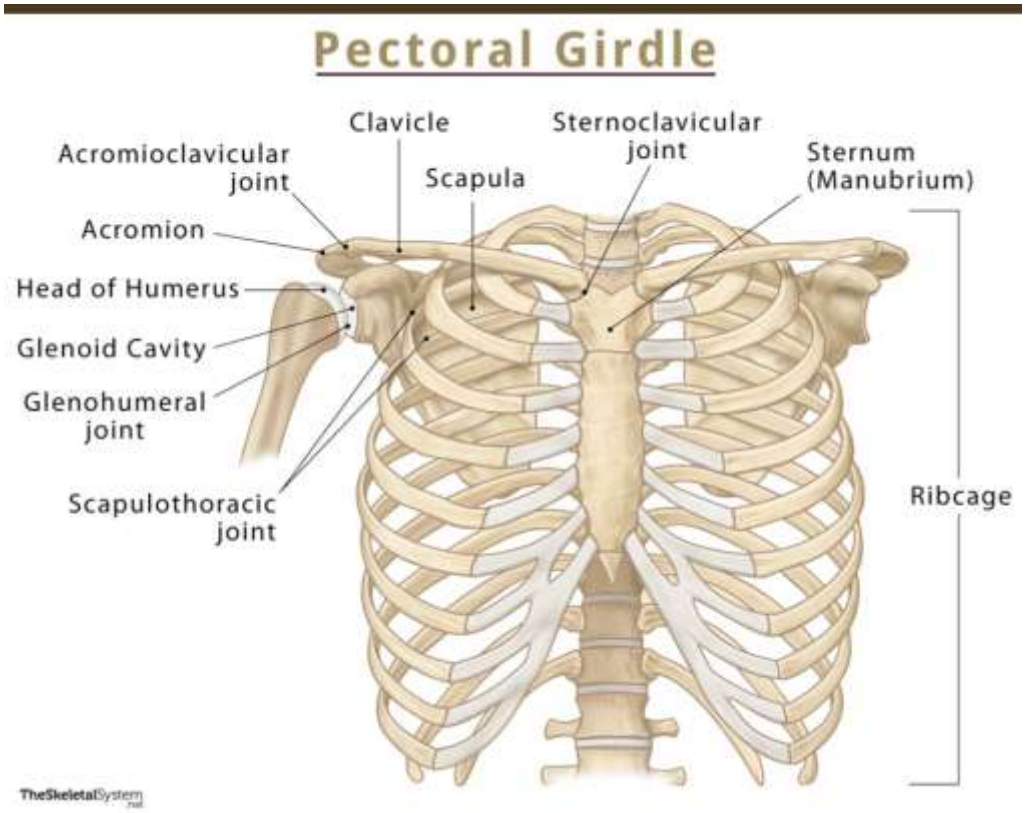
(c) Right scapula, posterior aspect

Figure 5.23c


A
B
C
D
A
E
F
G
H
I



a. Pectoral girdle, frontal view



NOTE

The pectoral girdle bones (scapula and clavicle) are **not fused** to the axial skeleton; they are **held by muscles**. This allows **flexibility** — the girdle can move through many angles — and helps to **absorb shock** when an animal lands on its fore limbs.

Note that In **adult mammals** the coracoid is **fused to the scapula** as the **coracoid process**, so each half of the pectoral girdle is usually described as **two bones: scapula and clavicle** (with the coracoid process on the scapula). If a question asks for "three bones", give **scapula, coracoid and clavicle**, as the textbook does. The clavicle is well developed in man but **reduced** in the rabbit.

Functions of the pectoral girdle

1. Attaches the **fore limbs** to the axial skeleton.
2. Provides the **glenoid cavity** for articulation with the head of the humerus (shoulder joint).
3. Provides surfaces (spine, acromion, metacromion, coracoid process) for **attachment of muscles** that move the fore limb.
4. Its flexible attachment allows a **wide range of movement** of the arm and absorbs shock.

THINK & ANSWER

Q: (a) What is the pectoral girdle? (b) Name three bones that form the pectoral girdle in mammals.

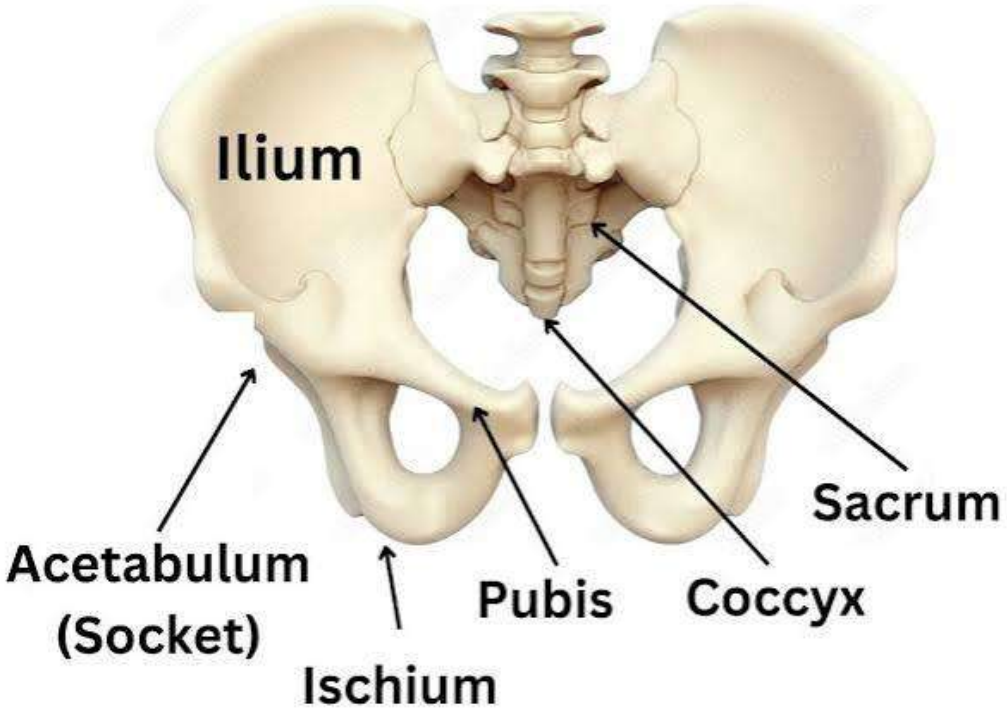
A: (a) The pectoral girdle is made up of bones which attach the **fore limb to the axial skeleton**. (b) **Scapula, coracoid and clavicle.**

9.2 The Pelvic Girdle

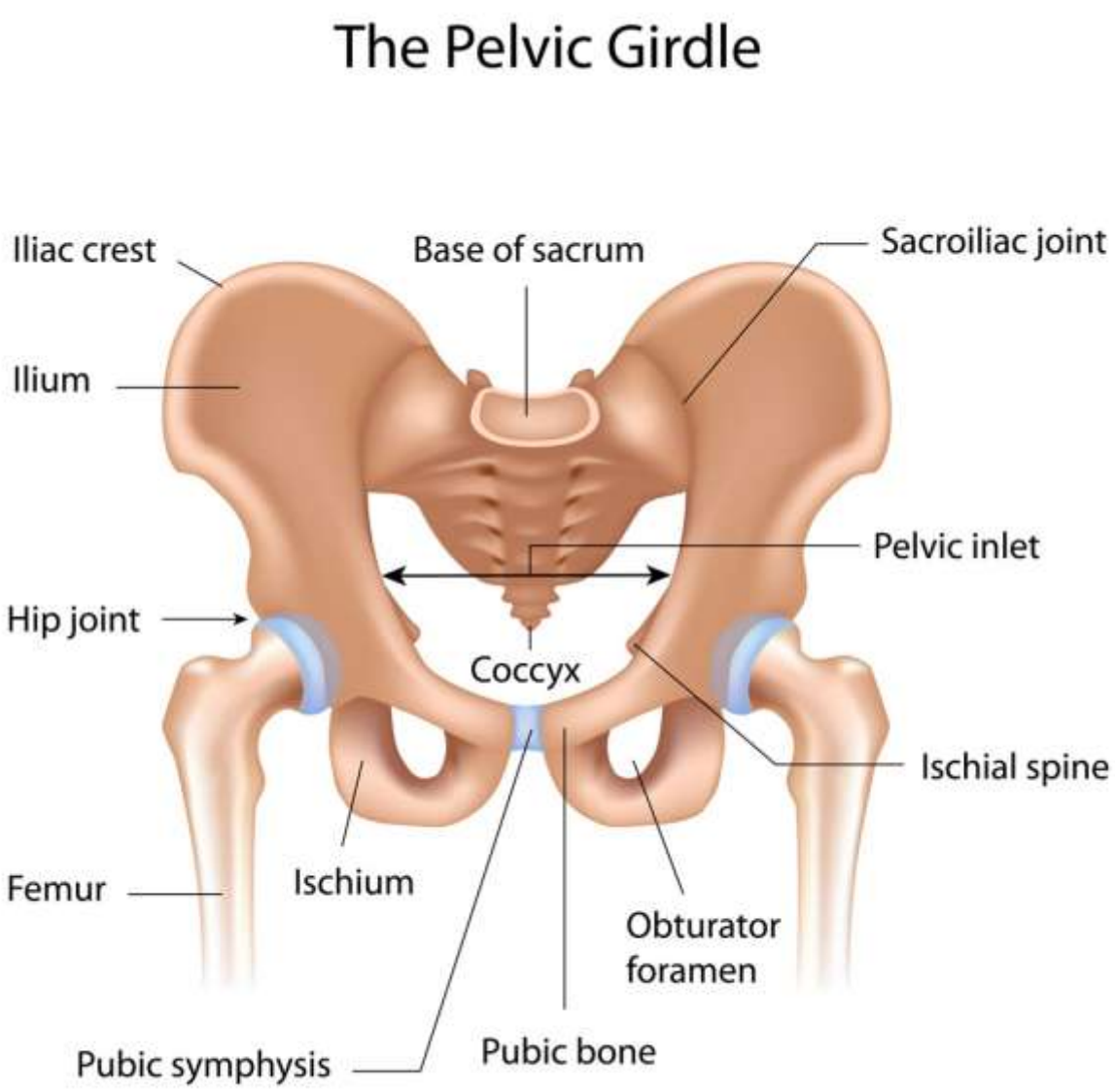
The pelvic girdle is made up of bones which attach the **hind limbs** to the axial skeleton. It also has **two halves**, but unlike the pectoral girdle, the two halves are **fused ventrally**. The line of fusion is called the **pubic symphysis**.

Each half of the pelvic girdle is known as the **innominate bone** (hip bone). The innominate bone is made of **three fused bones**: the **ilium**, the **ischium** and the **pubis**.

- The **ilium** is the **largest and longest** of the three, and it is the bone that **fuses the pelvic girdle to the vertebral column** (at the sacrum).
- The **ischium** and **pubis** fuse in such a way that they enclose a large hole between them called the **obturator foramen**. It allows the **passage of nerves and blood vessels** and makes the girdle **lighter**.
- At the point where all three bones meet is a deep hollow known as the **acetabulum**. This is where the **head of the femur fits**, joining the hind limb to the girdle (hip joint).
- The ilium has a **facet for the sacrum**, where it fuses with the sacral wings.



The pelvic girdle of humans



Remember: the two halves fuse at the **pubic symphysis** to form the pelvic girdle. The **pelvic girdle fused with the sacrum forms the pelvis**.

EXERCISE

Draw and label fully the human pelvis in your biological drawing book. Use a sharp HB pencil, draw with continuous lines (no shading), rule label lines that do not cross, and write a title below the drawing.

Functions of the pelvis

1. **Protects** the abdominal and pelvic organs such as the urinary bladder and the reproductive organs of the female.

2. **Supports and transmits the weight** of the upper body to the hind limbs.
3. The **obturator foramen** allows the passage of **nerves and blood vessels** and lightens the girdle.
4. The **acetabulum** provides a surface for **articulation with the head of the femur** (hip joint).
5. Provides surfaces for the **attachment of strong muscles** of the hind limbs and trunk.
6. In females, the wide pelvis forms the **birth canal** through which the baby passes.

9.3 Differences between the Pectoral and Pelvic Girdles

Pectoral girdle	**Pelvic girdle**
Flexibly attached to the axial skeleton by muscles.	**Fused** to the axial skeleton (sacrum).
Its bones are **not fused** together.	All its bones are **fused** (innominate bone).
The two halves are **not fused**.	The two halves are **fused** ventrally at the pubic symphysis.
Glenoid cavity is a **shallow** cavity.	**Acetabulum** is a **deep** cavity.
Made of scapula, coracoid and clavicle.	Made of ilium, ischium and pubis.
Attaches the **fore limbs**.	Attaches the **hind limbs**.
No obturator foramen.	Has an **obturator foramen**.
Allows a wide range of movement; bears less weight.	Rigid; supports the weight of the body.

EXAM TIP

Why is the glenoid cavity shallow but the acetabulum deep? A **shallow** socket gives the arm **freedom of movement** (throwing, climbing); a **deep** socket holds the femur **firmly** so the hip can bear the body's weight without dislocating. Structure always matches function!

9.4 Study Questions — Identifying Bones (with answers)

These questions from the textbook test whether you can link a named feature to its bone. Cover the right-hand column and test yourself.

Study Question 1

(a) The innominate bone is made up of three bones namely **ilium, ischium and pubis**. (b) The three bones of the innominate are fused at the **acetabulum**. (c) The two innominate bones fuse **ventrally** at the **pubic symphysis**.

Study Question 2: Identify the bone associated with each term

Term / feature	**Bone**
(a) Odontoid peg (process)	Axis

Term / feature	Bone
(b) Obturator foramen	Pelvic girdle
(c) Supratrochlear foramen	Humerus
(d) Glenoid cavity	Scapula
(e) Vertebrarterial canal	Cervical vertebra
(f) Cervical rib	Cervical vertebra
(g) Tuberculum (tubercle)	Rib
(h) Tubercular facet	Thoracic vertebra
(i) Occipital condyle	Skull
(j) Patella groove	Femur
(k) Olecranon fossa	Humerus
(l) Olecranon process	Ulna
(m) Acetabulum	Pelvic girdle
(n) Metacromion process	Scapula
(o) Condyles	Femur (also the lower jaw and the occipital condyles of the skull)
(p) Greater and lesser tuberosity	**Humerus**

MASTERMIND CORRECTION

The textbook hint gives "Femur" for **greater and lesser tuberosity**. This is wrong: the **tuberosities are on the humerus** (see the posterior view of the humerus in Part 1). The femur has **trochanters**, not tuberosities.

Study Question 3: Identify the bone associated with each feature

Term / feature	Bone
(a) Great trochanter	Femur
(b) Hypapophysis	Lumbar vertebra
(c) Coccyx	Caudal vertebrae (fused)
(d) Cranium	Skull
(e) Pelvis	Hip bone (pelvic girdle + sacrum)
(f) Pubic symphysis / ilium	Pelvic girdle
(g) Coracoid process	Scapula
(h) Trochlea	Humerus
(i) Deltoid ridge	Humerus

Term / feature	Bone
(j) Metapophysis	Lumbar vertebra
(k) Sutures	Skull
(l) Sigmoid notch	Ulna
(m) Cervical rib	Cervical vertebra
(n) Glenoid cavity	Scapula
(o) Acetabulum	Pelvic girdle

Study Question 4: Identify the bones with the following common names

Term / feature	Bone
(a) Shoulder blade	Scapula
(b) Collar bone	Clavicle
(c) Largest (thigh) bone in mammals	Femur
(d) Heel bone	Calcaneum — the largest tarsal
(e) Knee cap	Patella
(f) Wrist bone	Carpal
(g) Breastbone	Sternum
(h) Hip bone	Innominate bone

PART 10: JOINTS

A **joint** is a point where two or more bones meet. A joint may be **movable** or **immovable**.

10.1 Types of Joints

(a) Immovable (fixed / fibrous) joints

These joints **do not allow movement**. The bones are joined tightly by fibrous tissue or are fused.

Examples: (a) joints between the bones of the **cranium** (**sutures**); (b) joints between the fused bones of the **pelvic girdle** (ilium, ischium, pubis) and between the pelvic girdle and the sacrum.

(b) Slightly movable (cartilaginous) joints

These allow **only a little movement**. The bones are joined by a pad of cartilage. *Examples:* between the **centra of vertebrae** (intervertebral discs); at the **pubic symphysis**; between the **ribs and sternum** (costal cartilage).

(c) Movable (synovial) joints

These allow **free movement** of body parts. They include four types:

1. **Gliding (sliding) joint** — allows bones to **slide over one another** to a limited extent. *Examples:* the **wrist** (between carpals); the **ankle** (between tarsals); the **vertebral column** (between the zygapophyses of vertebrae).
2. **Pivot (rotating) joint** — allows **rotation of one part of the body on another**. *Example:* between the **atlas and axis** — the odontoid process of the axis fits into the neural canal of the atlas, and the atlas (with the skull) rotates around it; also between the radius and ulna (turning the palm).
3. **Hinge joint** — allows movement in **one plane / direction only**, like a hinged door. *Examples:* the **elbow** (humerus and ulna); the **knee** (femur and tibia); **finger and toe joints** (between the phalanges); the skull and atlas (nodding).
4. **Ball and socket joint** — allows movement in **all planes / directions** (and rotation). *Examples:* the **shoulder** (round head of the humerus in the glenoid cavity of the scapula); the **hip** (round head of the femur in the acetabulum of the pelvic girdle).

MASTERMIND CORRECTION

The textbook note says hinge and ball-and-socket joints are the synovial joints. In fact **all four** freely movable joints — gliding, pivot, hinge and ball-and-socket — are **synovial joints**, because each has a joint cavity with synovial fluid. Hinge and ball-and-socket joints are simply the **most freely movable** ones.

Type of joint	Degree of movement	Example (bones involved)
Immovable / fixed	None	Cranium (sutures); bones of pelvic girdle
Slightly movable	Little	Between vertebral centra; pubic symphysis
Gliding	Sliding, limited	Wrist (carpals); ankle (tarsals); zygapophyses
Pivot	Rotation of one bone around another	Atlas and axis
Hinge	One plane only	Elbow (humerus–ulna); knee (femur–tibia); fingers
Ball and socket	All planes; rotation	Shoulder (humerus–scapula); hip (femur–pelvic girdle)

THINK & ANSWER

Q: Name five types of joints and one example of the body part where each is found.

A: Immovable joint — cranium/skull; **pivot joint** — between the atlas and axis; **gliding joint** — wrist/ankle/vertebral column; **hinge joint** — knee/elbow; **ball and socket joint** — hip/shoulder.

10.2 Structure of a Typical Movable (Synovial) Joint

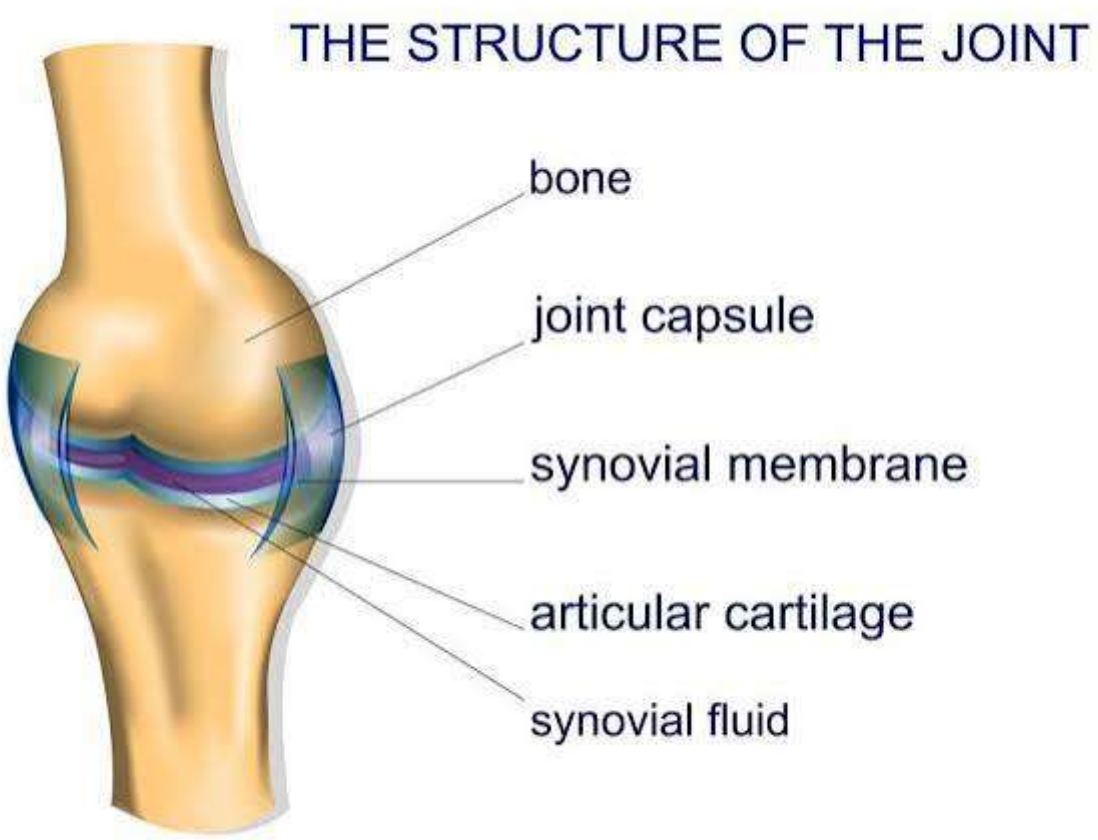
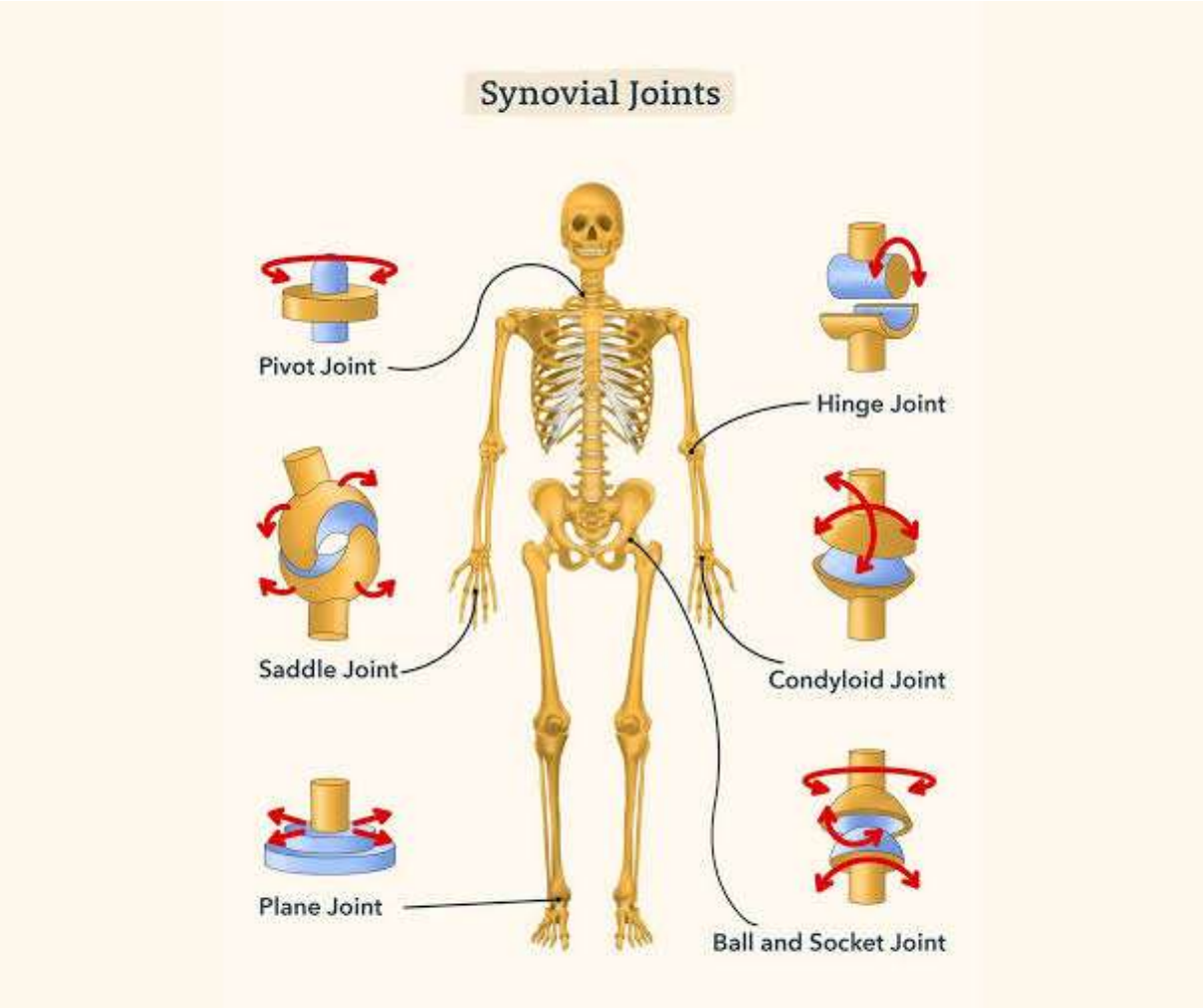
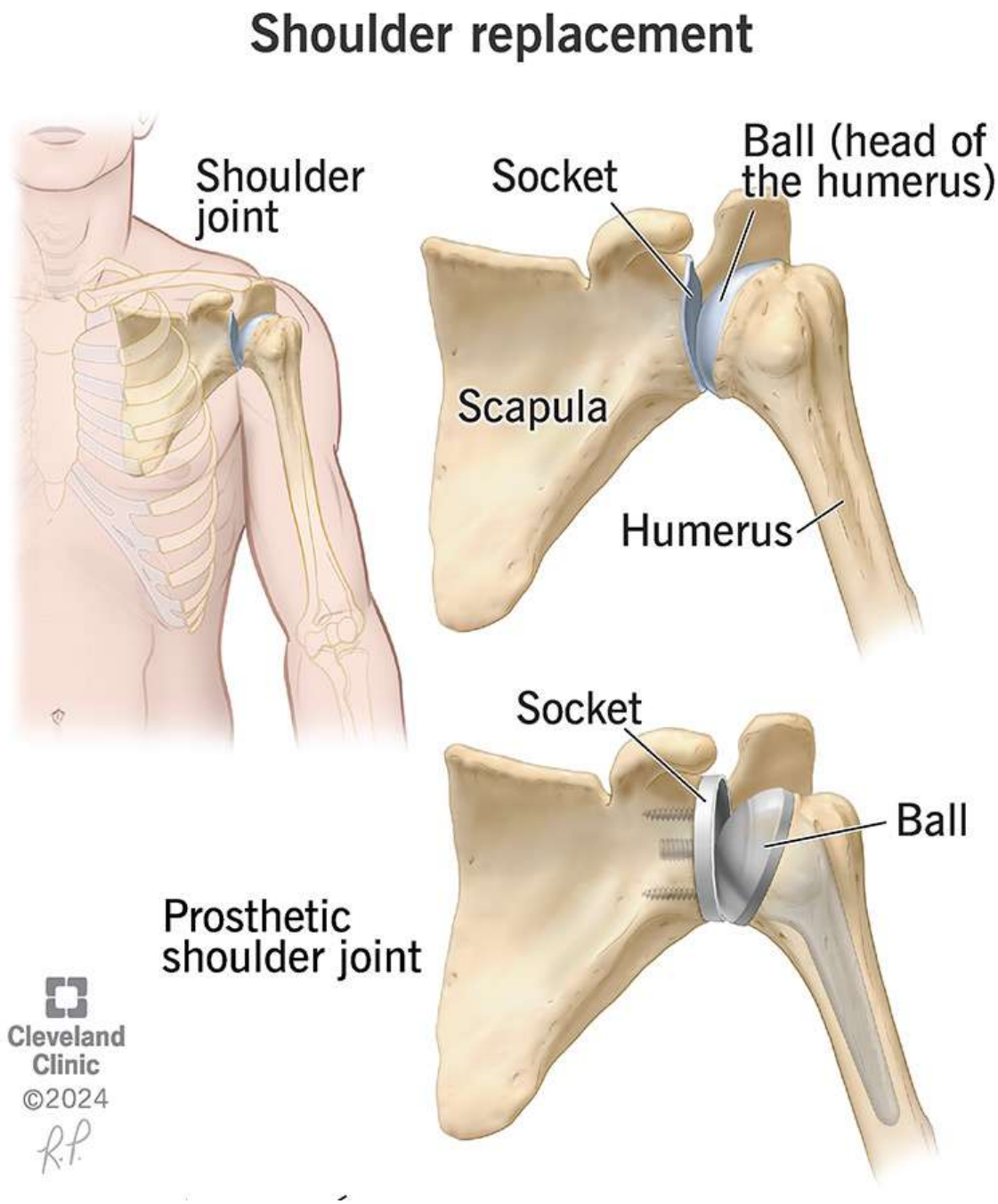


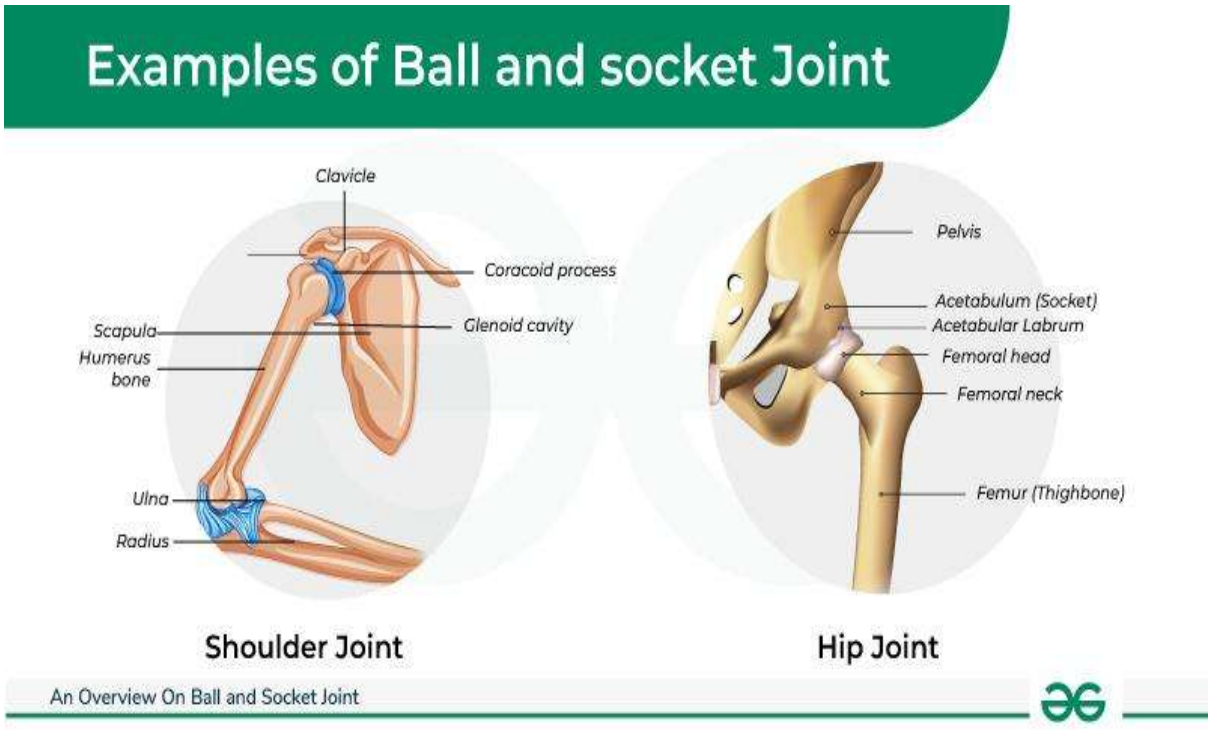
Fig. 34: A typical movable (synovial) joint — longitudinal section



Part	Description	Function
Ligaments (capsular / articular)	Tough bands of **elastic** connective tissue joining **bone to bone**; form the joint capsule.	Hold the bones in position and **prevent dislocation**, while allowing movement.
Articular cartilage	Smooth hyaline cartilage covering the ends (surfaces) of each bone at the joint.	Acts as a **shock absorber**; **reduces friction** so bone surfaces are not worn out.
Synovial membrane	Thin membrane lining the **inner surface of the ligament / capsule**.	**Secretes synovial fluid** into the synovial cavity.
Synovial cavity	The space enclosed by the synovial membrane.	Holds the synovial fluid.
Synovial fluid	Thick, slippery fluid in the cavity.	**Lubricates** the joint, **reduces friction** and nourishes the cartilage.

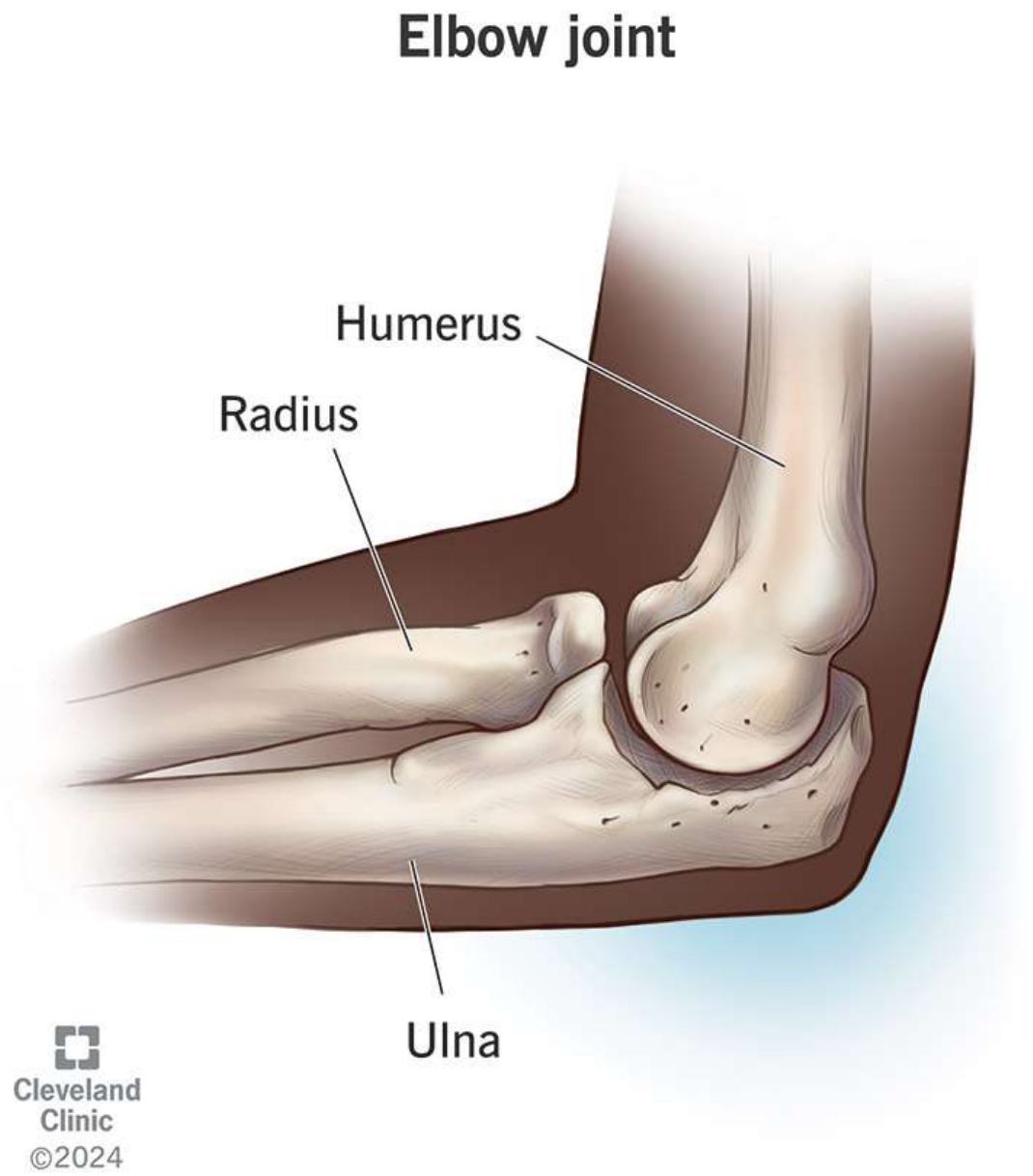
10.3 The Shoulder Joint (Ball and Socket)





- The shoulder joint is a **ball and socket joint** formed by the **pectoral girdle (scapula)** and the **humerus**.
- The **round head of the humerus** fits like a **ball** into the **glenoid cavity** of the scapula, which acts as the **socket**.
- **Capsular ligaments** join bone to bone. At the shoulder they are **loose**, allowing movement in all directions (but making the shoulder easier to dislocate).
- **Articular cartilage** covers the articulating surfaces. Its flexibility makes it act like a **cushion** when bones touch and rub, reducing friction and preventing wearing off of the bones.
- The inner surface of the ligament is lined by the **synovial membrane**, which secretes **synovial fluid** into the synovial cavity to lubricate the joint and reduce friction.

10.4 The Elbow Joint (Hinge)



- The elbow joint is a **hinge joint** between the **humerus and the ulna**.
- The **trochlea** of the humerus articulates with the **sigmoid (trochlear) notch** of the ulna.
- **Capsular ligaments** join bone to bone; articular cartilage, synovial membrane and synovial fluid work as in the shoulder joint.
- The **olecranon process** of the ulna locks into the olecranon fossa of the humerus, so the arm **cannot bend backwards** — movement is in one plane only.
-

Shoulder joint vs elbow joint

Shoulder joint	**Elbow joint**
Ball and socket joint.	Hinge joint.
Between the humerus and the scapula (glenoid cavity).	Between the humerus (trochlea) and the ulna (sigmoid notch).
Movement in **all planes**, including rotation.	Movement in **one plane** only.
Capsular ligaments are loose.	Capsular ligaments are tighter, with strong side ligaments.
No bony stop to movement.	Olecranon process stops backward bending.

NOTE

Arthritis: breakdown (wearing away) of the articular cartilage causes **arthritis**, which is marked by joint **pain, swelling and stiffness**, because the bones rub directly against each other.

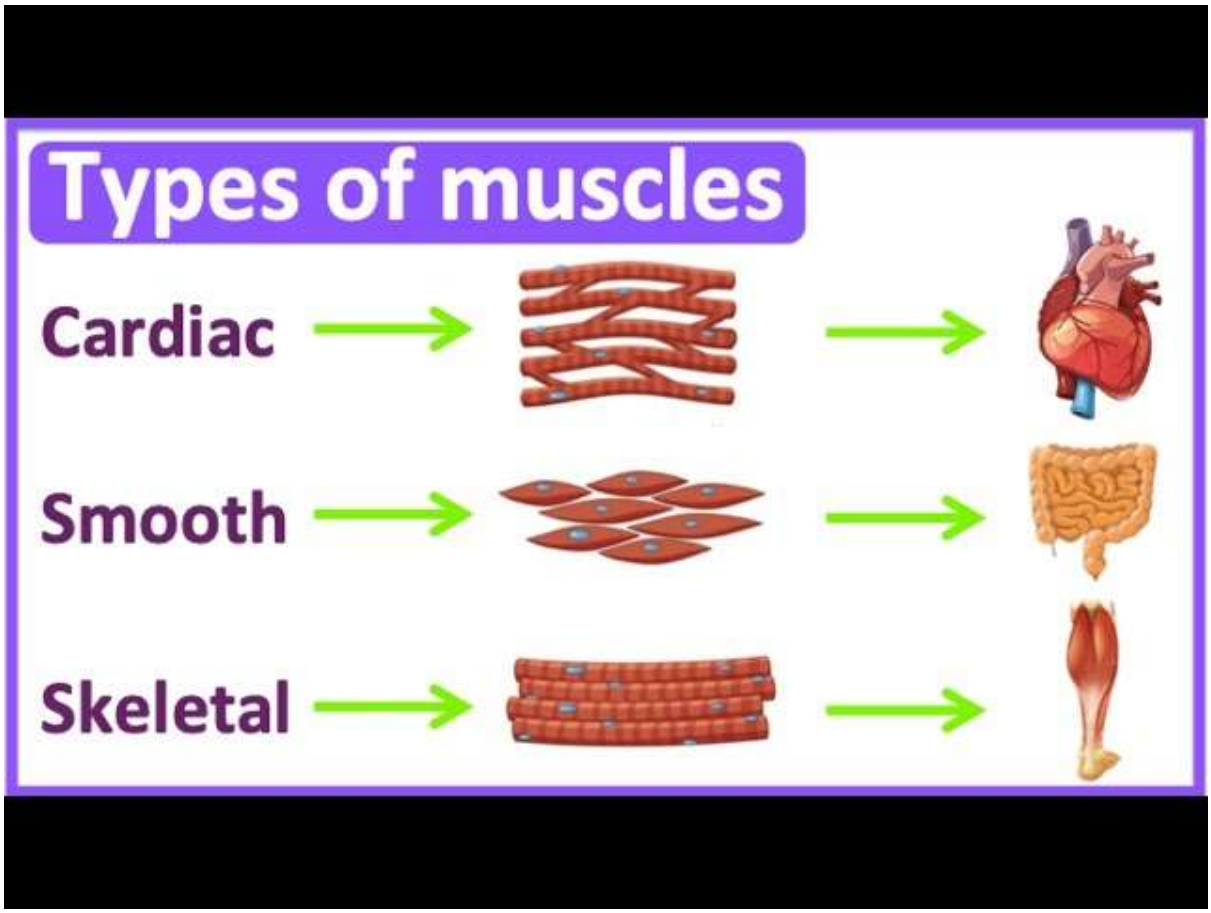
PART 11: MUSCLES

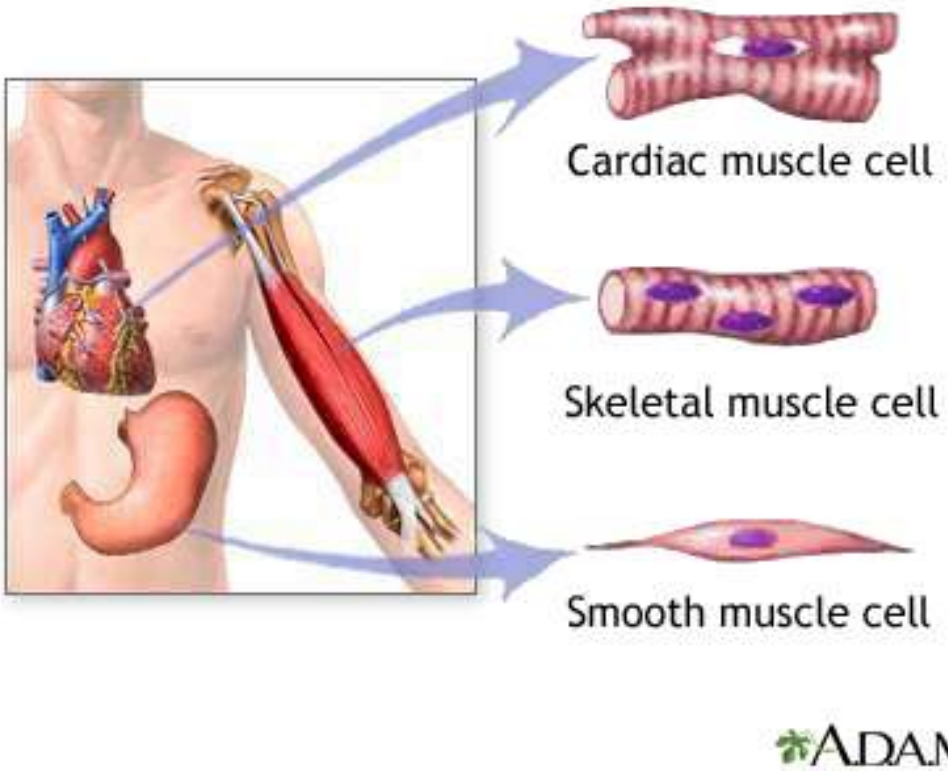
11.1 Overview

Muscles are tissues made of elongated cells called **muscle fibres**, which are capable of **contracting** to bring about movement of the body or body parts.

- Some muscle fibres have transverse bands called **striations** running across (perpendicular to) their length — these are **striated** muscle fibres.
- Some muscle fibres **lack striations** — these are **smooth** muscle fibres.
- Some muscles are controlled by the **voluntary** (conscious) actions of the nervous system; others work **involuntarily**.

There are **three types of muscle**: **skeletal**, **cardiac** and **smooth**. Only one of these — skeletal muscle — is under the control of the will; it is called **voluntary muscle**. The other two are **involuntary**: **smooth muscle** (the only unstriated one) and **cardiac muscle** (found in the heart; involuntary but striated).


Types of muscles
Cardiac
Smooth
Skeletal


Cardiac muscle cell
Skeletal muscle cell
Smooth muscle cell
A.D.A.M.

11.2 Skeletal (Voluntary / Striated) Muscle

Skeletal muscles are controlled by the **conscious will**. They surround the skeleton and are attached to it by **tendons**; they are concerned with the **movement of the skeleton**. They consist of striated muscle fibres arranged in **bundles**, each bounded by a sheath of connective tissue.

Characteristics

1. Controlled by the **conscious will** (voluntary).
2. Made of **striated** (striped) muscle fibres.
3. Each fibre is long, cylindrical, **unbranched** and has **many nuclei** (multinucleate) lying near the edge.
4. Contracts **quickly and powerfully** but cannot remain contracted for a long time.
5. Is **easily tired (fatigued)**.

Where found: attached to the skeleton (e.g. biceps, triceps, intercostal muscles, jaw muscles).

Functions: (1) breathing movements, e.g. intercostal muscles and diaphragm; (2) movement of limbs; (3) movement of the jaws (chewing); (4) maintenance of body posture; (5) production of heat by shivering.

11.3 Cardiac Muscle

Cardiac muscle is found **only in the heart**. It has the ability to **sustain activity** throughout life.

Characteristics

1. **Involuntary** — it contracts on its own (**myogenic**); nerves only speed it up or slow it down.
2. Made of **striated** muscle fibres.
3. Fibres are **branched** and **short**, joined end to end by **intercalary (intercalated) discs**.
4. Each cell has **one centrally placed nucleus**.
5. Contracts **rapidly and rhythmically**.
6. **Does not get tired** — it keeps up a rhythmic pumping action.

Where found: the heart only. **Function:** pumping blood around the body.

MASTERMIND CORRECTION

The textbook says cardiac muscle "does not require nervous stimulation". More precisely, the heartbeat **starts within the heart muscle itself** (the pacemaker / SA node), so it is **myogenic**. Nerves do not start each beat, but they **do change the rate** of beating (e.g. faster during exercise).

11.4 Smooth (Involuntary / Unstriated / Visceral) Muscle

Smooth muscles **lack striations** and form the walls of internal body organs such as the gut.

Characteristics

1. **Involuntary** — not under conscious control.

2. **Lack striations** (unstriped).
3. Fibres are **spindle-shaped** (elongated with tapered ends), each with **one central nucleus**.
4. Contract **slowly**, and can stay contracted for a **long time without fatigue**.

Where found: the walls of internal organs — the **gut** (alimentary canal), **blood vessels**, the **uterus and vagina**, the **urinary bladder and urethra**, the iris of the eye.

Functions: peristalsis in the gut; changing the diameter of blood vessels; contractions of the uterus during birth; emptying the bladder.

11.5 Comparing the Three Types of Muscle

Feature	Skeletal	Cardiac	Smooth
Control	Voluntary	Involuntary (myogenic)	Involuntary
Striations	Present	Present	Absent
Shape of fibre	Long, cylindrical, unbranched	Short, branched	Spindle-shaped, tapered
Nuclei	Many per fibre, at the edge	One, central	One, central
Intercalary discs	Absent	Present	Absent
Speed of contraction	Fast	Rapid, rhythmic	Slow
Fatigue	Tires easily	Never tires	Does not tire easily
Location	Attached to skeleton	Heart only	Walls of internal organs

Differences between skeletal and cardiac muscle (textbook table)

Skeletal muscle	Cardiac muscle
1. Voluntary.	1. Involuntary.
2. Unbranched fibres.	2. Branched fibres.
3. Long fibres.	3. Shorter fibres.
4. No intercalary discs.	4. Intercalary discs present.
5. Each fibre contains more than one nucleus.	5. Each cell contains a single nucleus.
6. Tires easily.	6. Does not tire.

PART 12: HOW SKELETAL MUSCLES BRING ABOUT MOVEMENT

12.1 Origin and Insertion

Muscles are attached to bones by **tendons** (strong, **inelastic** cords). For a muscle to bring about movement, it must be attached to **two bones**: a **fixed (immovable)** bone and a **movable** bone. *Example:* the scapula of the shoulder is fixed, since it is attached to the axial skeleton, while the radius in the forearm is movable.

- The point of attachment of a muscle to the **fixed** bone is the **origin**.
- The point of attachment to the **movable** bone is the **insertion**.

12.2 Why Muscles Work in Pairs

Muscles can **contract and relax**, and skeletal muscles work by **pulling** on bones. When a muscle contracts, it becomes **shorter and fatter** and pulls the bone into which it is inserted. A muscle can pull a bone in **one direction only** — it cannot push it back, because relaxing does not return the bone to its original position.

For the bone to regain its original position, **another muscle on the opposite side must contract** and pull it back. Muscles therefore act in **pairs**: one contracts to bend the body part and the other contracts to straighten it. A pair of muscles acting in turn, opposite to each other, to bring about movement is called **antagonistic muscles**.

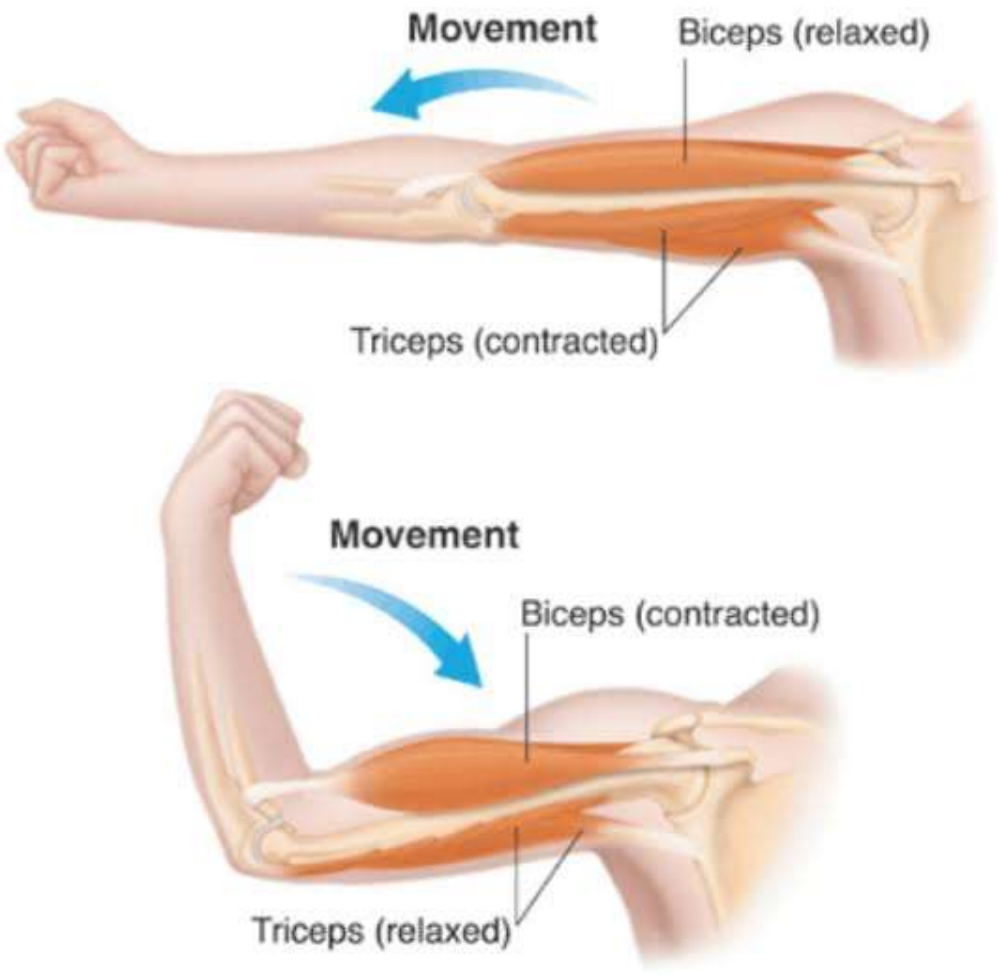
12.3 Flexors and Extensors

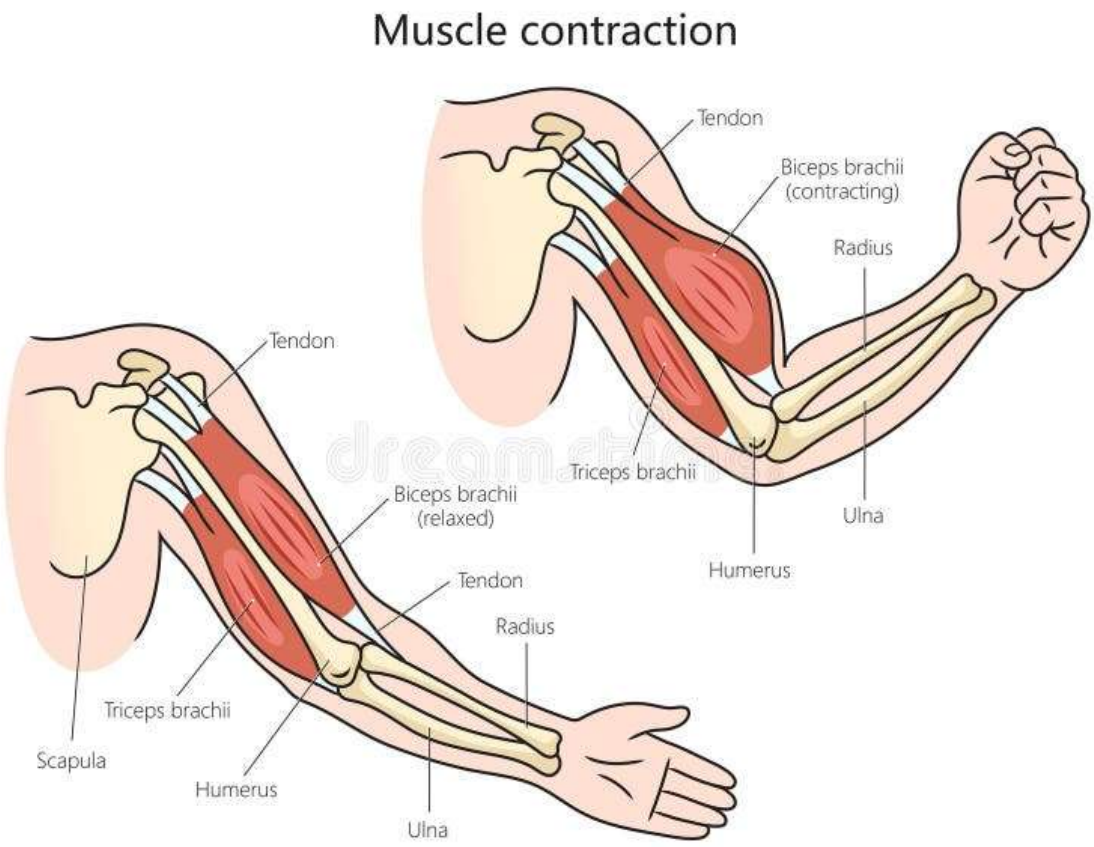
- **Flexor:** the muscle of an antagonistic pair which contracts to **bend** a limb by pulling on it. In the human fore limb, the flexor is the **biceps** ("bi-" = it has **two origins**).
- **Extensor:** the muscle which contracts to **straighten** a limb by pulling on it. In the fore limb, the extensor is the **triceps** ("tri-" = it has **three origins**).

Attachments of the biceps and triceps

Muscle	**Origin (fixed bone)**	**Insertion (movable bone)**	**Action**
Biceps (flexor)	**Two** origins, both on the **scapula**	**Radius**	Bends (flexes) the arm
Triceps (extensor)	**Three** origins: **one on the scapula** and **two on the humerus**	**Ulna** (at the olecranon process)	Straightens (extends) the arm

The humerus forms a **hinge joint** with the ulna at the **elbow**, which serves as the **fulcrum (pivot)**. The bones act as **levers**.


Movement
Biceps (relaxed)
Triceps (contracted)
Movement
Biceps (contracted)
Triceps (relaxed)



12.4 Describing Bending and Straightening of the Arm in Exams

MODEL EXAM ANSWER

(a) Bending the arm (flexion)

1. Bending is brought about by a **pair of skeletal muscles** — biceps and triceps — which work **antagonistically**.
2. It is started by **nervous impulses** carried to the muscle by a **motor neurone**.
3. The **biceps (flexor) contracts** — it shortens and becomes fatter.
4. At the same time the **triceps (extensor) relaxes**.
5. Contraction of the biceps provides a **pulling force on the radius** (its insertion).
6. The forearm is raised about the **elbow joint (fulcrum)** and the arm **bends**.

(b) Straightening the arm (extension)

1. Straightening is also brought about by the antagonistic pair, started by **nervous stimulation through a motor neurone**.
2. The **triceps (extensor) contracts**.
3. At the same time the **biceps relaxes**.
4. Contraction of the triceps provides a **pulling force on the ulna** at the **olecranon process**.
5. The forearm is pulled down about the elbow and the arm **straightens**.

MASTERMIND CORRECTION

Note that : The two muscles work **in coordination but opposite to each other**: when one **contracts**, the other **relaxes** at the same time. They do **not both contract** together — if they did, the arm would not move (see "muscle pull" below). The textbook's printed notes for straightening also repeat "to bend the arm" — it should read **"to straighten the arm"**.

NOTE

Muscles need **energy** for contraction. This energy is obtained from **tissue (cellular) respiration** in the form of **ATP**. During very hard exercise, muscles may respire **anaerobically**, producing **lactic acid**, which causes **fatigue and cramps** and builds up an **oxygen debt**.

THINK & ANSWER

Q: What are antagonistic muscles?

A: A **pair of muscles which work opposite to each other** to bring about movement: when one contracts, the other relaxes. *Example:* the biceps (flexor) and triceps (extensor) of the upper arm.

12.5 Muscle Pull

The textbook describes **muscle pull** as a condition that occurs when a pair of antagonistic muscles **contract at the same time** — both pull, so the person finds it difficult to move the limb in any direction.

NOTE

In everyday and medical use, a "pulled muscle" (**muscle strain**) means **over-stretching or tearing of muscle fibres**, usually through sudden or excessive effort. It causes pain, swelling and difficulty using the muscle. For WASSCE, give the textbook definition, and you may add the strain meaning.

12.6 Tissues and Structures Involved in Movement in Mammals

Tissue / structure	Role in movement
1. **Bone**	Rigid levers on which muscles pull; forms the framework.
2. **Cartilage**	Covers bone ends at joints; reduces friction; absorbs shock.
3. **Ligament**	Joins bone to bone at joints; holds bones in place, prevents dislocation.

Tissue / structure	Role in movement
4. **Muscle**	Contracts to pull on bones and produce movement.
5. **Tendon**	Joins muscle to bone; transmits the pull of the muscle to the bone.
6. *Nerve* (added)	Carries impulses that stimulate and coordinate muscle contraction.

Tendons versus ligaments

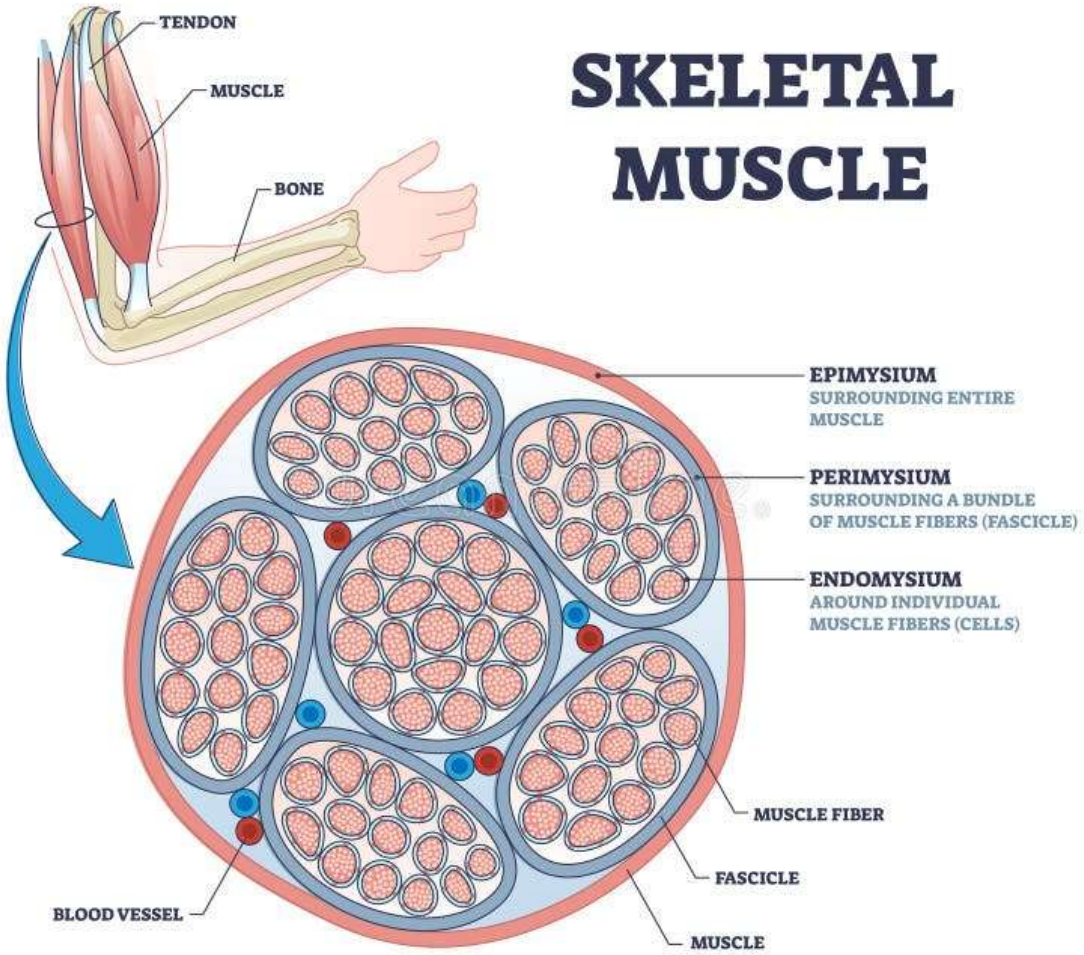
Tendon	Ligament
Joins **muscle to bone**.	Joins **bone to bone**.
Inelastic — transmits the whole pull of the muscle.	**Elastic** (slightly stretchy) — allows movement at the joint.
Made mainly of white **collagen** fibres.	Contains yellow **elastic** fibres as well as collagen.

PART 13: STRUCTURE OF SKELETAL MUSCLE AND THE SLIDING FILAMENT MODEL

13.1 Structure of a Skeletal Muscle

A skeletal muscle is a **tissue** made up of a bundle of similar cells. The cells are called **muscle fibres**. From the largest to the smallest:

Muscle → bundles of muscle fibres → muscle fibre (cell) → myofibril → myofilaments (actin and myosin)



A muscle fibre is a typical animal cell, but its parts have **special names**:

Ordinary cell	**Muscle fibre**
Cytoplasm	**Sarcoplasm**
Cell membrane	**Sarcolemma**
Endoplasmic reticulum	**Sarcoplasmic reticulum** (stores calcium ions)
Nucleus	Many nuclei (multinucleate)

Each muscle fibre contains long protein structures called **myofibrils** in its sarcoplasm. There are also many **mitochondria** between the myofibrils to supply ATP. Each myofibril is made up of two types of **contractile protein filaments**:

- **Myosin** — the **thick** filament, with protein **heads** that project from it.
- **Actin** — the **thin** filament.

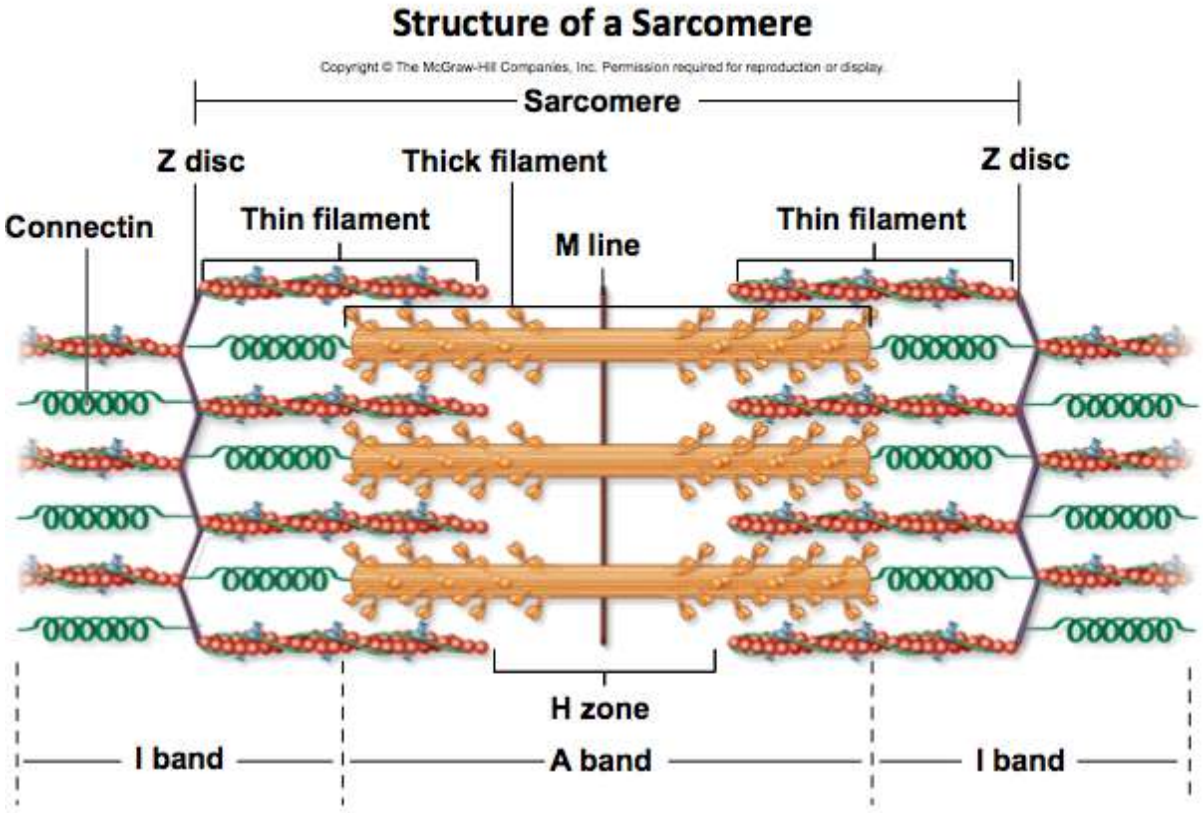
Myosin and actin are arranged **in parallel** and **overlap**, with actin filaments on each side of the myosin filaments.

13.2 The Sarcomere

The **sarcomere** is the **functional (smallest contracting) unit** of a myofibril. A myofibril is made up of many sarcomeres joined end to end. The overlapping actin and myosin divide the sarcomere into **bands, zones and lines**:

Region	What it is	Appearance
A band	The **entire length of the thick myosin** filaments (including where actin overlaps).	Dark (anisotropic)
I band	The part of the thin **actin that does not overlap** with myosin. There are **two** I bands, one on each side of the A band (each shared with the next sarcomere).	Light (isotropic)
H zone	The central part of the **myosin that does not overlap** with actin.	Lighter region in the middle of the A band
M line	A line of protein in the centre that **joins all the myosin filaments** together.	Thin dark line in the H zone
Z line	A line which **joins (anchors) the actin filaments** of adjacent sarcomeres.	Dark line at each end

A sarcomere therefore extends **from one Z line to the next Z line**. The myosin and actin are linked at the region of overlap by **cross bridges** formed from the **protein heads of myosin**. The overlap regions appear as dark bands; the I band is light — this pattern gives skeletal muscle its **striations**.



EXAM TIP

Memory aids: the **A** band is d**A**rk and is the length of myosin (**A** never ch**A**nges); the **I** band is l**I**ght; the **H** zone is in the middle (German *hell* = bright); **Z** lines are at the ends (Z is the end of the alphabet); **M** line is in the **M**iddle.

13.3 The Sliding Filament Model of Muscle Contraction

What brings about contraction? Contraction is caused by a **reduction in the length of the sarcomeres**. For a sarcomere to shorten, the **Z lines must move closer together**. To achieve this, the **actin filaments slide over the myosin filaments** towards the centre (M line). Because the actin filaments are attached to the Z lines, the Z lines are pulled inwards and the sarcomere shortens. This happens in all sarcomeres at once, so the whole myofibril, muscle fibre and muscle shorten. **The filaments themselves do not shorten — they slide.**

What makes the actin move?

- The **protein heads of myosin** attach to **binding (active) sites** on the actin filament, forming a **cross bridge** (the **actomyosin / actinomyosin complex**).
- The myosin heads then **tilt (swivel)** and pull the actin filaments **inwards** — this is the **power stroke**.
- At rest, the binding sites on actin are **covered** by a protein called **tropomyosin** (held in place by **troponin**), so cross bridges cannot form.
- **Calcium ions (Ca^{2+})** uncover the binding sites. Calcium ions are stored in the **sarcoplasmic reticulum** and are released only when the muscle is stimulated by a **motor neurone**.
- **ATP** is needed to **detach** the myosin head from actin and to **re-cock** it ready for the next stroke.

MODEL ANSWER — Describe the sliding filament model of muscle contraction (typical compulsory question)

1. A **nerve impulse** arrives along a **motor neurone** at the **neuromuscular junction**.
2. A **neurotransmitter** (acetylcholine) is released and **diffuses across the junction** to the **sarcolemma**.
3. The neurotransmitter **excites (depolarises) the muscle fibre**; the impulse spreads into the fibre.
4. The excitation makes the **sarcoplasmic reticulum release calcium ions** into the sarcoplasm.
5. Calcium ions bind to **troponin**, which moves **tropomyosin** away and **exposes the binding (active) sites** on the actin filament.
6. The **myosin heads attach** to the exposed sites, forming **cross bridges** (actomyosin complex).

7. The myosin heads **tilt**, pulling the **actin filaments inwards** towards the M line (**power stroke**); actin **slides over** myosin.
8. **ATP binds** to the myosin head, causing it to **detach** from actin (the cross bridge breaks).
9. The ATP is **hydrolysed to ADP + inorganic phosphate (Pi)**, releasing energy that **re-cocks** the myosin head.
10. The head **re-attaches further along** the actin and pulls again. The cycle is **repeated many times** (a "ratchet" action).
11. Actin moves inwards → **Z lines are pulled closer** → **I band and H zone shorten** → **sarcomere shortens** → myofibril, fibre and whole **muscle contract**.

Relaxation: when stimulation stops, calcium ions are **actively pumped back** into the sarcoplasmic reticulum (using ATP); tropomyosin covers the binding sites again, cross bridges cannot form, and the filaments slide back apart.

MASTERMIND CORRECTION

The textbook says ATP hydrolysis releases energy "to detach the heads". More precisely: **binding of ATP** makes the myosin head **detach** from actin, and **hydrolysis of ATP** releases energy to **re-cock** the head for the next stroke. Both are needed for the cycle to continue. (This is why muscles lock stiff after death — **rigor mortis** — when ATP runs out and the heads cannot detach.) The textbook also says calcium ions react with "a chemical" covering the active site; name it as **tropomyosin (with troponin)**.

Recap — the cross-bridge cycle in 7 steps

Step	Event
1	Calcium ions expose the binding sites on actin.
2	Myosin heads attach — cross bridge formed.
3	Power stroke — heads tilt and pull actin inwards.
4	ATP binds to the myosin head.
5	Actomyosin complex breaks — head detaches.
6	ATP is hydrolysed (ADP + Pi) — head is re-cocked.
7	Cross bridge re-forms further along; strokes continue.

13.4 Changes in the Sarcomere during Contraction

Part of sarcomere	Change during contraction	Reason
Z lines	Move **closer together** (pulled inwards)	Actin attached to Z lines is pulled inwards
Sarcomere	**Shortens**	Z lines move closer
A band	**Remains the same** length	Myosin filaments do not change length
I band	**Shortens**	More actin overlaps myosin
H zone	**Shortens / may disappear**	Actin slides into the central zone
Actin and myosin filaments	**Stay the same** length	They slide; they do not shorten

PART 14: DAMAGE AND ACCIDENTS INVOLVING BONES, JOINTS AND MUSCLES

14.1 Fracture

A **fracture** is a condition in which a **bone is broken** (cracked or broken through).

Symptoms of a fracture

1. **Swelling** of the affected part.
2. **Abnormal appearance** of the affected part, e.g. bending in the wrong position, or an apparent increase/decrease in length.
3. **Pain** at the affected part, especially on movement.
4. **Difficulty in using** the part (loss of function).
5. Sometimes bruising, or a grating feeling/sound when the broken ends rub.

Types of fracture

Type	Description
Simple (closed) fracture	The bone breaks but does **not pierce the skin**.
Compound (open) fracture	The broken bone **pierces through the skin** — high risk of infection.
Greenstick fracture	The bone **bends and cracks on one side only**; common in children, whose bones are still flexible.
Comminuted fracture	The bone is **shattered into several pieces**.

First aid for a fracture

1. Keep the casualty **calm and still**; do not try to push the bone back.
2. **Immobilise** the part with a **splint** (e.g. a straight stick or board padded with cloth) and bandage it, or support an arm in a **sling**.
3. For an open fracture, **cover the wound** with a clean dressing to reduce infection and control bleeding.

4. Apply a cold compress to reduce swelling.
5. Take the casualty to a **hospital** quickly for an X-ray and proper setting (plaster cast).

14.2 Other Disorders of the Skeleton and Muscles

Condition	What happens	Cause / prevention
Dislocation	A bone is forced **out of its joint** (e.g. shoulder).	Sudden twist or fall; set by a medical worker.
Sprain	**Ligaments** at a joint are over-stretched or torn (e.g. ankle).	Sudden twisting; rest, ice, compression, elevation.
Arthritis	Articular cartilage wears away; joints become **painful, swollen and stiff**.	Ageing, injury, infection.
Rickets (children) / osteomalacia (adults)	Soft, weak, **bent bones** (bow legs).	Lack of **vitamin D**, calcium or phosphorus; prevented by sunlight, milk, fish.
Osteoporosis	Bones become **thin, porous and brittle**; break easily.	Ageing (esp. women after menopause), lack of calcium; prevented by calcium-rich diet and exercise.
Muscle cramp	Sudden, painful, sustained **contraction** of a muscle.	Fatigue, lactic acid, salt loss in sweat.
Muscle strain (pulled muscle)	Muscle fibres **over-stretched or torn**.	Sudden effort; warm up before exercise.

EXAM TIP

To keep bones and muscles healthy: eat foods rich in **calcium** (milk, small fish like "kobi" and anchovies, green vegetables) and **vitamin D** (sunlight, eggs, fish oil); **exercise regularly**; keep good **posture**; and **warm up** before sports.

14.3 Trial Questions on Muscle Structure (with answers)

No.	Question	Answer
1	The contractile unit of the myofibril is the ……	Sarcomere
2	The smallest (cellular) unit of a muscle is the ……	Muscle fibre
3	Inside the …… are myofibrils.	Sarcoplasm of the muscle fibre
4	Arrange from largest to smallest: fibre, myofibril, muscle, myofilament.	Muscle > fibre > myofibril > myofilament
5	A myofibril is made up of two types of myofilaments, namely …… and ……	Myosin and actin
6	The functional unit of the myofibril is ……	Sarcomere
7	A myofibril is divided into sarcomeres by …… lines.	Z lines
8	Adjacent actin filaments are held by a ……	Z line

No.	Question	Answer
9	Myosin filaments in a sarcomere are held by ……	M line
10	A contracting muscle …… in length.	Reduces (shortens)
11	When a muscle contracts, the length of the …… band remains unchanged.	A band
12	The …… of the myosin filament is the A band.	Entire length
13	The …… of the actin filament is the I band.	Part that does not overlap with myosin
14	The …… of myosin is the H zone.	Part that does not overlap with actin

QUICK REVISION GLOSSARY

Term	Meaning
Limb girdle	Bones which attach the limbs to the axial skeleton.
Innominate bone	One half of the pelvic girdle: fused ilium, ischium and pubis.
Pubic symphysis	Line of fusion of the two halves of the pelvic girdle.
Obturator foramen	Hole between ischium and pubis for nerves and blood vessels.
Pelvis	Pelvic girdle fused with the sacrum.
Synovial joint	Freely movable joint with a cavity containing synovial fluid.
Origin / insertion	Attachment of a muscle to a fixed bone / to the movable bone.
Flexor / extensor	Muscle that bends / straightens a limb.
Antagonistic muscles	Pair of muscles working opposite each other.
Sarcolemma, sarcoplasm, sarcoplasmic reticulum	Membrane, cytoplasm and endoplasmic reticulum of a muscle fibre.
Myofibril	Long contractile structure in a muscle fibre, made of sarcomeres.
Sarcomere	Contracting unit of a myofibril, from Z line to Z line.
Cross bridge	Attachment of a myosin head to actin (actomyosin complex).
Neuromuscular junction	Junction between a motor neurone and a muscle fibre.
Fracture	A broken bone.

SECTION A: OBJECTIVE TEST (40 QUESTIONS)

Instructions: Choose the correct option (A, B, C or D) for each question.

1. Bones which attach the limbs to the axial skeleton are called
A. limb girdles
B. phalanges
C. ribs
D. vertebrae

2. Which of the following bones is part of the pectoral girdle?
A. Ischium
B. Scapula
C. Ilium
D. Pubis

3. The collar bone is also known as the
A. sternum
B. scapula
C. coracoid
D. clavicle

4. The head of the humerus articulates with the
A. obturator foramen
B. sigmoid notch
C. glenoid cavity
D. acetabulum

5. The acromion process provides a surface for articulation with the
A. humerus
B. clavicle
C. radius
D. sternum

6. The pectoral girdle is attached to the axial skeleton by
A. fusion with the sacrum
B. muscles
C. cartilage only
D. the pubic symphysis

7. Each half of the pelvic girdle is called the
A. sacrum
B. coccyx
C. scapula
D. innominate bone

8. The innominate bone is made up of the
A. ilium, ischium and pubis
B. scapula, coracoid and clavicle
C. femur, tibia and fibula
D. sacrum and coccyx

9. The bone that fuses the pelvic girdle to the vertebral column is the
A. ilium
B. femur
C. pubis
D. ischium

10. The two halves of the pelvic girdle fuse ventrally at the
A. acetabulum
B. pubic symphysis
C. obturator foramen
D. glenoid cavity

11. The hole between the ischium and pubis is the
A. obturator foramen
B. acetabulum
C. neural canal
D. supratrochlear foramen

12. The pelvic girdle fused with the sacrum forms the
A. pelvis
B. rib cage
C. pectoral girdle
D. vertebral column

13. Which of the following is a difference between the pectoral and pelvic girdles?
A. The glenoid cavity is shallow while the acetabulum is deep
B. Both are fused to the vertebral column
C. The pectoral halves are fused ventrally
D. The pelvic girdle attaches the fore limbs

14. Greater and lesser tuberosities are found on the
A. ulna
B. scapula
C. humerus
D. femur

15. The metacromion process is found on the
A. ilium
B. scapula
C. clavicle
D. humerus

16. A point where two or more bones meet is a
A. girdle
B. ligament
C. joint
D. tendon

17. The joints between the bones of the cranium are
A. hinge joints
B. immovable joints
C. pivot joints
D. gliding joints

18. The joints between the carpals of the wrist are
A. hinge joints
B. gliding joints
C. pivot joints
D. ball and socket joints

19. The joint between the atlas and the axis is a
A. hinge joint
B. pivot joint
C. fixed joint
D. gliding joint

20. The knee joint is an example of a
A. ball and socket joint
B. pivot joint
C. gliding joint
D. hinge joint

21. Movement in all planes is possible at the
A. knee joint
B. elbow joint
C. finger joint
D. hip joint

22. At the elbow, the trochlea of the humerus articulates with the
A. sigmoid notch of the ulna
B. head of the radius
C. acetabulum
D. glenoid cavity

23. The fluid that lubricates a movable joint is secreted by the
A. synovial membrane
B. tendon
C. ligament
D. articular cartilage

24. The function of the articular cartilage is to
A. secrete synovial fluid
B. absorb shock and reduce friction
C. join bone to bone
D. join muscle to bone

25. Ligaments join
A. muscle to muscle
B. muscle to bone
C. nerve to muscle
D. bone to bone

26. Breakdown of articular cartilage leading to painful, stiff joints is called

A. arthritis B. rickets
C. fracture D. cramp

27. Muscle found only in the heart is

A. smooth muscle B. cardiac muscle
C. voluntary muscle D. skeletal muscle

28. Intercalary discs are a feature of

A. smooth muscle B. skeletal muscle
C. cardiac muscle D. tendons

29. Which type of muscle is responsible for peristalsis in the gut?

A. Cardiac muscle B. Smooth muscle
C. Voluntary muscle D. Skeletal muscle

30. Skeletal muscle is described as

A. involuntary and striated B. voluntary and unstriated
C. voluntary and striated D. involuntary and unstriated

31. The point of attachment of a muscle to a fixed bone is the

A. insertion B. tendon
C. origin D. fulcrum

32. The biceps muscle is inserted on the

A. radius B. scapula
C. humerus D. ulna

33. The triceps has

A. four origins B. one origin
C. three origins D. two origins

34. When the arm is straightened, the

A. both muscles relax B. both muscles contract
C. biceps contracts and triceps relaxes D. triceps contracts and biceps relaxes

35. A muscle which contracts to bend a limb is a

A. extensor B. tendon
C. ligament D. flexor

36. The cytoplasm of a muscle fibre is called the

A. sarcolemma B. sarcoplasm
C. myofibril D. sarcomere

37. The functional contracting unit of a myofibril is the

A. myosin B. Z line
C. sarcomere D. muscle fibre

38. The thick filament of a myofibril is

A. collagen B. actin
C. myosin D. tropomyosin

39. During muscle contraction, the band that remains unchanged in length is the

A. A band
B. H zone
C. sarcomere
D. I band

40. Calcium ions for muscle contraction are released from the

A. sarcoplasmic reticulum
B. Z line
C. nucleus
D. sarcolemma

SECTION B: THEORY QUESTIONS

Instructions: Answer all questions. Draw diagrams neatly in pencil, label them fully and give each a title.

Question 1

(a) What is a limb girdle? **[2 marks]**

(b) Describe the structure of the pectoral girdle of a mammal. **[6 marks]**

(c) Make a large, labelled drawing of the pelvic girdle of a rabbit. **[7 marks]**

(d) State four differences between the pectoral girdle and the pelvic girdle. **[4 marks]**

(e) State three functions of the pelvis. **[3 marks]**

Question 2

(a) Define a joint. **[1 mark]**

(b) Name five types of joints and give one example of each. **[10 marks]**

(c) Make a labelled diagram of a typical synovial joint. **[6 marks]**

(d) State the function of each of the following: (i) ligament; (ii) articular cartilage; (iii) synovial membrane; (iv) synovial fluid. **[4 marks]**

Question 3

(a) Describe the shoulder joint and the elbow joint. **[8 marks]**

(b) State three differences between the shoulder joint and the elbow joint. **[3 marks]**

(c) Explain why the hip joint is less easily dislocated than the shoulder joint. **[3 marks]**

Question 4

(a) Name the three types of muscle and state where each is found in the body. **[6 marks]**

(b) State four characteristics each of (i) skeletal muscle; (ii) cardiac muscle. **[8 marks]**

(c) State four differences between skeletal muscle and smooth muscle. **[4 marks]**

Question 5

(a) What are antagonistic muscles? **[2 marks]**

(b) Distinguish between (i) origin and insertion; (ii) flexor and extensor. **[4 marks]**

(c) With the aid of labelled diagrams, describe how the arm is bent and straightened. **[10 marks]**

(d) State three differences between tendons and ligaments. **[3 marks]**

Question 6 (Compulsory-type)

(a) Make a labelled diagram of a sarcomere. **[6 marks]**

(b) Describe the sliding filament model of muscle contraction. **[10 marks]**

(c) State what happens to each of the following during contraction: (i) A band; (ii) I band; (iii) H zone; (iv) Z lines. **[4 marks]**

Question 7

(a) What is a fracture? **[1 mark]**

(b) State four symptoms of a fracture. **[4 marks]**

(c) Distinguish between a simple fracture and a compound fracture. **[2 marks]**

(d) Describe the first aid given to a person with a fractured arm. **[4 marks]**

(e) Name the vitamin whose deficiency causes rickets, and state two sources of it. **[3 marks]**

ANSWER SECTION (DETACH BEFORE ISSUING TO STUDENTS IF REQUIRED)

Answer Key — Objective Test

No.	Ans.	Reason	No.	Ans.	Reason
1	**A**	Definition.	21	**D**	Ball and socket.
2	**B**	Shoulder girdle bone.	22	**A**	Elbow hinge.
3	**D**	Common name.	23	**A**	Secretes fluid.
4	**C**	Shoulder joint socket.	24	**B**	Cushion.
5	**B**	Joins clavicle.	25	**D**	Bone–bone.
6	**B**	Not fused.	26	**A**	Cartilage worn.
7	**D**	Hip bone half.	27	**B**	Heart only.
8	**A**	Three fused bones.	28	**C**	Join cardiac cells.
9	**A**	Largest; fuses to sacrum.	29	**B**	Involuntary gut wall.
10	**B**	Ventral line of fusion.	30	**C**	Under will; striped.
11	**A**	Nerves and vessels pass.	31	**C**	Fixed bone.
12	**A**	Pelvis = girdle + sacrum.	32	**A**	Flexor pulls radius.
13	**A**	Structure fits function.	33	**C**	Scapula + 2 humerus.
14	**C**	Textbook hint wrong.	34	**D**	Extensor contracts.
15	**B**	Rabbit scapula.	35	**D**	Bends.
16	**C**	Definition.	36	**B**	Special name.
17	**B**	Sutures.	37	**C**	Z to Z.
18	**B**	Slide over each other.	38	**C**	Thick.
19	**B**	Rotation.	39	**A**	Myosin length fixed.
20	**D**	One plane.	40	**A**	Ca^{2+} store.

Answer distribution — A: 12 B: 12 C: 9 D: 7

Model Answers — Theory

Question 1

(a) A limb girdle is a set of bones which **attach the limbs to the axial skeleton**.

(b) Two halves, each made of scapula, coracoid and clavicle. Scapula: flat, triangular, with a spine; glenoid cavity at its apex for the head of the humerus; acromion and metacromion processes for muscle attachment (acromion also articulates with the clavicle); coracoid fused at the apex as the coracoid process. Clavicle joins the acromion to the sternum. The girdle is held to the sternum ventrally and to the vertebral column dorsally by **muscles**, not fused.

(c) Diagram as Fig. 31. *Marks:* title (1); ilium (1); acetabulum (1); obturator foramen (1); pubic symphysis (1); ischium and pubis (1); accuracy and neatness (1).

(d) Any four rows of the girdles table (flexibly attached/fused; bones not fused/fused; halves not fused/fused at pubic symphysis; shallow glenoid/deep acetabulum; no/has obturator foramen).

(e) Protects abdominal and pelvic organs (bladder, female reproductive organs); supports and transmits body weight to hind limbs; acetabulum articulates with femur; obturator foramen lets nerves and blood vessels pass; attachment of muscles; forms birth canal in females.

Question 2

(a) A joint is a point where **two or more bones meet**.

(b) Immovable — cranium sutures; slightly movable — between vertebral centra; gliding — wrist/ankle; pivot — atlas and axis; hinge — elbow/knee; ball and socket — shoulder/hip. *(1 mark name, 1 mark example; any five.)*

(c) Diagram as Fig. 34: bone ×2, ligament/capsule, synovial membrane, synovial fluid (cavity), articular cartilage; title.

(d) (i) Holds bones together and prevents dislocation. (ii) Absorbs shock and reduces friction. (iii) Secretes synovial fluid. (iv) Lubricates the joint and reduces friction.

Question 3

(a) Shoulder: ball and socket joint; round head of humerus fits into the shallow glenoid cavity of the scapula; loose capsular ligaments join the bones; articular cartilage covers surfaces; synovial membrane lines the capsule and secretes synovial fluid. **Elbow:** hinge joint between humerus and ulna; trochlea of humerus fits into the sigmoid notch of ulna; capsular ligaments, articular cartilage, synovial membrane and fluid as above; olecranon process prevents backward bending.

(b) Ball-and-socket/hinge; all planes/one plane; humerus + scapula/humerus + ulna; loose/tighter ligaments; no bony stop/olecranon stop.

(c) The acetabulum is a **deep** socket that holds the head of the femur firmly, and the hip is surrounded by very strong ligaments and muscles. The glenoid cavity is **shallow** and the shoulder ligaments are **loose**, so the humerus can slip out more easily.

Question 4

(a) Skeletal — attached to the skeleton (e.g. biceps); cardiac — heart only; smooth — walls of internal organs such as gut, blood vessels, uterus, bladder.

(b)(i) Voluntary; striated; long, unbranched, multinucleate fibres; contracts quickly and powerfully; tires easily. **(ii)** Involuntary (myogenic); striated; short branched fibres joined by intercalary discs; one central nucleus; contracts rhythmically; never tires.

(c) Skeletal voluntary/smooth involuntary; striated/unstriated; long cylindrical multinucleate fibres/spindle-shaped with one nucleus; fast contraction and tires easily/slow sustained contraction without fatigue; attached to bones/walls of organs.

Question 5

(a) A pair of muscles which work opposite to each other to bring about movement — when one contracts, the other relaxes (e.g. biceps and triceps).

(b)(i) Origin — attachment of a muscle to a fixed bone; insertion — attachment to the movable bone. **(ii)** Flexor — contracts to bend a limb (biceps); extensor — contracts to straighten a limb (triceps).

(c) Diagrams as Figs 38 and 39. *Bending:* impulse via motor neurone; biceps contracts, triceps relaxes; biceps (origins on scapula) pulls the radius; forearm rises about the elbow (fulcrum). *Straightening:* triceps contracts, biceps relaxes; triceps pulls the ulna at the olecranon process; arm straightens. Muscles only pull, so they work as an antagonistic pair.

(d) Tendon joins muscle to bone/ligament joins bone to bone; tendon inelastic/ligament elastic; tendon mainly collagen/ligament has elastic fibres.

Question 6

(a) Diagram as Fig. 41: Z lines, M line, A band, I bands, H zone, actin, myosin, myosin heads (cross bridges); title.

(b) Award 1 mark per point up to 10, following the Model Answer in Section 13.3: impulse along motor neurone → neurotransmitter released at neuromuscular junction → diffuses to sarcolemma → fibre excited → sarcoplasmic reticulum releases Ca^{2+} → Ca^{2+} binds troponin, tropomyosin moves, binding sites exposed → myosin heads attach (cross bridges) → power stroke pulls actin inwards → ATP binds, head detaches → ATP hydrolysed, head re-cocked, reattaches further along → Z lines pulled closer, sarcomere shortens, muscle contracts.

(c) (i) A band — remains the same. (ii) I band — shortens. (iii) H zone — shortens / disappears. (iv) Z lines — move closer together.

Question 7

(a) A fracture is a condition in which a bone is broken or cracked.

(b) Swelling; abnormal appearance (bending in wrong position, change in length); pain at the affected part; difficulty in using the part; bruising or grating sound.

(c) In a simple (closed) fracture the broken bone does not pierce the skin; in a compound (open) fracture the broken bone pierces the skin.

(d) Keep the casualty still and calm; do not try to straighten the bone; support the arm with a padded splint and bandage, and place it in a sling; cover any wound with a clean dressing; apply cold compress; send to hospital quickly.

(e) Vitamin D. Sources: sunlight on the skin, fish/fish-liver oil, eggs, milk, butter.